

FOSS Bonsai & Blender

Maarten Vroegindeweij

3BM Consulting Engineers



THUAS

THE HAGUE

23 – 25 APRIL 2025

Agenda

Introduction

Free Open Source Bonsai | What & Why?

Project Context

Approach

Modeling & Drawing Generation

Collaboration

Challenges & Solutions

Future of Bonsai

Questions



About the speaker



Ingenieursbureau 3BM
Co-founder



Domera
Founder



DORDRECHT ACADEMY

BBE: Maths, Mechanics



Maarten Vroegindeweij

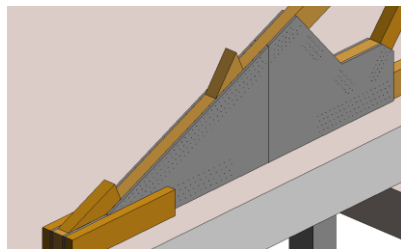
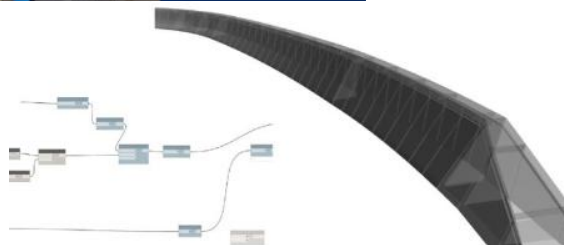
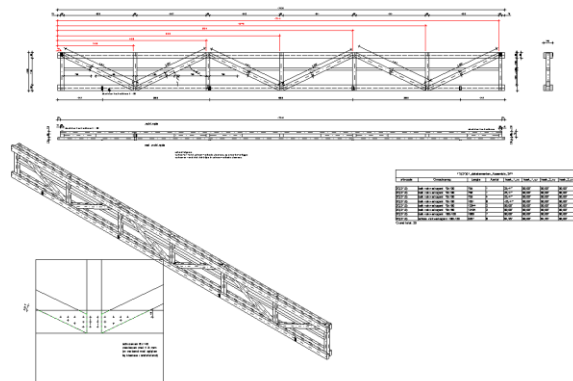
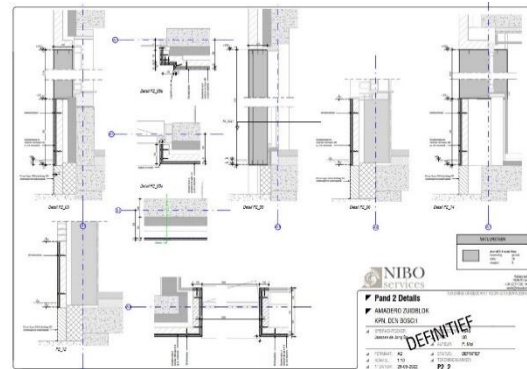
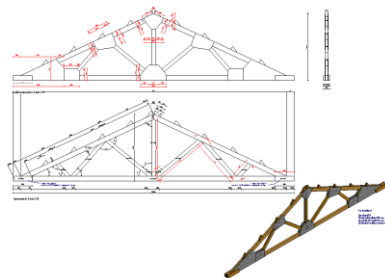
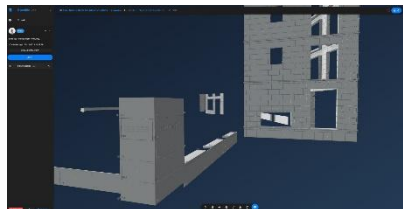
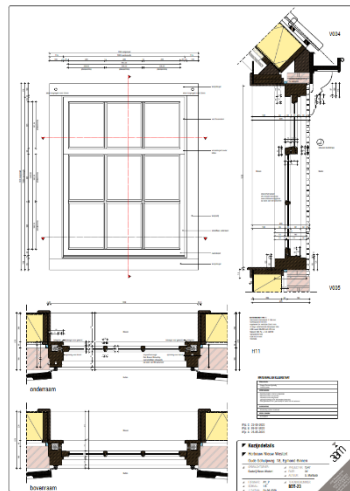
<https://www.linkedin.com/in/maarten-vroegindeweij-652ab418a/>



3BM Consulting Engineers



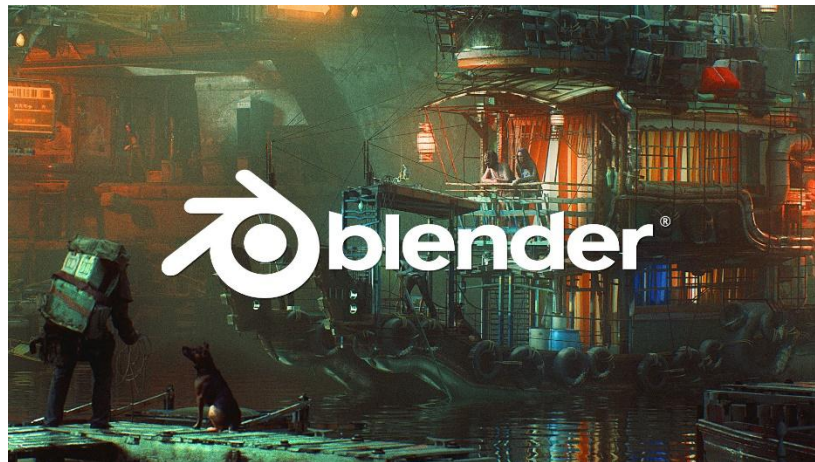
3BM



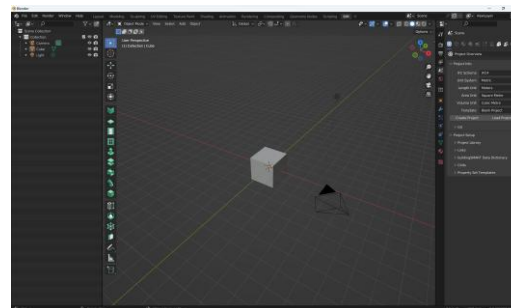
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Blender

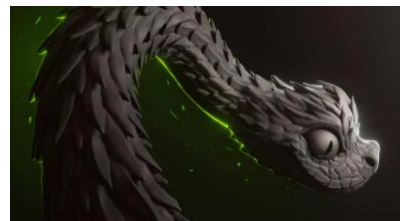
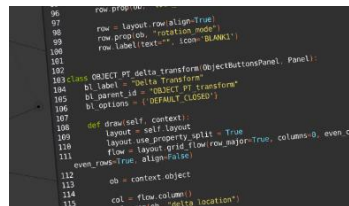
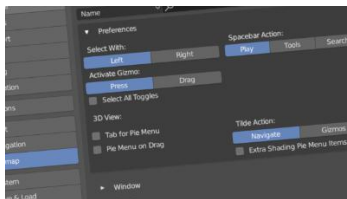


 Julian Eisel Blender Developer Germany	 Julien Kaspar 3D Artist - Blender Studio Germany	 Lukas Tönne Blender Developer Germany
 Maaïke Kleverlaan Filmmaker The Netherlands	 Nathan Vegdahl Blender Developer United States	 Pablo Fournier Animator - Blender Studio Spain
 Pablo Vazquez Design and Communication Argentina	 Rik Schutte Lead Animator - Blender Studio The Netherlands	 Sebastian Parborg Blender Developer Sweden
 Sergey Sharybin Principal Engineer Russia	 Simon Thommes 3D Artist - Blender Studio Germany	 Sybren Stüvel Blender Developer The Netherlands
 Ton Roosendaal CEO The Netherlands	 Vivien Lulkowski Concept Artist - Blender Studio Germany	 Weizhen Huang Blender Developer China



Blender

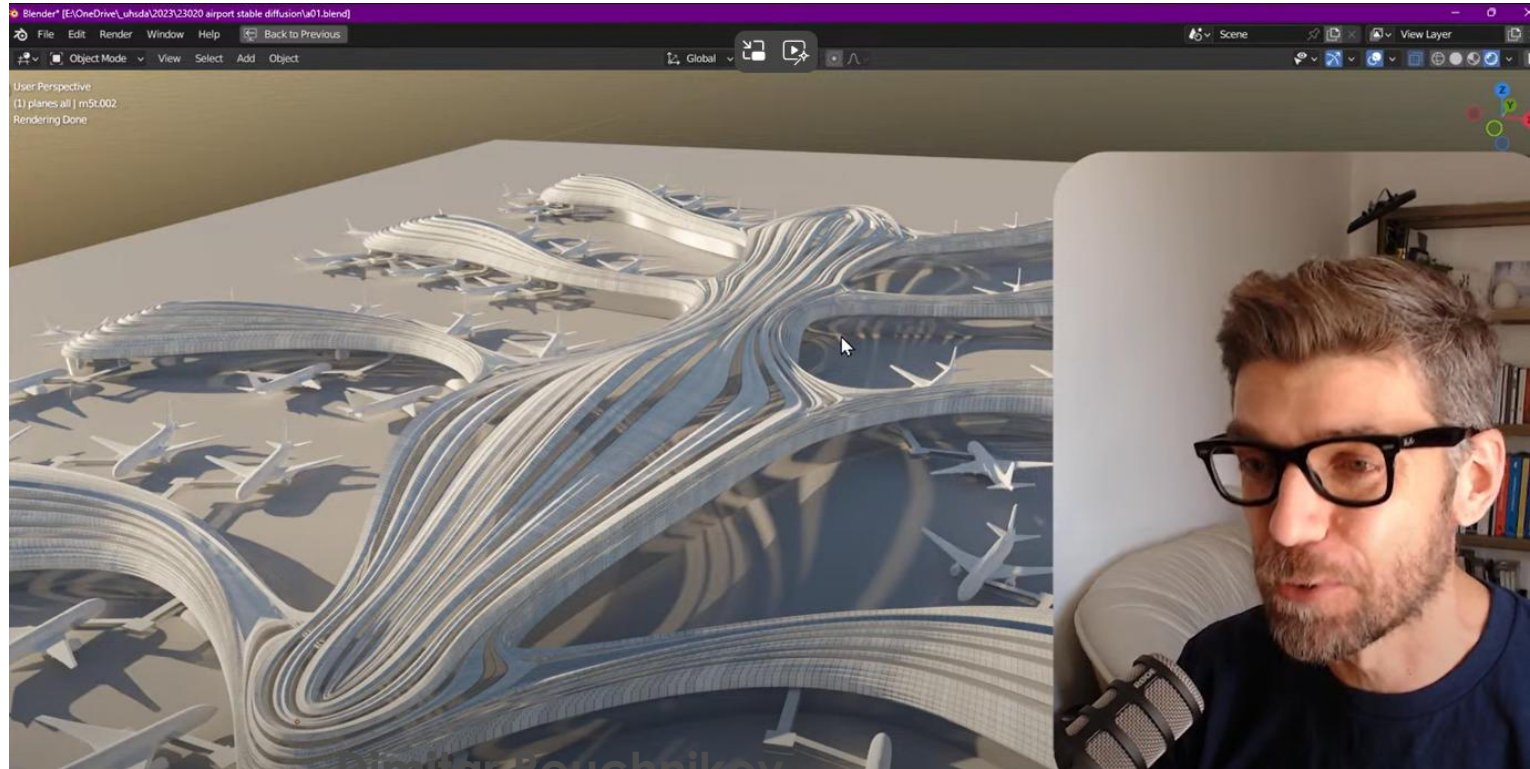
- Rendering
- Modelling
- Sculpting
- Animation & Rigging
- Story Artist
- VFX
- Simulation
- Pipeline
- Video Editing
- Scripting: Python API
- Interface fully customizable



Blender for architecture



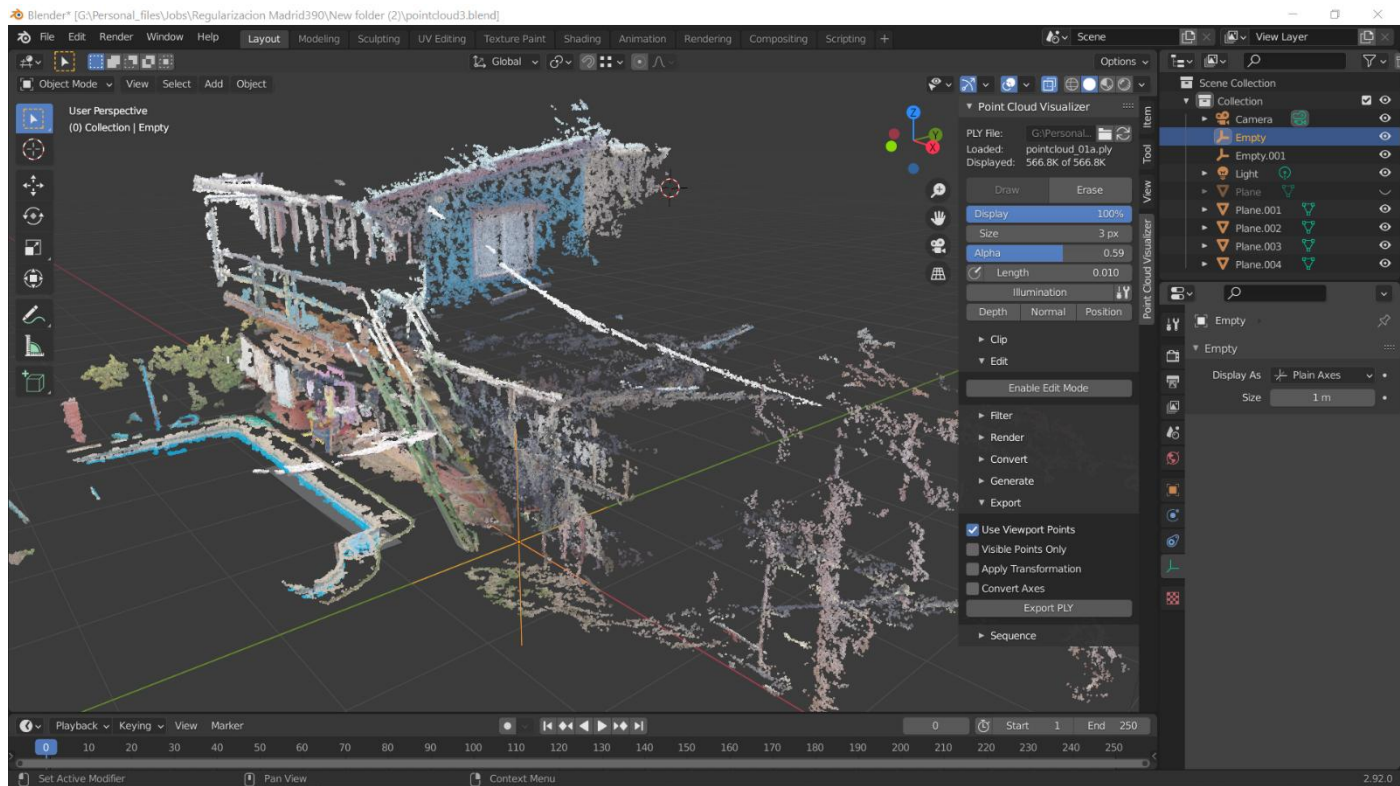
Blender for architecture



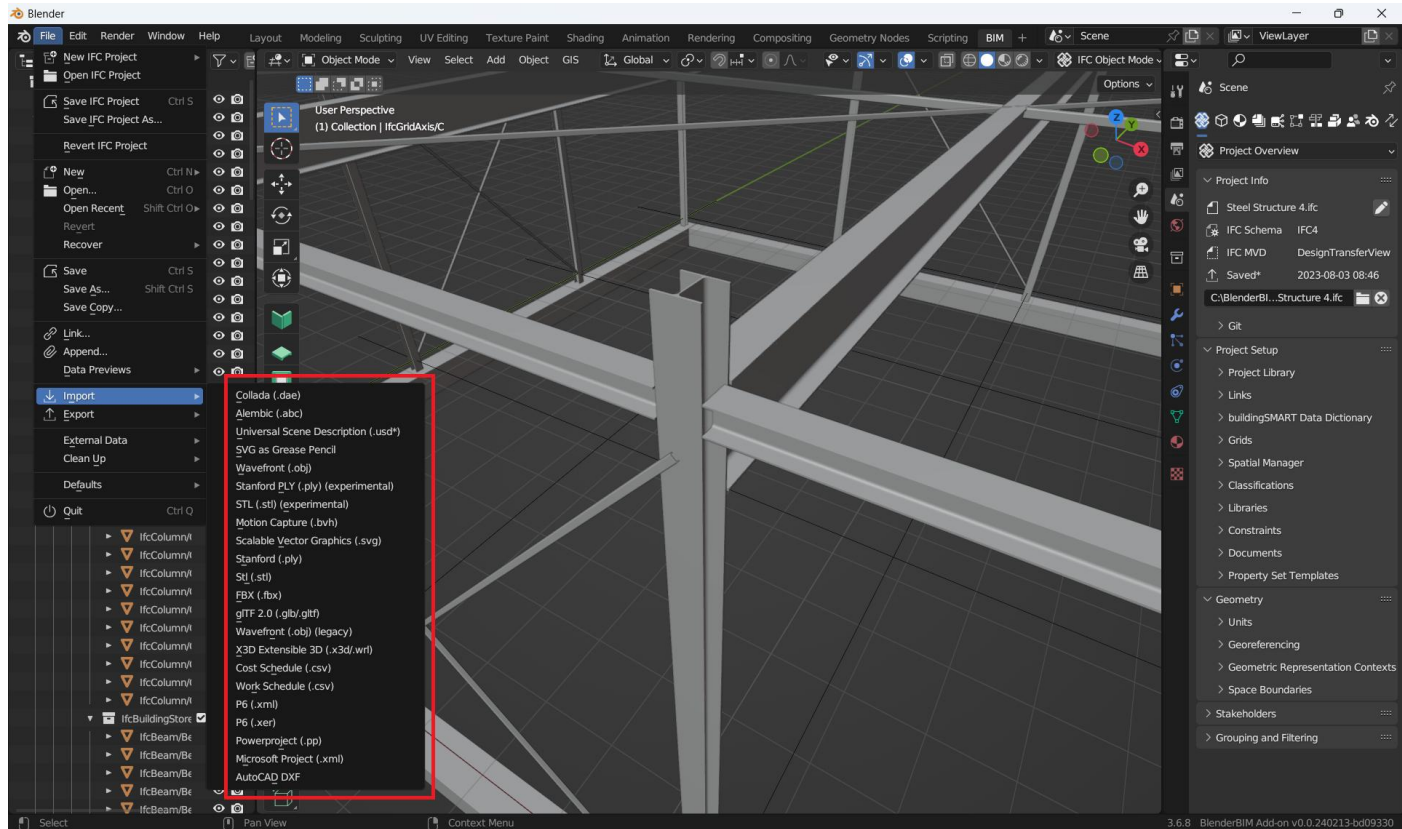
Dimidat Pouchnikov

[illegible]

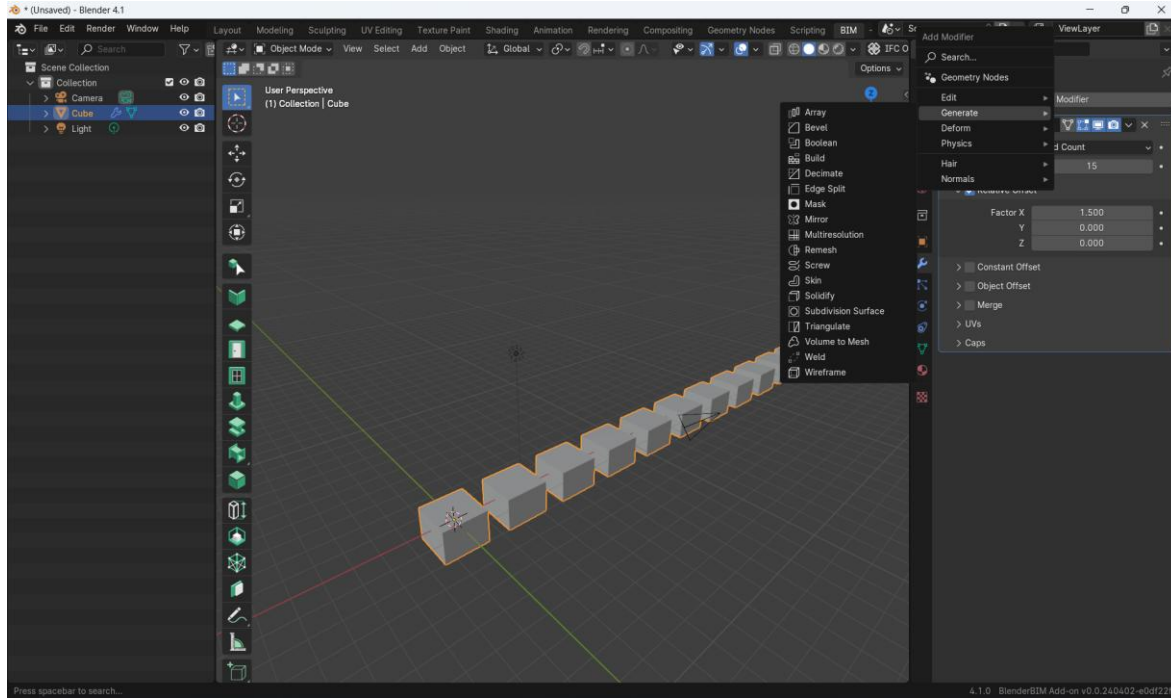
Point cloud



Blender Import/Export

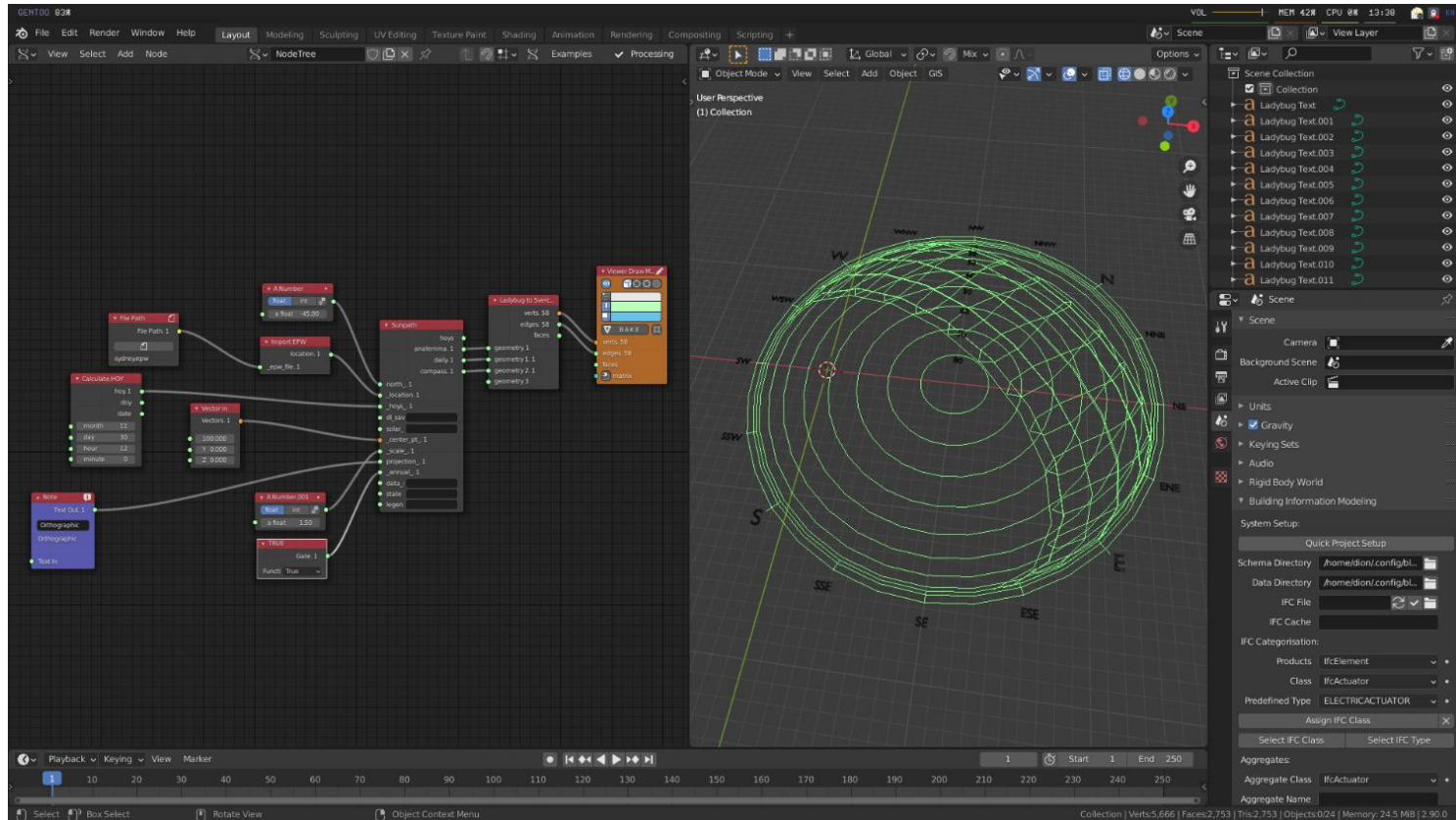


Blender modifiers

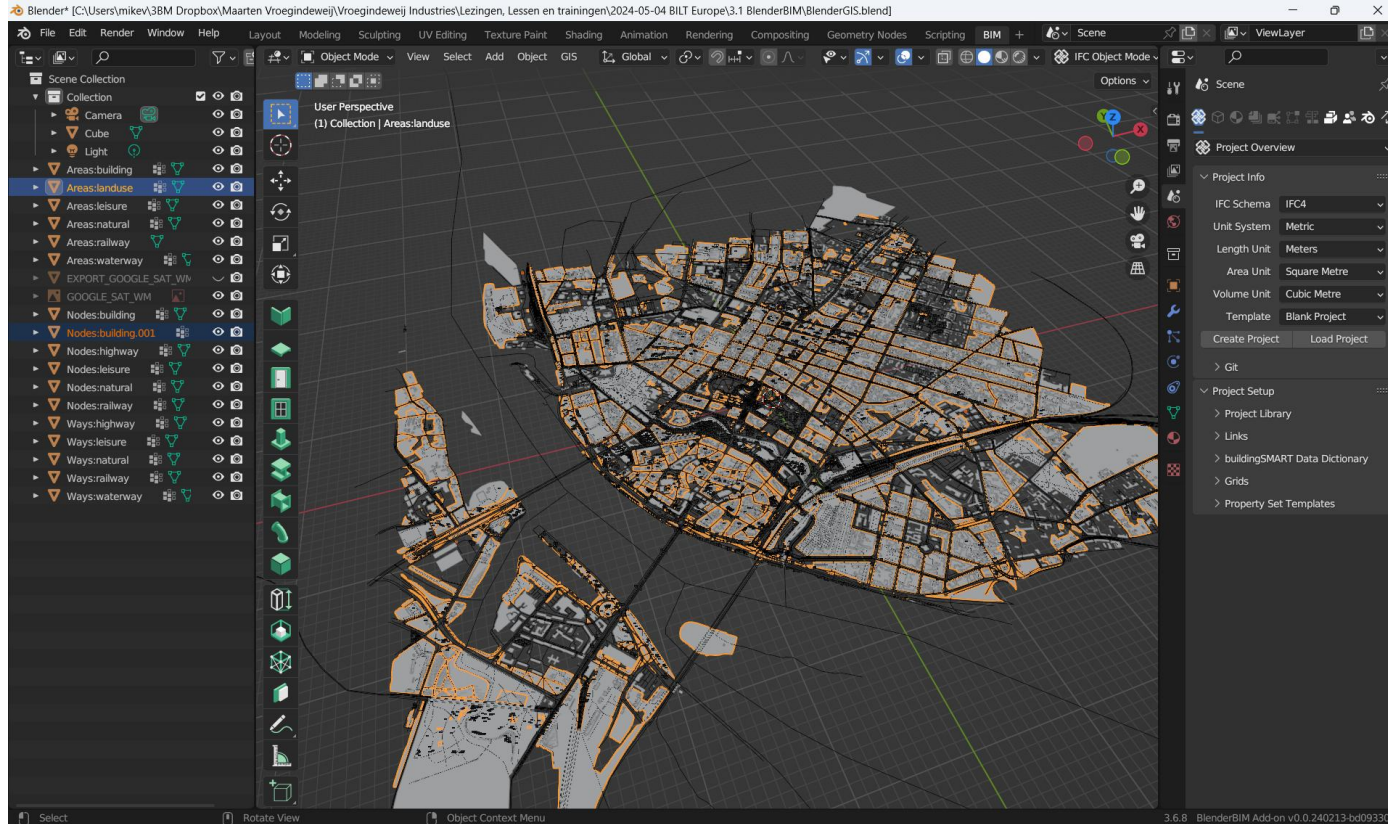


<https://www.youtube.com/watch?app=desktop&v=idcFMhoSdIc>

Blender Sverchok



BlenderGIS



Blender & CalcPAD?

CalcPAD VM 7.1.9 - Week 2 Fundering op staal.txt

File Edit Insert Output Help Numbers: Real Complex Angles: D R G Non-metric units: UK US Equation format: Professional Inline Max output count: 20 (5-100)

Code

```

10 "Fundatieberekening aandeel cohesie"
11 N_c = 20.6" (uit tabel)"
12 b_acc = 1" (breedte fundering)"
13 l_acc = 2" (lengte fundering)"
14 s_c = 1 + 0.2*(b_acc/l_acc)
15 i_c = 1" (hellingfactor)"
16 cohesie = b_acc*l_acc*(c_gem,d*N_c*s_c*i_c)*0kPa|kN/m^2
17
18 "Fundatieberekening aandeel dekking"
19 grond.dek = -1*(ok.fund.tov.PEIL - maaiveld)
20 u = 0kN/m^2
21 a_v.z.o.d = y_gronddekking*grond.dek - u
22 N_q = 10.6
23 s_q = 1 + (b_acc/l_acc)*sin(phi_gem,d)
24 i_q = 1
25 dekking = i_q*s_q*N_q*a_v.z.o.d|kN/m^2
26
27 "Fundatieberekening aandeel funderingsbreedte"
28
29
30

```

Output

```

u = 0kN/m^2
phi_v.z.o.d = y_gronddekking*grond.dek - u = 16.4kN/m^2*0.7m - 0kN/m^2 = 11.48kPa
N_q = 10.6
s_q = 1 + b_acc/l_acc*sin(phi_gem,d) = 1 + 1/2*sin(25°) = 1.21
i_q = 1
dekking = i_q*s_q*N_q*a_v.z.o.d = 1.21*10.6*11.48kPa = 147.4kN/m^2

```

Fundatieberekening aandeel funderingsbreedte

```

N_y = 9
s_y = 1 - 0.3*b_acc/l_acc = 1 - 0.3*1/2 = 0.85
i_y = 1
funderingsbreedte = 0.5*y_gem,d*N_y*s_y*i_y*b_acc*1m = 0.5*17kN/m^2*9*0.85*1kNm^2

```

Totale draagkracht fundering

```

a_max,d = cohesie + dekking + funderingsbreedte = 0kN/m^2 + 147.4kN/m^2
= 212.43kN/m^2
A_acc = b_acc*l_acc = 12 = 2
F_v.z.d = A_acc*a_max,d*1m^2 = 2*212.43kN/m^2*1m^2 = 424.85kN

```

Round 2 | ☒ Zero small elem. ☒ Substitute Plot: ☒ Adaptive ☐ Shadows: North Palette: Rainbow ☐ Smooth Ext. browser: Chrome

CalcPAD VM 7.1.9 - Deep Beam FEA.cpd

File Edit Insert Output Help Numbers: Real Complex Angles: D R G Non-metric units: UK US Equation format: Professional Inline Max output count: 20 (5-100)

Code

```

12 'style="height:160pt; margin-left:20pt;"/></td></tr></table>
13 'Support length -'d = 0.4'm
14 'Support elastic stiffness -'k_s = 50000'MN/m^2
15 'Support function -'s(x) = k_s*t*(x <= d & x >= l - d)
16 'Material properties</b></p>
17 'Modulus of elasticity -'E = 20000'MPa
18 'Poisson's ratio -'v = 0.2
19 'Finite element mesh</h3>
20 'We will use rectangular finite element with n = 8'DOFs
21 'Number of elements along <var>l</var> and <var>h</var>
22 'n_l = 20', 'n_h = 10
23 'Total number of elements -'n_e = n_l*n_h
24 'Total number of joints -'n_j = (n_l+1)*(n_h+1)
25 'Element dimensions -'l_1 = l/n_l', 'h_1 = h/n_h
26 x_j = vector(n_j)', y_j = vector(n_j)

```

Output

```

2.25 4.84 9.37 -13.16 16.26 -19.3 -23.25 -27.63 -27.94 -16.37 0 16.37 27.94 27.63 23.25 19.3 16.26 13.16 9.37 4.84 -2.25
-0.9 -2.1 -4.48 -6.68 -8.38 -8.95 -13.2 -18.7 -20.41 -13.02 0 13.02 20.41 18.7 13.2 8.95 8.38 6.68 4.48 2.1 -0.9
-0.378 -0.832 -1.86 -3.06 -3.85 -4.56 -5.67 -15.47 -23.98 -12.91 -3.41 -10^-10 12.91 23.98 15.47 5.67 4.56 3.85 3.06 1.86 0.832 -0.378

```

Max. value at 3/4 of span - $N_y(3/4, h/2) = N_y(0.4, 2/2) = 29.95 \text{ kN/m}$

Principal membrane forces, kN/m

$N_x(x, z) = 0.5 (N_y(x, z) + N_y(x, z))$

$\Delta N_x(x, z) = 0.5 (N_y(x, z) - N_y(x, z)) + 4 N_y(x, z)$

$N_{tot}(x, z) = N_y(x, z) + \Delta N_x(x, z) \text{ kN/m}$

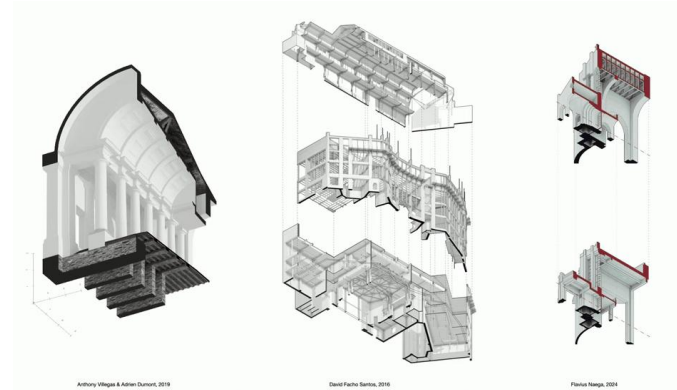
Max. value at 3/4 of span - $N_{tot}(3/4, h/2) = \Delta N_x(3/4, h/2) = 29.95 \text{ kN/m}$

Round 2 | ☒ Zero small elem. ☒ Substitute Plot: ☒ Adaptive ☐ Shadows: North Palette: Rainbow ☐ Smooth Ext. browser: Chrome

Blender in construction industry

Paweł Czernecki

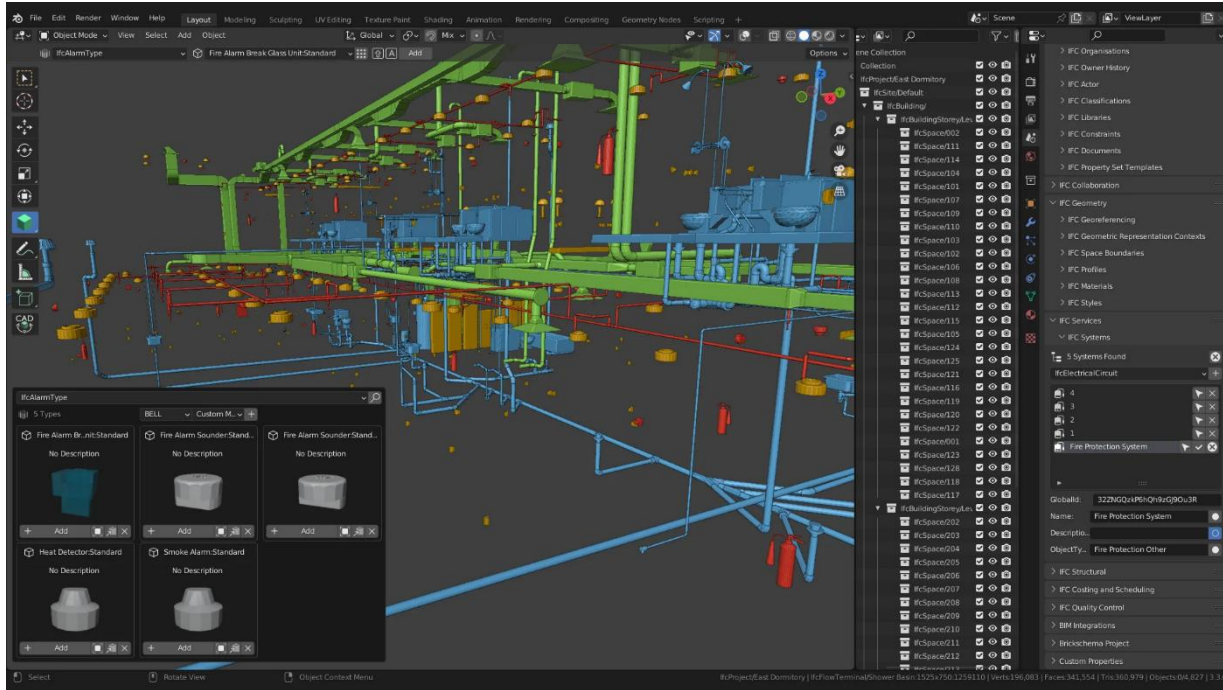
ERBUD, Poland



FOSS Bonsai

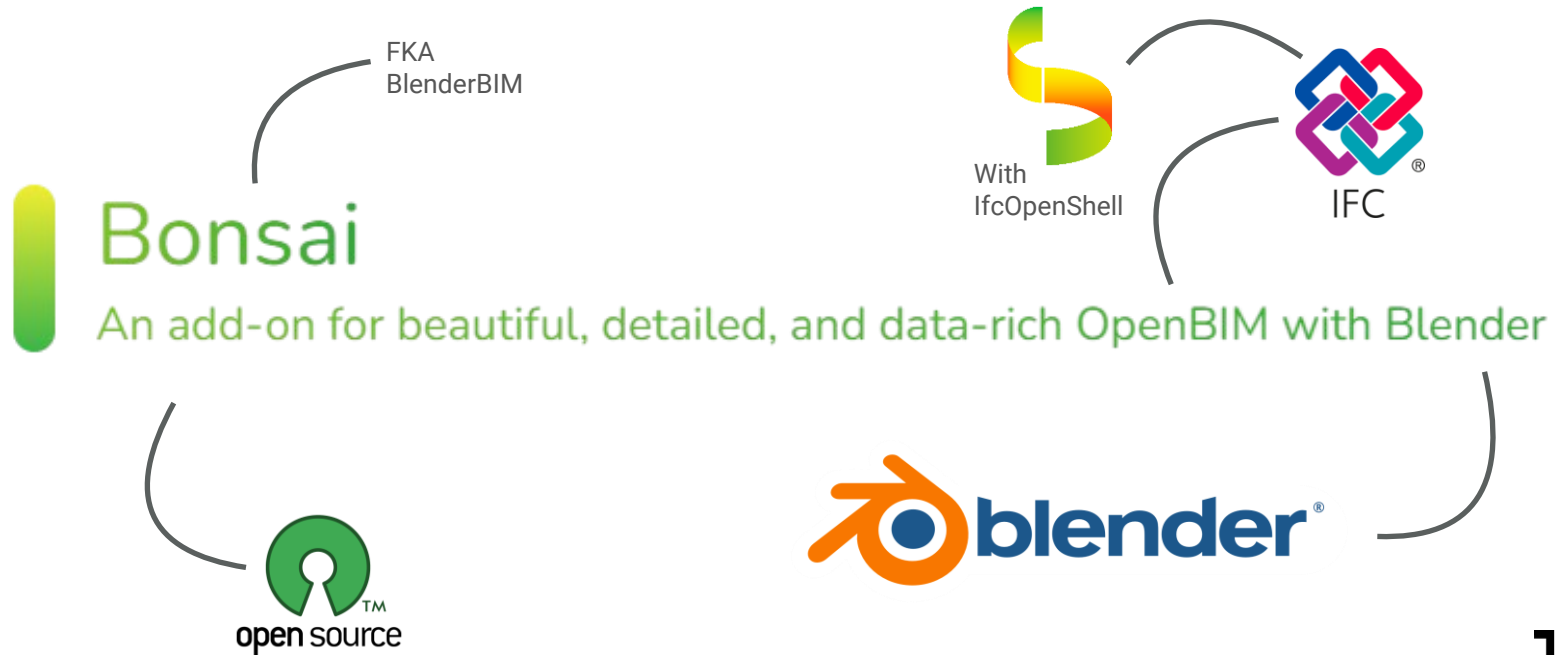


Bonsai | What & Why?



- IFC-viewer
- IFC editor
- 3D modeling
- 2D drawings
- Construction calculations
- HVAC, MEP, Electrical
- 4D planning, cost estimation
- IDS-checker
- Clash detection
- Full Python API
- BCF
- Git
- Solid and meshmodeller
- IFC editing

Bonsai | What & Why?



Bonsai | What & Why?

BLENDER

BONSAI

IfcOpenShell

*.blend
Mesh, Nurbs
Bmesh etc

User



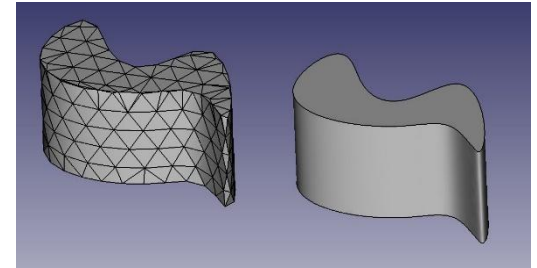
API



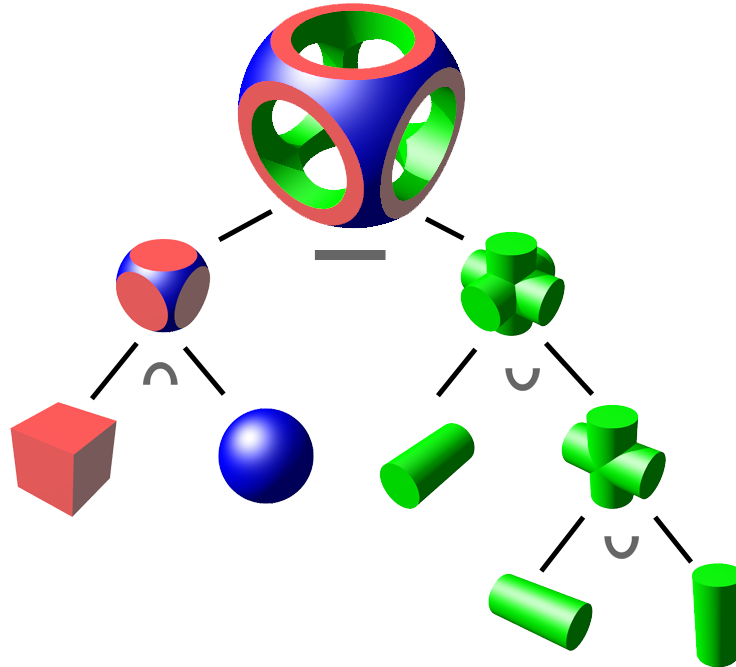
IFC Data



*.ifc
Mesh, Brep,
CSG/Boolean



CSG/Boolean



Bonsai, the community > 200 contributors



Thomas Krijnen(aothms)

IfcOpenShell

<https://github.com/aothms>



Dion Moul

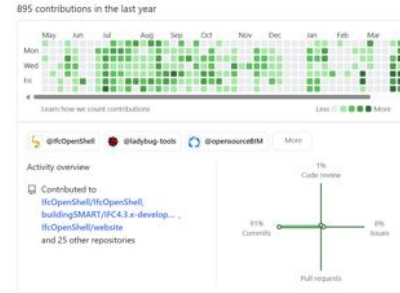
BlenderBIM/Bonsai

<https://github.com/Moul>



Andrej

<https://github.com/Andrej730>



OSArch Forum

Home · General

IfcEarthworksCut/Fill now with automatic QTO



steverugi

April 21 · edited April 21

Another magic performed by @Massimo today:

In IFC4x3 we can use IfcEarthworksCut (and Fill), which can be for both infrastructure and residential projects.

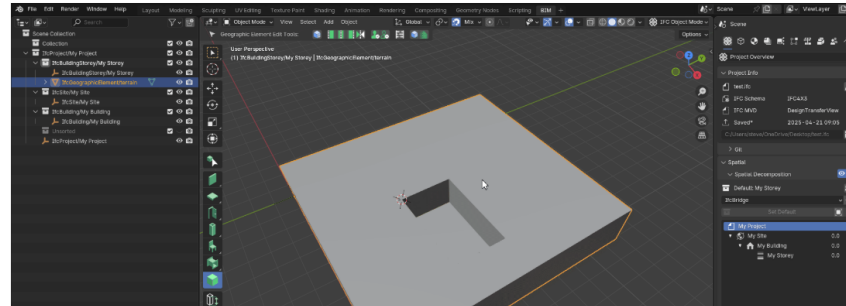
With today's release it's now possible to automatically generate Qto_EarthworksCutBaseQuantities

In the gif below you can see the following steps:

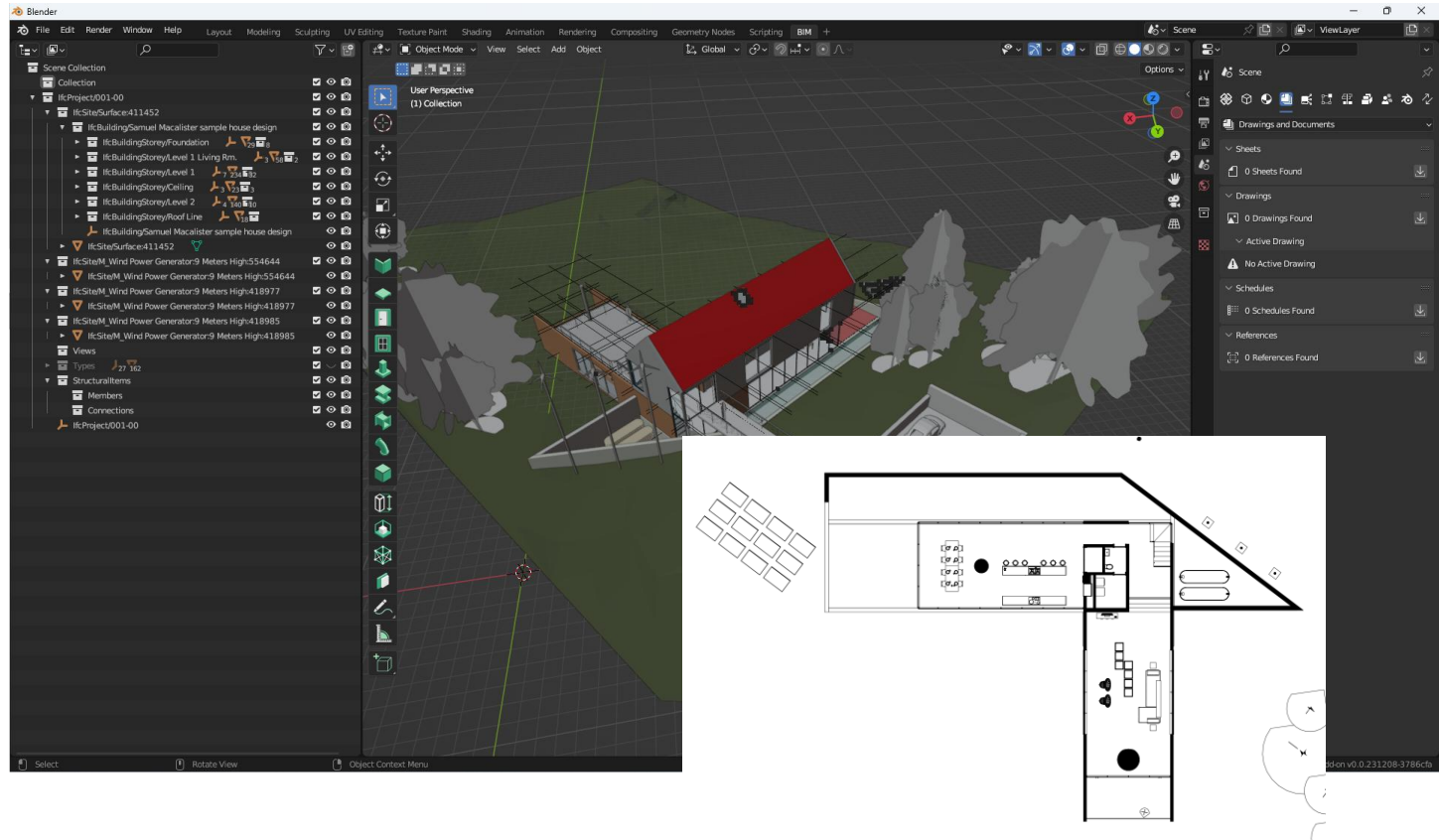
- have a solid (IfcGeographicElement.Terrain) able to receive a void
- add an IfcFeatureElement IfcEarthworksCut using the terrain as Featured Object
- use Custom Extruded Solid as representation (but you can use "Tessellation from Object" if you have one)
- edit the void using "tab" twice to go to Ifc Item Mode > Ifc Edit mode
- adjust the polyline defining the boundary of your cut
- check the Depth, "tab" again to enter to "Ifc Item Mode" if necessary
- Alt-O, with the terrain selected to exit and visualize the "excavation"

Quantity take-off is as usual, select Blender engine to perform it, it also applies to IfcEarthworksFill.

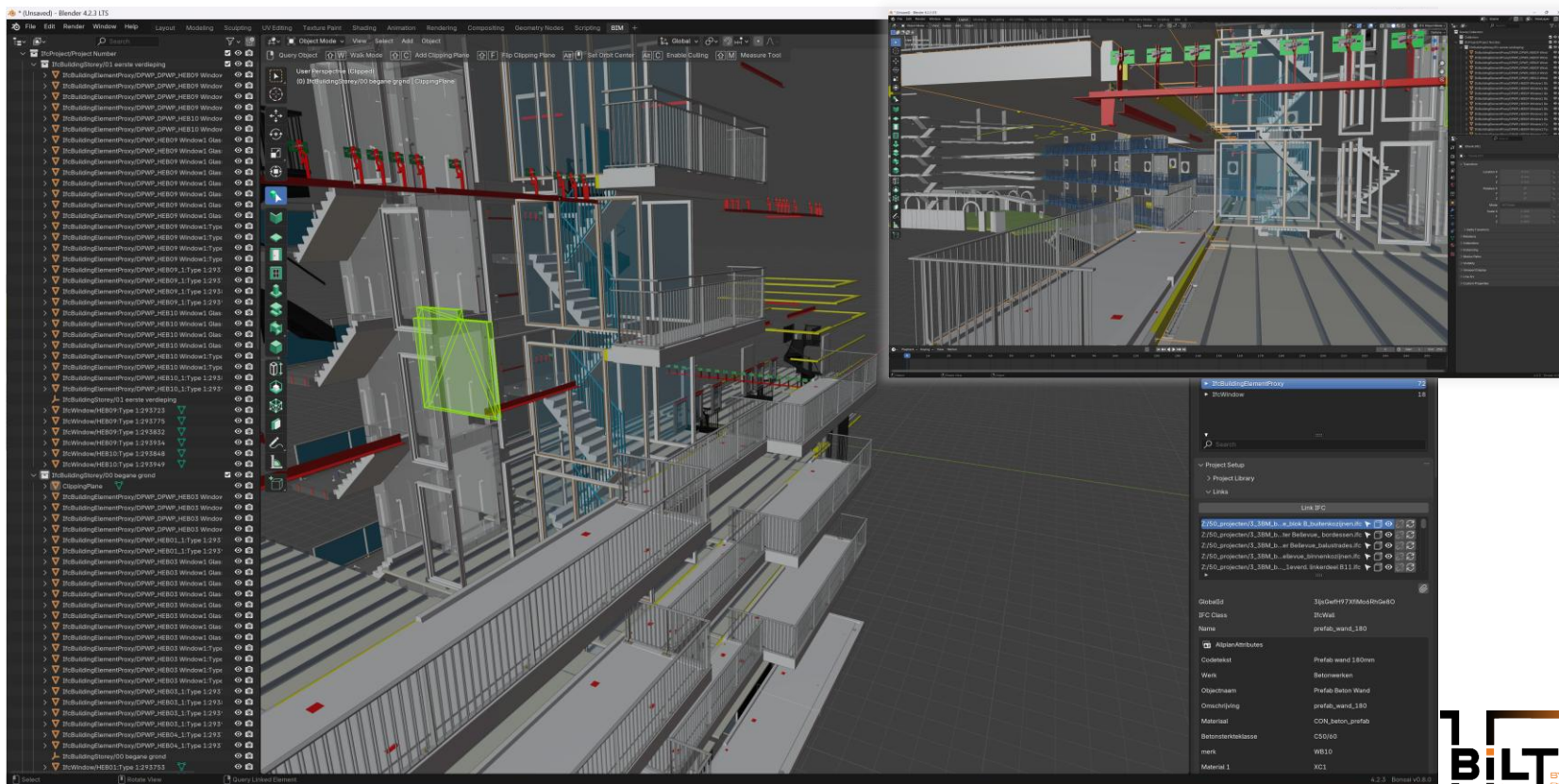
Nice one @Massimo, many thanks again



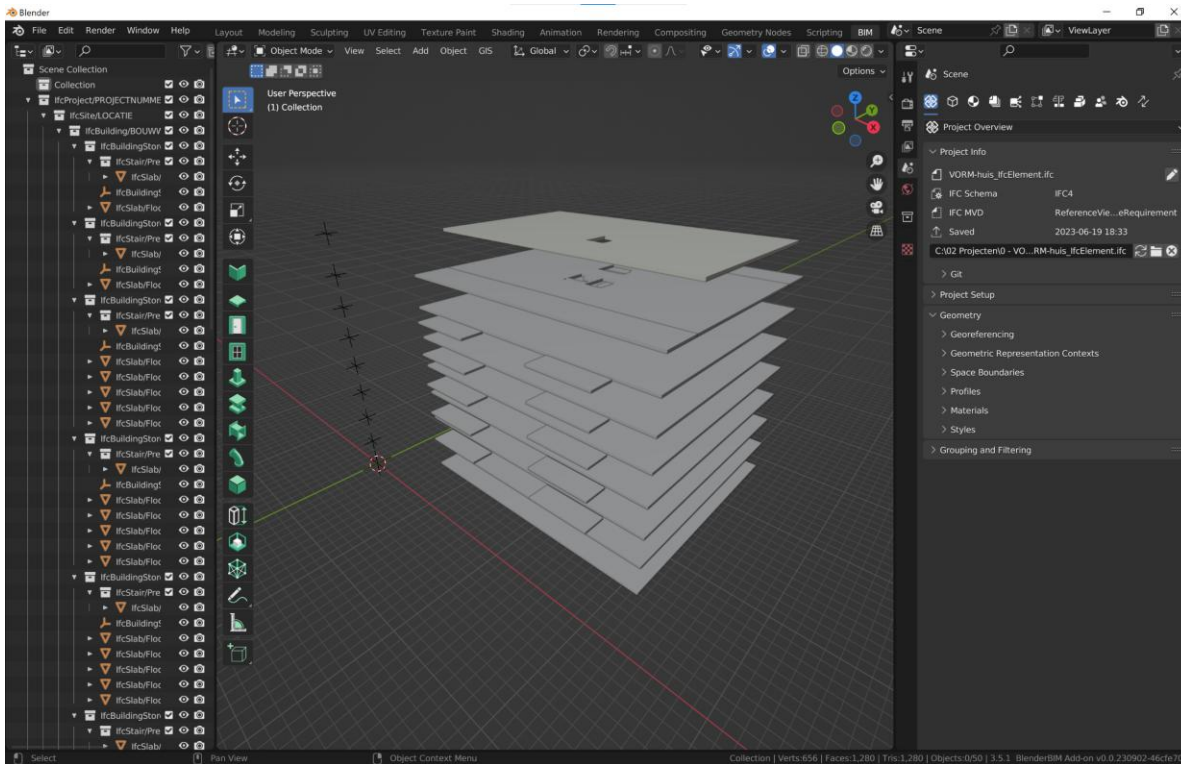
Open IFC



BIM Coordination

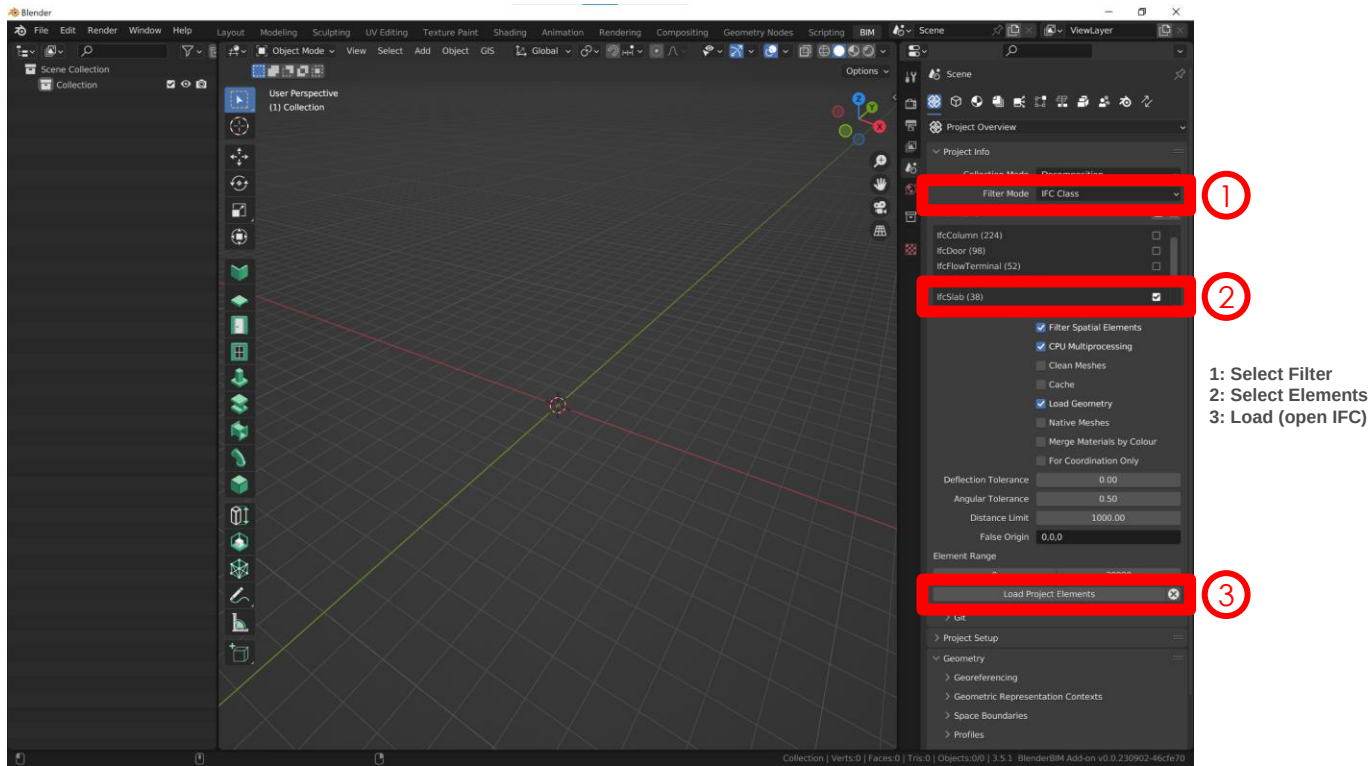


Open IFC partially

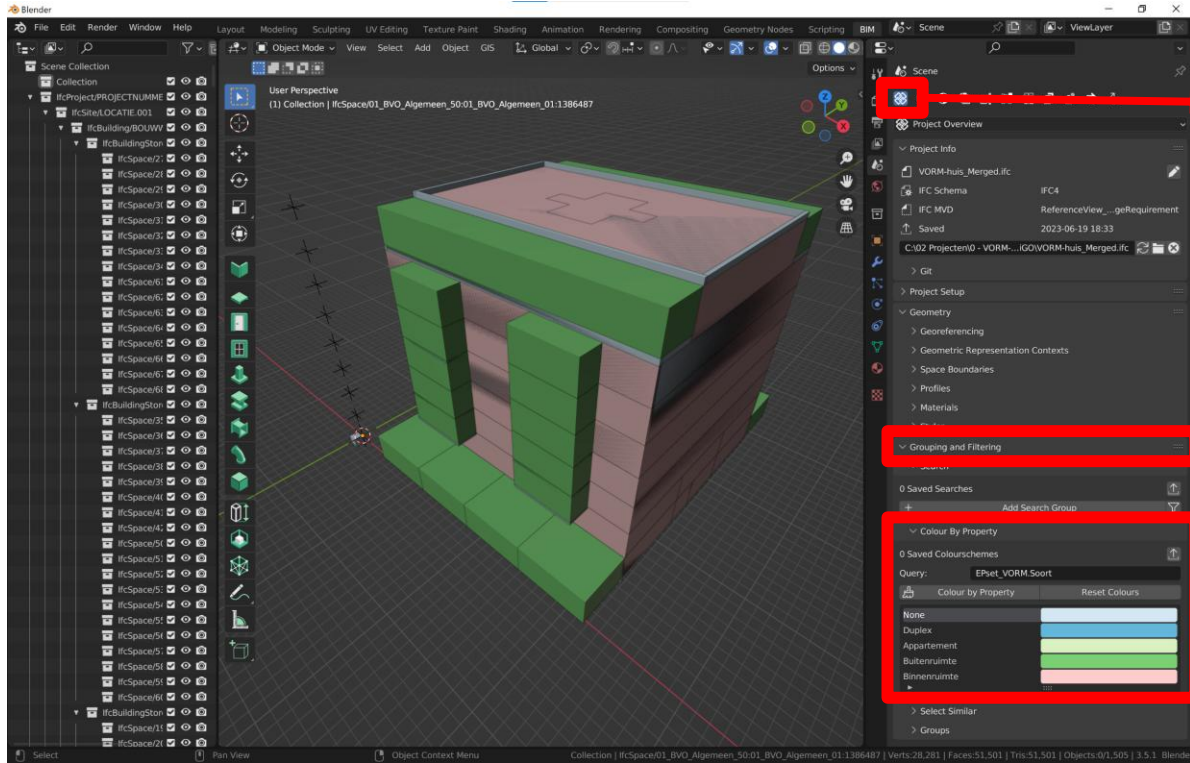


Only the IfcSlab is loaded.

Open IFC partially



Grouping and filtering

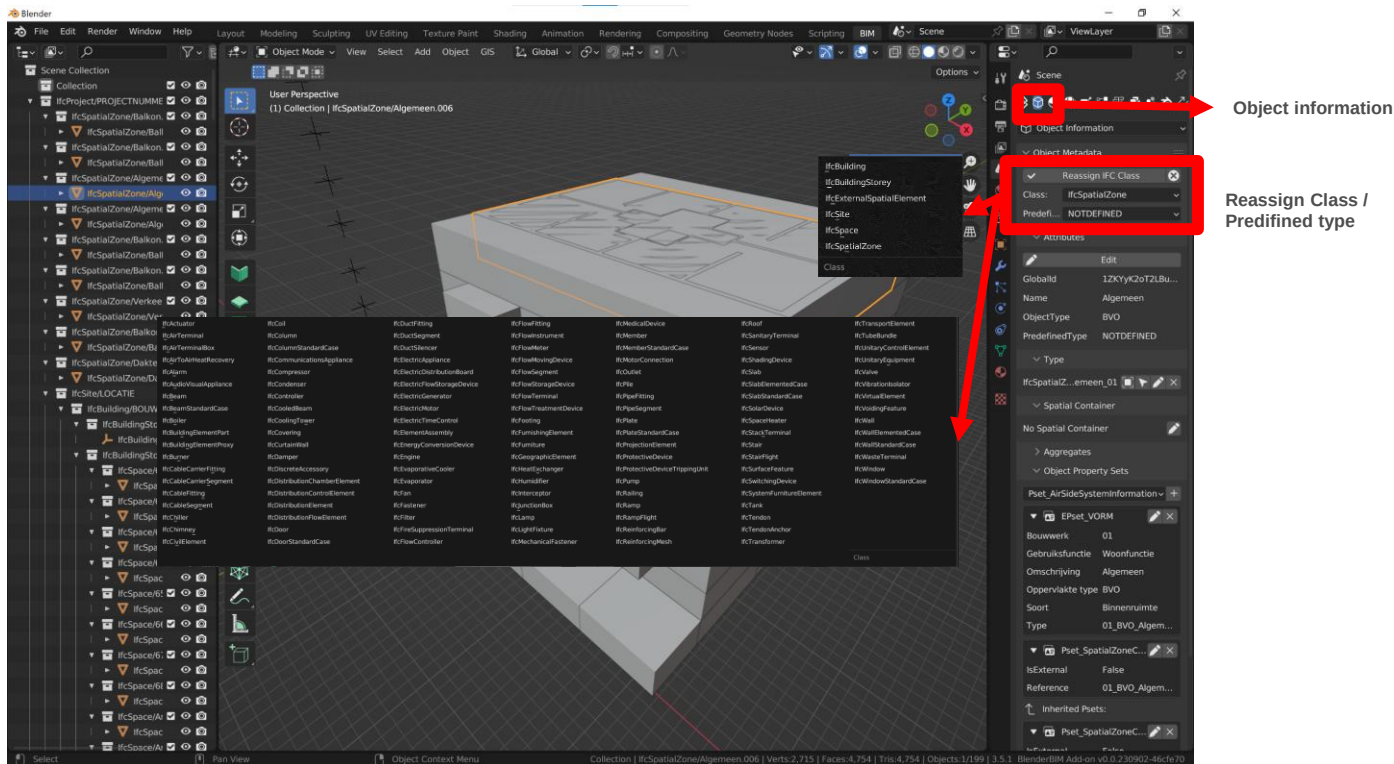


The image shows the Blender 2.80.1 interface. The central 3D Viewport displays a perspective view of a 3D model of a building, which is a cube with a cross on top, colored green and brown. The left sidebar shows the Outliner and Properties panels. The right sidebar shows the Project Overview panel, which is highlighted with a red box. The Project Overview panel has a 'Project Overview' tab selected, and a red arrow points to it with the text 'Project overview'. Below the Project Overview panel, the 'Grouping and Filtering' section is highlighted with a red box. This section contains a 'Colour By Property' panel, which is also highlighted with a red box. The 'Colour By Property' panel shows a list of properties with corresponding color swatches. The text 'Color on a value of a Pset' is next to this panel.

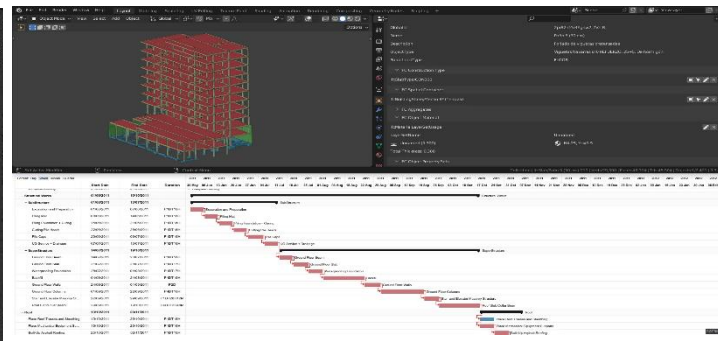
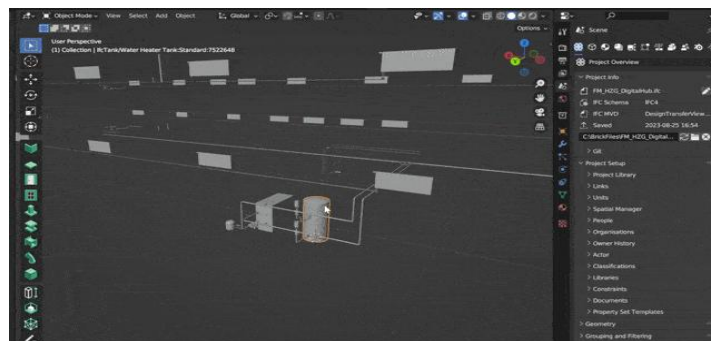
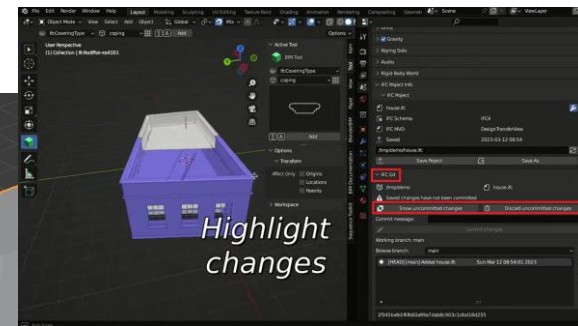
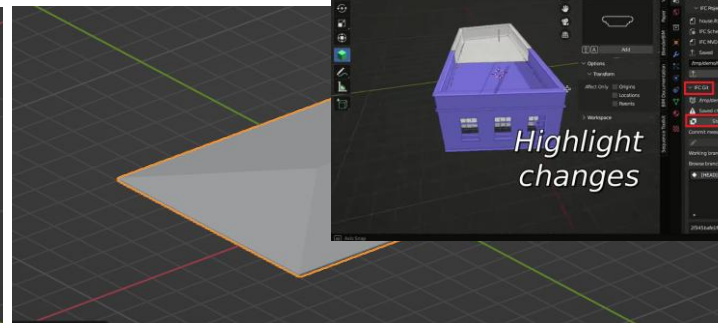
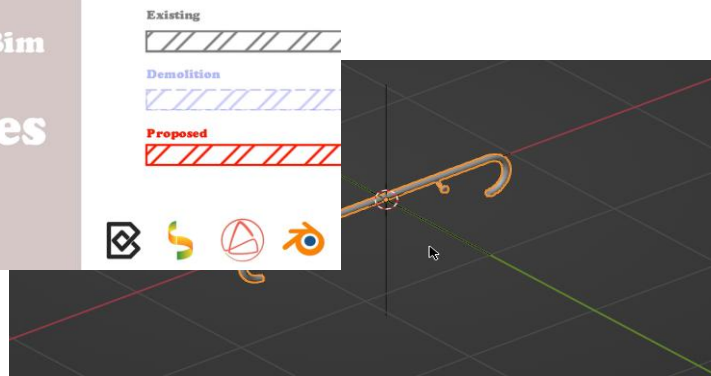
Project overview

Color on a value of a Pset

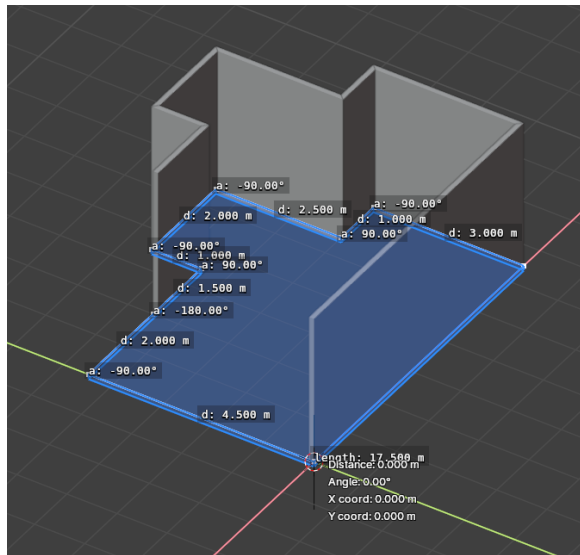
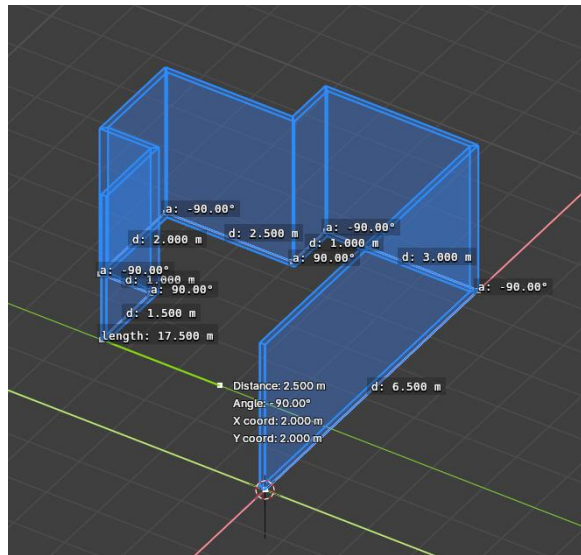
Ifc Classes



Bonsai | What & Why?

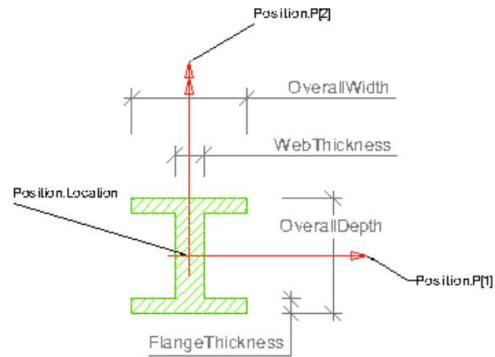


Bonsai | What & Why?

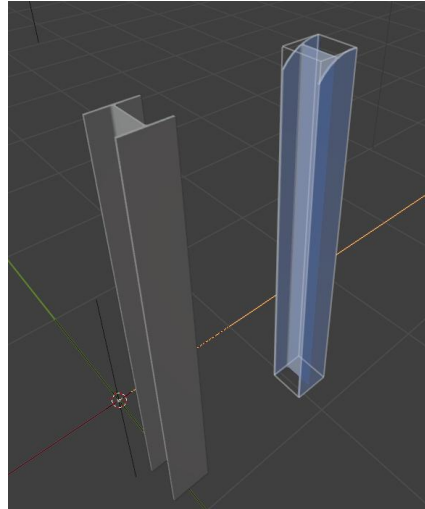


Bonsai |

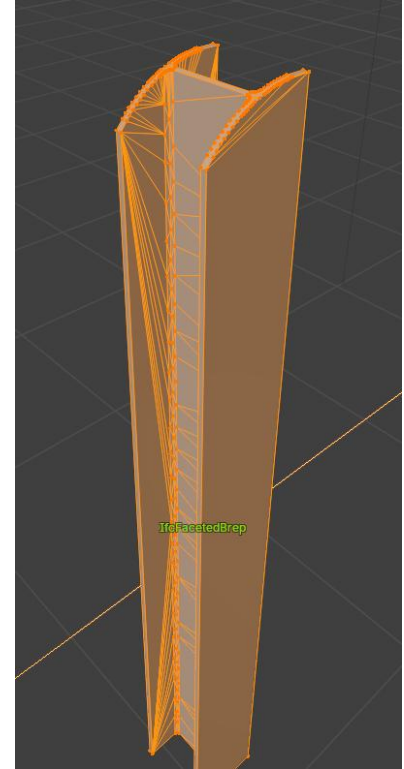
IfcIShapeProfileDef



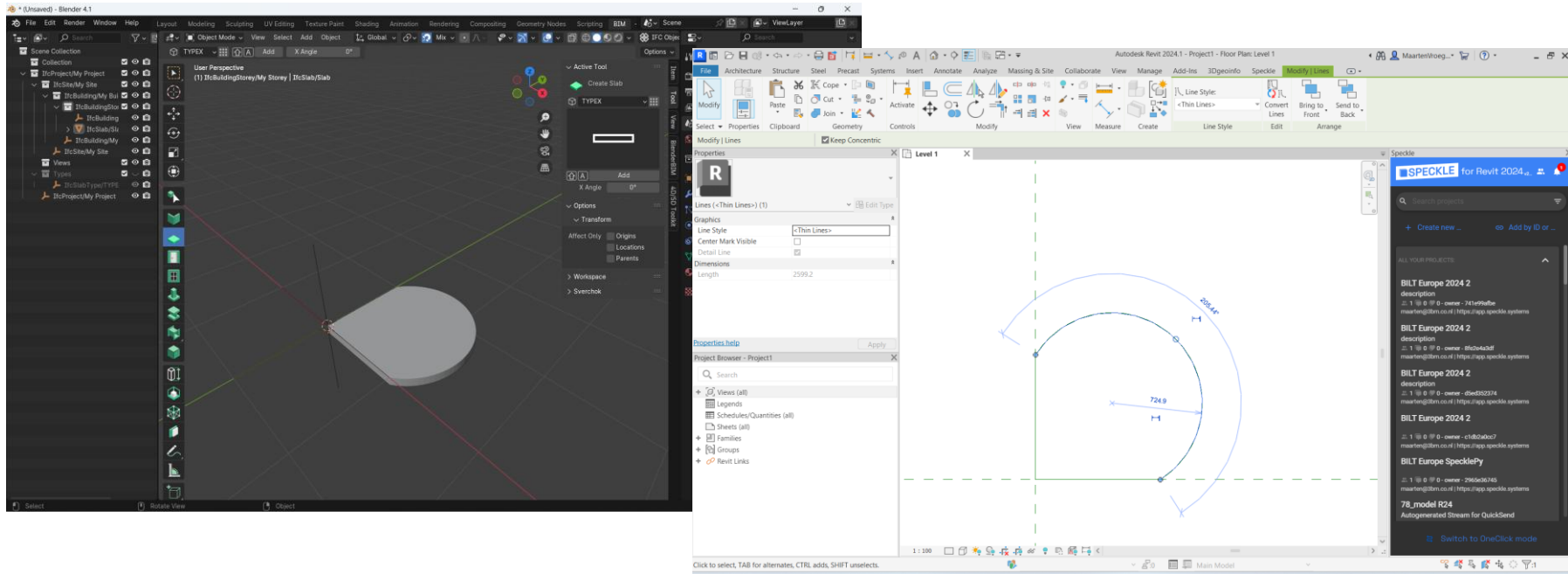
IfcExtrudedAreaSolid



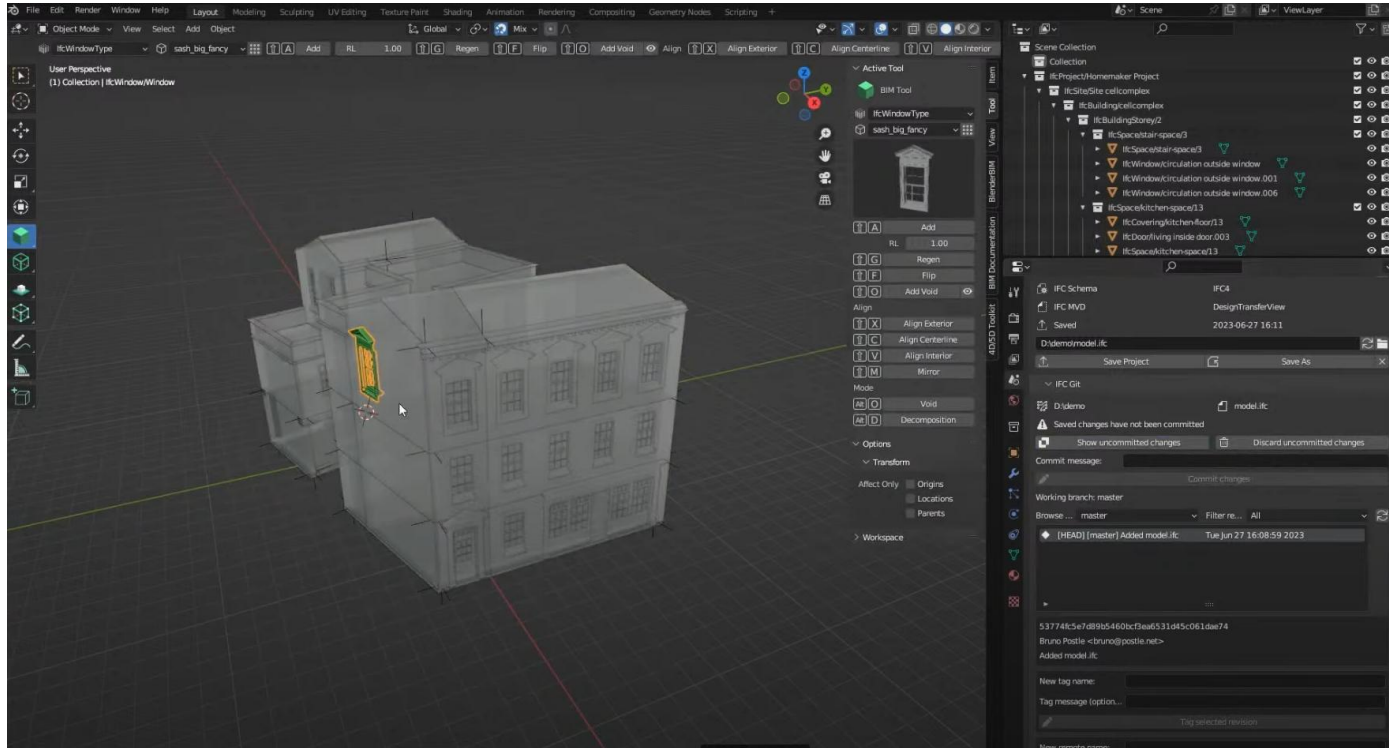
IfcFacetedBrep



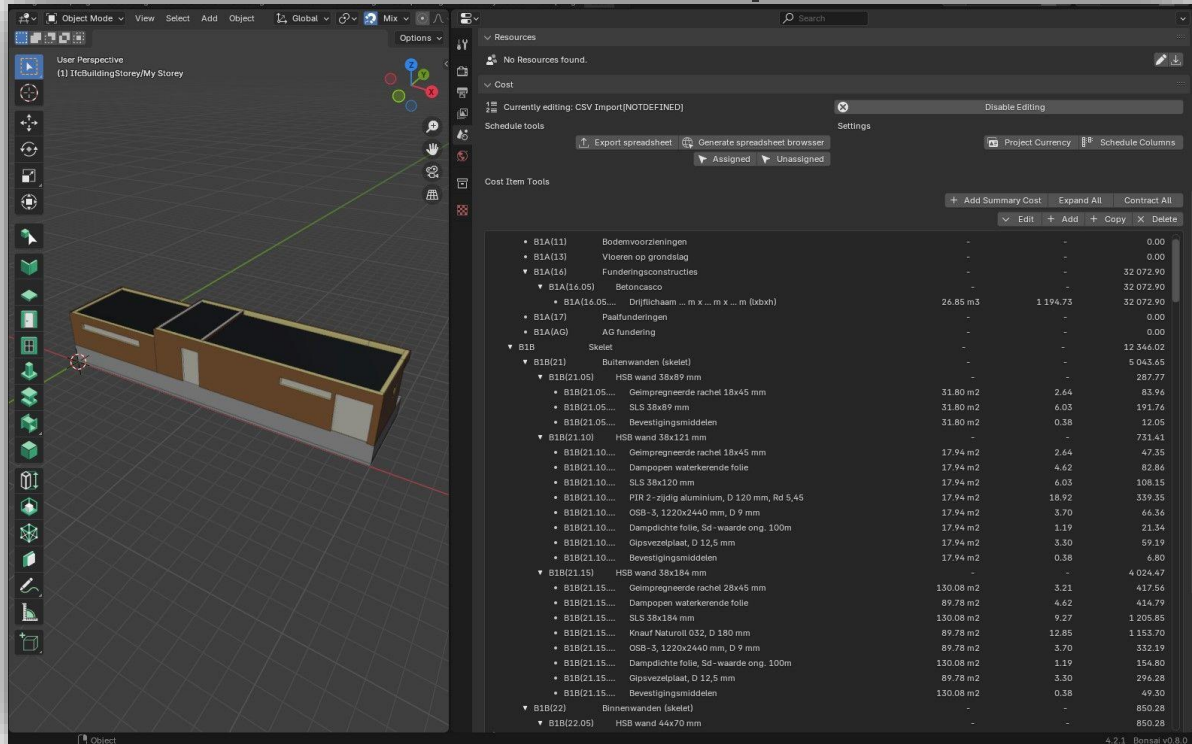
Bonsai



Bonsai | What & Why?



Bonsai | What & Why?



```
1 import ifcopenshell
2 import pdfkit
3 import base64
4 import os
5
6 # Laad het IFC-bestand
7 ifc_file = ifcopenshell.open("C:/Users/rickd/Dropbox/pc/Begroting_Gouda.ifc")
8
9
10 def encode_image_to_base64(image_path):
11     with open(image_path, "rb") as image_file:
12         return base64.b64encode(image_file.read()).decode('utf-8')
13
14 # Functie om HTML-content te genereren met een specifieke afwerking
15 def generate_html_content(element_name, layer_set_description, layer_descriptions, base64_image):
16     # Encode afbeelding naar Base64
17     base64_image = encode_image_to_base64(image_path)
18
19     # Voor de beschrijving van de MaterialLayerSet zonder bullet en met nieuwe regel
20     if layer_set_description:
21         layer_set_desc_html = "<div>" + "</div><div>".join([desc.strip() for desc in layer_descriptions])
22     else:
23         layer_set_desc_html = ""
24
25     # Voor de beschrijvingen van de individuele lagen
26     description_list = "<li>".join([f"<li>{sub_desc.strip()}</li>" for sub_desc in layer_descriptions])
27
28     return f"""
29     <tr>
30         <td style="padding-left: 10px; padding-right: 10px; vertical-align: top;">
31             <div style="vertical-align: top;">
32                 {layer_set_desc_html}
33             </div>
34             <ul style="padding-left: 15px; margin-right: 10px;">{description_list}</ul>
35         </td>
36         <td style="padding: 10px; text-align: center;">
37             
38         </td>
39     </tr>
40     """
41
42 # Functie om alle eigenschapswaarden uit een PropertySet te extraheren
43 def get_property_set_values(ifc_file, property_set_name):
44     property_values = []
45     # Loop door alle PropertySets in het IFC-bestand
46     for entity in ifc_file.by_type('IfcPropertySet'):
47         if entity.Name == property_set_name:
48             for prop in entity.HasProperties:
49                 # Verkrijg de property naam en waarde
50                 property_name = prop.Name
51                 property_value = prop.NominalValue.wrappedValue if hasattr(prop.NominalValue, 'wrappedValue') else prop.NominalValue
52                 if property_value:
53                     property_values.append((property_name, property_value))
54     return property_values
55
```


Bonsai | What & Why?

Begroting voor Waterwoning

Naam: Naam klant
Omschrijving: Nieuwbouw woonboot straatnaam + plaatsnaam

Betreft: Begroting waterwoning
Datum: 2024-10-23
Begroting nummer: 12345

HasParNr	Omschrijving	Aantal	Eenheid	Prijs	Subtotaal	Totaal
TOTAAL	Totale bouwkosten					
B1A	Fundering					
B1A(11)	Bodemvoorzieningen					
B1A(13)	Vloeren op grondslag					
B1A(16)	Funderingsconstructies					
B1A(16.05)	Betons casco					
B1A(16.05.05)	Drijfschaam ... m x ... m x ... m (bxbh)	26.85	m²	€1.194,73	€32.072,90	
B1A(17)	Paalfunderingen					
B1A(AG)	AG fundering					
B1B	Skelet					
B1B(21)	Buitenwanden (skelet)					
B1B(21.05)	HSB wand 38x89 mm					
B1B(21.05.05)	Geïmpregneerde rachel 18x45 mm	31.80	m²	€2,64	€83,96	
B1B(21.05.10)	SLS 38x89 mm	31.80	m²	€4,47	€142,15	
B1B(21.05.15)	Bewestigingsmiddelen	31.80	m²	€1,19	€37,95	
B1B(21.05.20)	Arbeid	1	stuks			
B1B(21.10)	HSB wand 38x121 mm					
B1B(21.10.05)	Geïmpregneerde rachel 18x45 mm	17.94	m²	€2,64	€47,35	
B1B(21.10.10)	Dampopen waterkerende folie	17.94	m²	€4,62	€82,86	
B1B(21.10.15)	SLS 38x120 mm	17.94	m²	€6,03	€108,15	
B1B(21.10.20)	PIR 2-zijlig aluminium, D 120 mm, Rd 5,45	17.94	m²	€18,92	€339,35	
B1B(21.10.25)	OSB-3, 1220x440 mm, D 9 mm	17.94	m²	€3,70	€66,26	
B1B(21.10.30)	Dampdichte folie, Sd-waarde ong. 100m	17.94	m²	€1,19	€21,34	
B1B(21.10.35)	Gipsvezelplaat, D 12,5 mm	17.94	m²	€3,30	€59,19	
B1B(21.10.40)	Bewestigingsmiddelen	17.94	m²	€1,19	€21,40	
B1B(21.10.45)	Arbeid	1	stuks			
B1B(21.15)	HSB wand 38x184 mm					
B1B(21.15.05)	Geïmpregneerde rachel 28x45 mm	130.08	m²	€3,21	€417,56	
B1B(21.15.10)	Dampopen waterkerende folie	89.78	m²	€4,62	€414,79	
B1B(21.15.15)	SLS 38x184 mm	130.08	m²	€9,27	€1.205,85	

Offerte voor Waterwoning

Naam: Naam klant
Adres: Adres klant
Postcode en plaats: Postcode+Plaatsnaam

Betreft: Offerte waterwoning
Datum: 2024-10-23
Offerte nummer: 12345

Technische specificaties

De constructie bestaat uit CLT. De gevelbekleding bestaat uit hout, Aluminium kozijnen, bitumen dakbedekking en EPS-isolatie zijn onderdeel van het project.

Bouwwerk voorzieningen

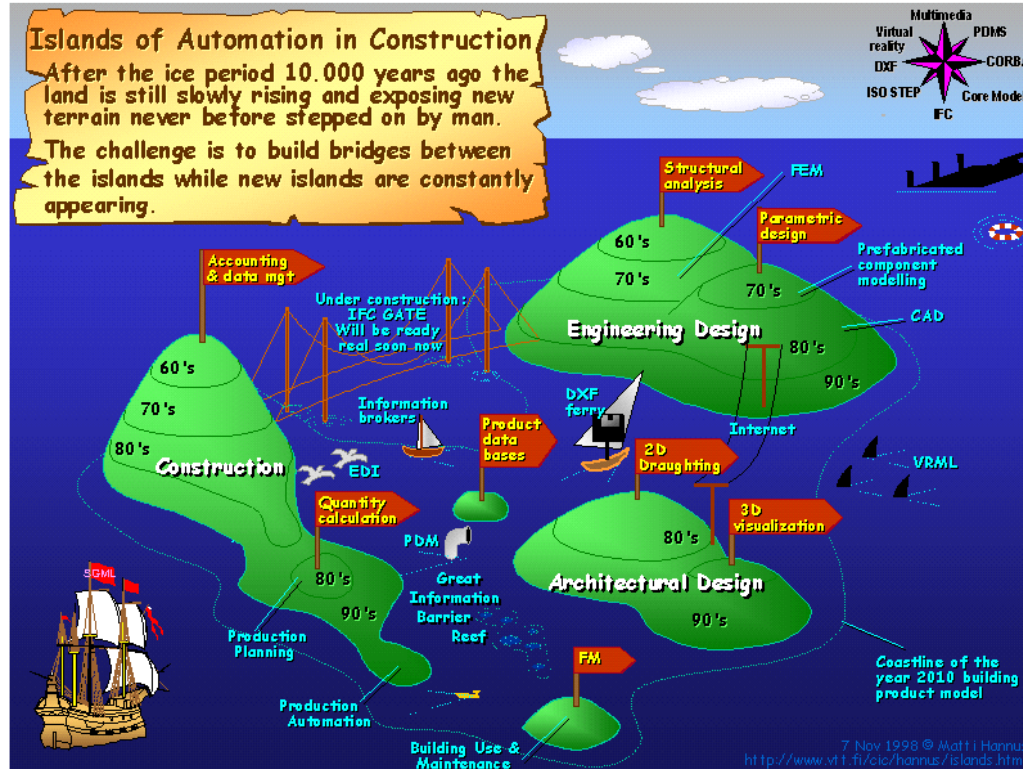
Element Naam	Beschrijving	Afbeelding
Helling	- Zidagen - Hellingen - Ligtdagen	
Bouwplaats	- Afvoeren van het bouw- en sloofafval door de aannemer - Betonbak hellingen op de scheepswerf bij Domera bij gereed - Plaatsen van stalen steiger rondom de gehele woning - Gebruik van bouwstroom en water voor rekening aannemer - Gebruik van bouwstroom en water op locatie voor rekening opdrachtgever	
Afval scheiding	- Plastic - Kunststof - Hout - Steen - Isolatie - Aluminium - Staal - Restafval	
Stroom	Benodigde energie dat nodig is bij het bouwen van de woning wordt mvv Hollandse groene stroom opgewekt	

Algemeen

Element Naam	Beschrijving	Afbeelding
Rc: waarden (minimaal)	- Voor vloeren: 3,7 m2KW - Voor gevels: 4,7 m2KW - Voor daken: 6,3 m2KW	

Gewapend beton betonsasco	Betonbak - Ontwerplevensduur 100 jaar - De wand en de vloer worden in één geheel gestort - Aanbrengen van wapening in de wanden en de vloer met voldoende dekking volgens berekening constructeur - Beton C35/45 - Naturel Grijs - Paschal bekisting - Ruwe afwerking	
HSB 38x89 mm	Voorzetwand betonbak Houtskeletbouw 89 mm	
HSB 38x121 mm	Voorzetwand betonbak Buiten->Binnen - Regelwerk bv ventilatie tussen HSB wand en betonwand - Dampopen waterkerende folie - Houtskeletbouw 121 mm met PIR isolatie D 120 mm - Watervast OSB III D 9 mm - Dampdichte folie - Gipsvezelplaat D 12,5 mm	
HSB 38x184 mm	Vloer Binnen->Buiten - Underlayment D 18 mm - Houtskeletbouw 184 mm met Naturol 032 dikte 180 mm - Dampopen waterkerende folie - Eenkel regelwerk	
HSB 38x184 mm	Buitenwand Buiten->Binnen - Dubbele regelwerk geïmpregneerd - Dampopen waterkerende folie - Houtskeletbouw 184 mm met Naturol 032 dikte 180 mm - Watervast OSB III D 9 mm - Dampdichte folie - Gipsvezelplaat D 12,5 mm	
HSB 44x70 mm	Binnenwand - Gipsvezelplaat D 12,5 mm - Watervast OSB III D 9 mm - Houtskeletbouw 70 mm met daartussen glaswol D 70 mm - Watervast OSB III D 9 mm - Gipsvezelplaat D 12,5 mm	

Revolution



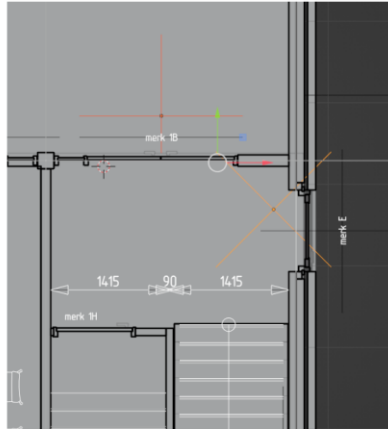
Bonsai | What & Why

Home - General

BBIM - Rotation of IfcSymbol

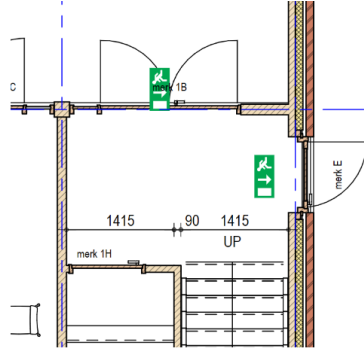
jan_f
March 13 Flag

Hi,
Is it possible to make symbols rotate via rotating the BlenderBBIM cross thing? Currently this doesn't work for my symbol and I have to rotate it in the svg. Problem is that if you want another orientation you have to duplicate it and rotate it in svg again. Thus creating double symbol types just for rotation issues.



Issue

13 March 2024



```
<!-- Noodlitzgang symbol -->
<g id="g10">
  <transform="scale(0.01) rotate(90)">
    <path>
      d="M 0 0 V 524.43 H 1133.86 V 0 H 0"
      style="fill:#00a650;fill-opacity:1;fill-rule:nonzero;stroke:none"
      id="path12" />
    <path>
      d="M 650.906 77.9805 786.453 211.008 H 677.859 V 430.012 H 623.95
      style="fill:#ffffff;fill-opacity:1;fill-rule:nonzero;stroke:none"
      id="path14" />
    <path>
      d="M 1076.02 78.8398 V 433.82 H 872.656 V 78.8398 h 203.364"
      style="fill:#ffffff;fill-opacity:1;fill-rule:nonzero;stroke:none"
      id="path16" />
    <path>
      d="m 355.508 117.063 v 72.617 l -75.203 75.199 21.879 21.875 46.0
      style="fill:#ffffff;fill-opacity:1;fill-rule:nonzero;stroke:none"
      id="path18" />
    </g>
  </g>
```

I already tried it without the rotate function, but then it just defaults to standard position.

Might be very slightly related to this rotation issue: <https://community.bimba.org/discussion/2006/bim-extruded-arc-profile-doesnt-save-rotation-and-instance-not-found-error#latest>

Tagged @BlenderBBIM, @Annotations

Andrei730
March 14 Flag

You used IfcAnnotation[TEXT] and I used IfcAnnotation[SYMBOL], as it is only a symbol. Should I use text classification but then without text?

I've totally missed that we have SYMBOL/MULTI_SYMBOL types also. SYMBOL should support scaling and rotation after [this commit](#), you can try it out.

When you open the Ifc and then the plan view, it automatically decides to hide the red instances:

Looks like a bug, need to investigate further.

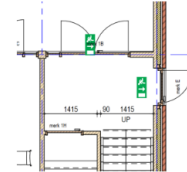
Quote

Solved

14 March 2024

Some situations where cut objects with holes (e.g. hollow sectional profiles) might not have their holes displayed are now fixed.

You can now manage and generate drawings from detached windows. Blender lets you create and split new windows, so this is useful especially for users with multi-monitor setups. Activating drawings now also works when you have objects isolated. Text symbols and multi symbol objects can be rotated in the viewport too. Multisymbol annotations are also now decorated to make it easier to see.



A small but significant usability improvement is that you can now select elements in any order when assigning an annotation to a product for smart annotations instead of the eyedropper. Also built-in text annotation assets can now be used in any view, not just plan view. Angles now display using the degrees symbol, a small but significant change in documentation quality. Drawing include/exclude filters now have a new friendly UI which also lets you use saved searches.

Generating drawings is now a bit more stable. Users would sometimes hit frustrating bugs where drawings would simply fail to generate or activate, but a number of these with various reasons have been solved. There are still more to solve, but progress is being made.

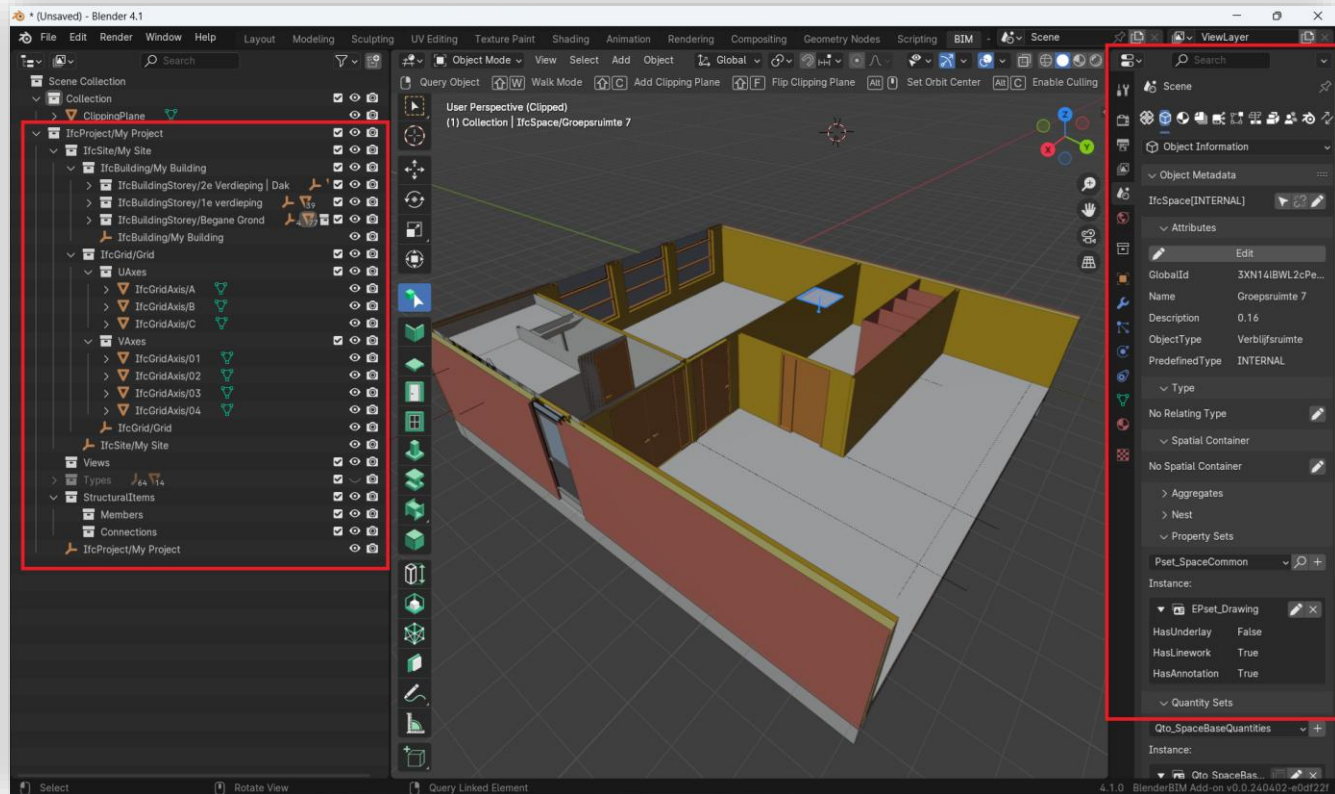
Releasenotes

2 April 2024



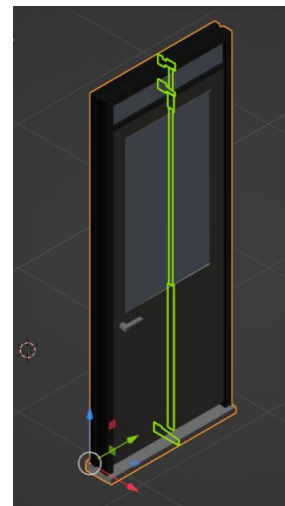
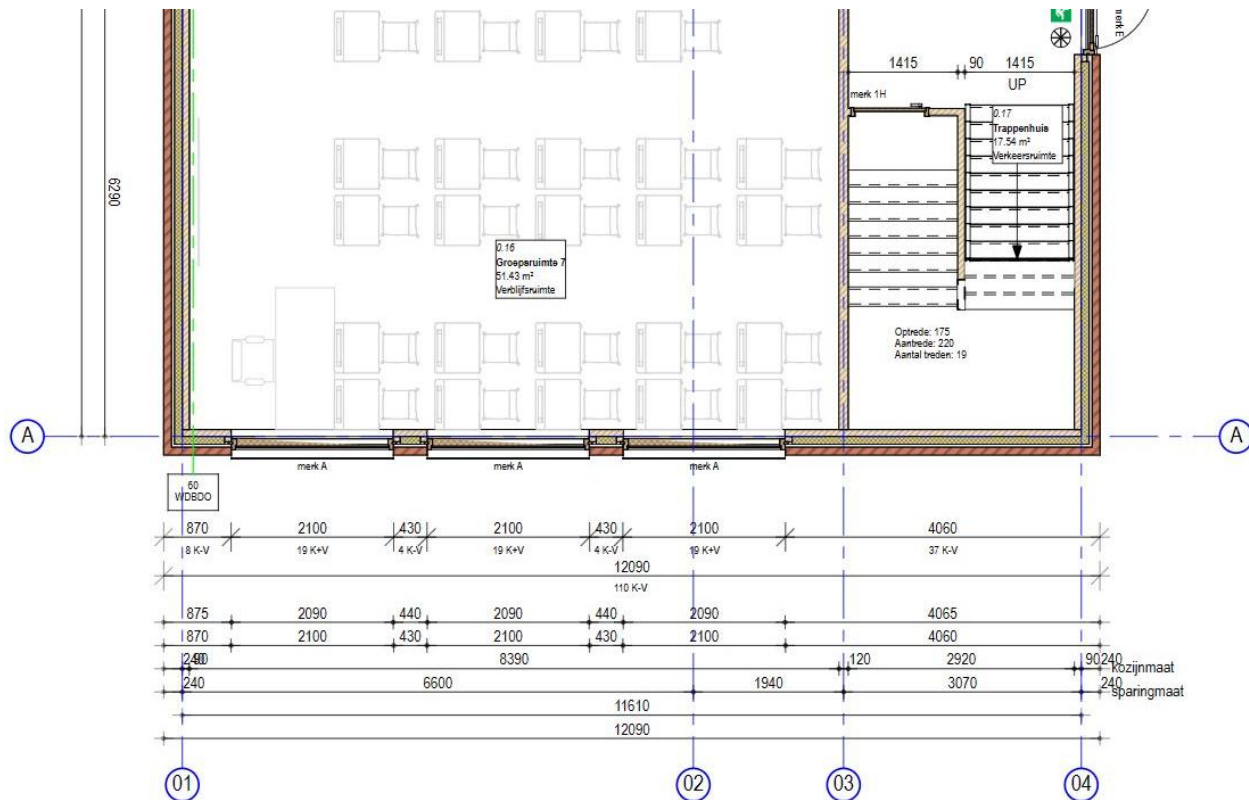
Bonsai Add-on v0.8.1 has been released with 1.902(!!) new features and fixes

Previous projects



Remember from last year? ;)

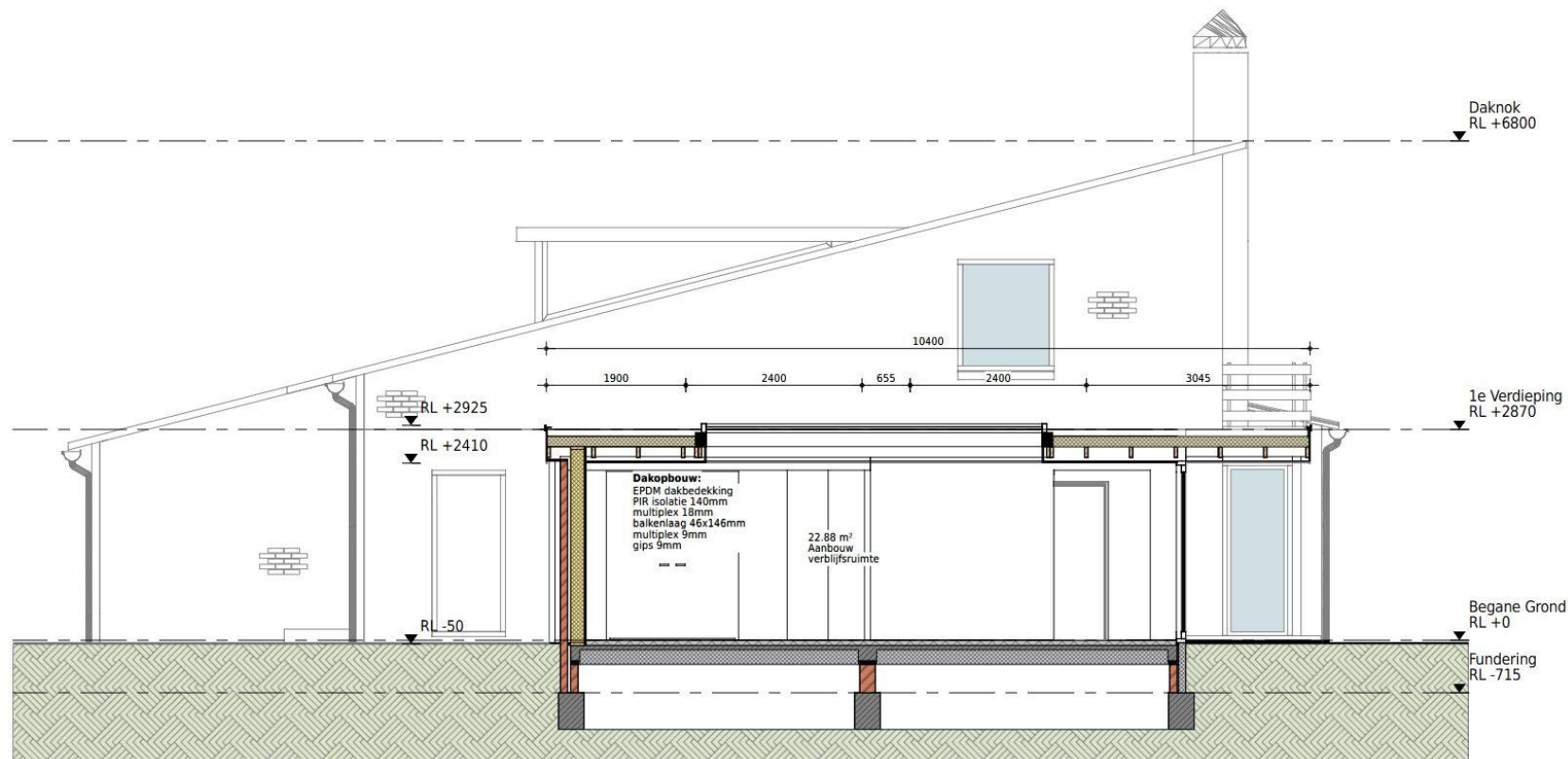
Previous projects



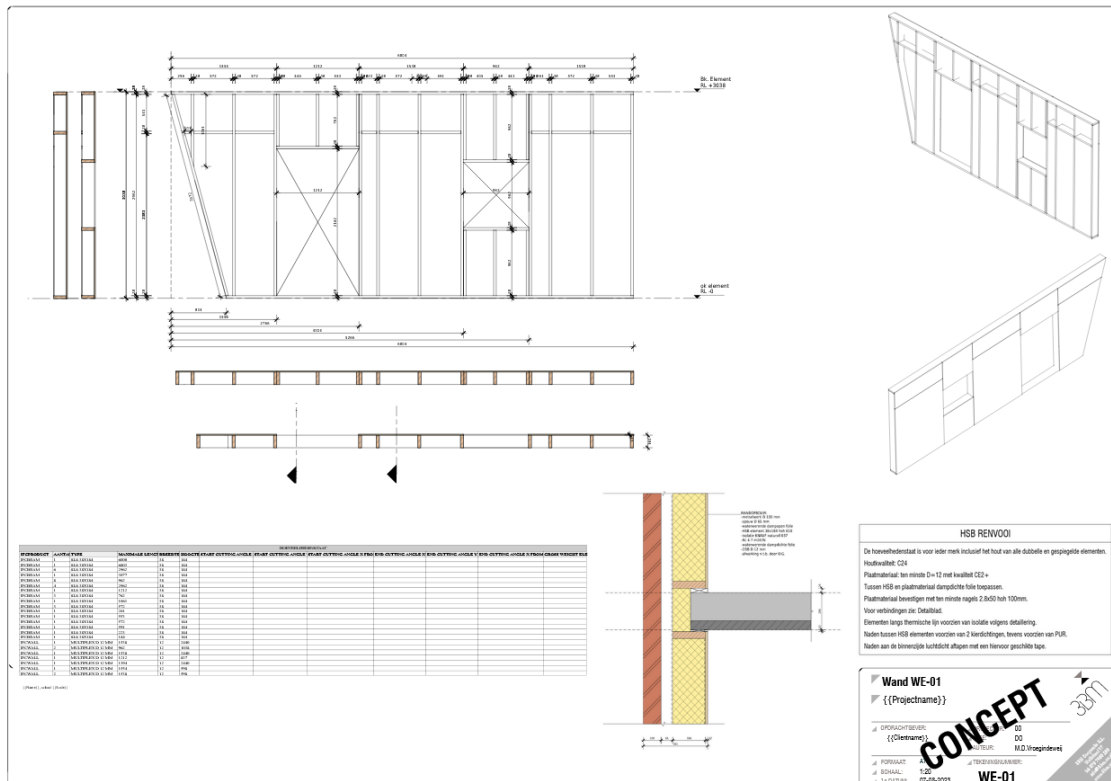
Previous projects



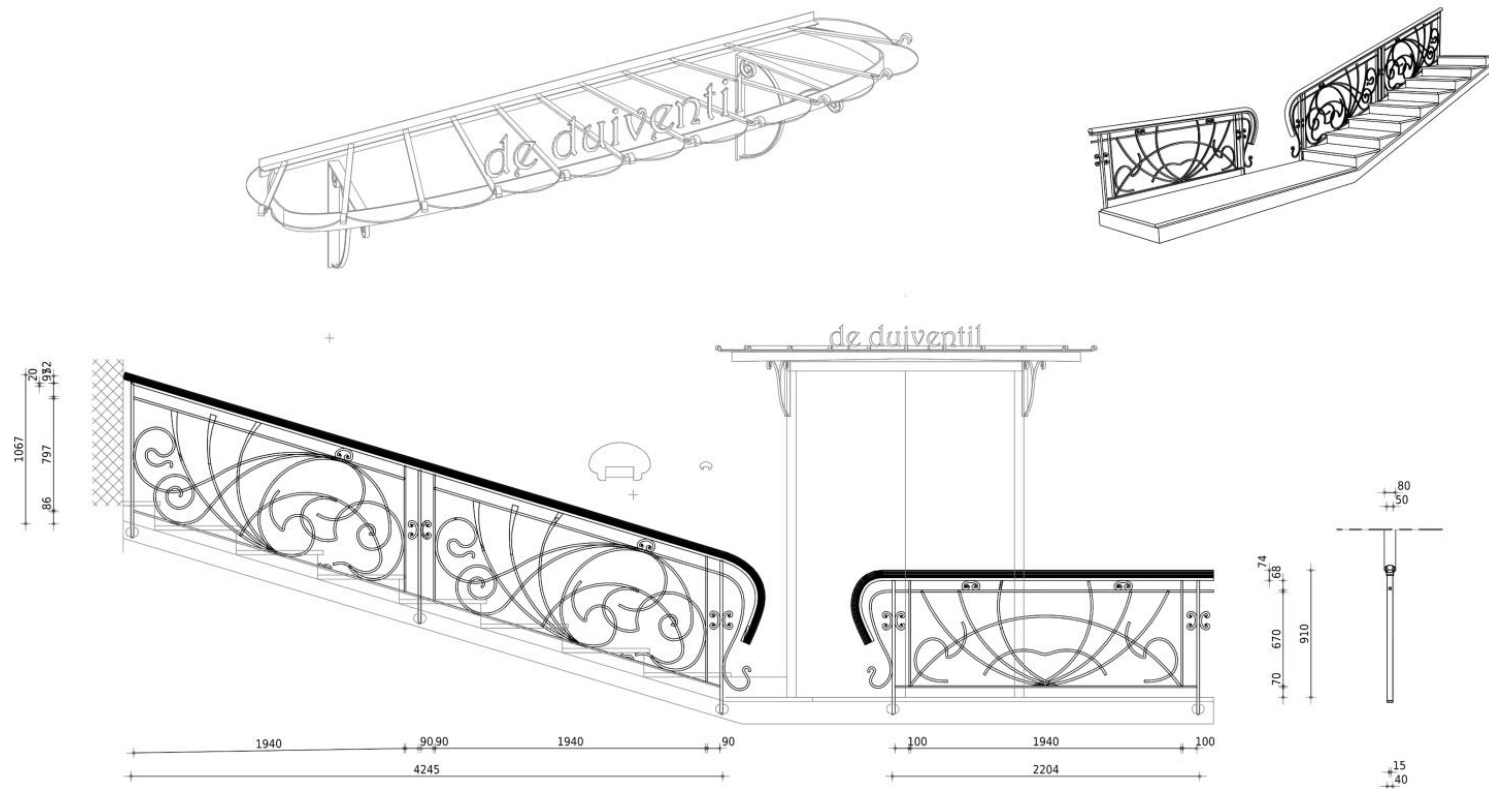
Previous projects



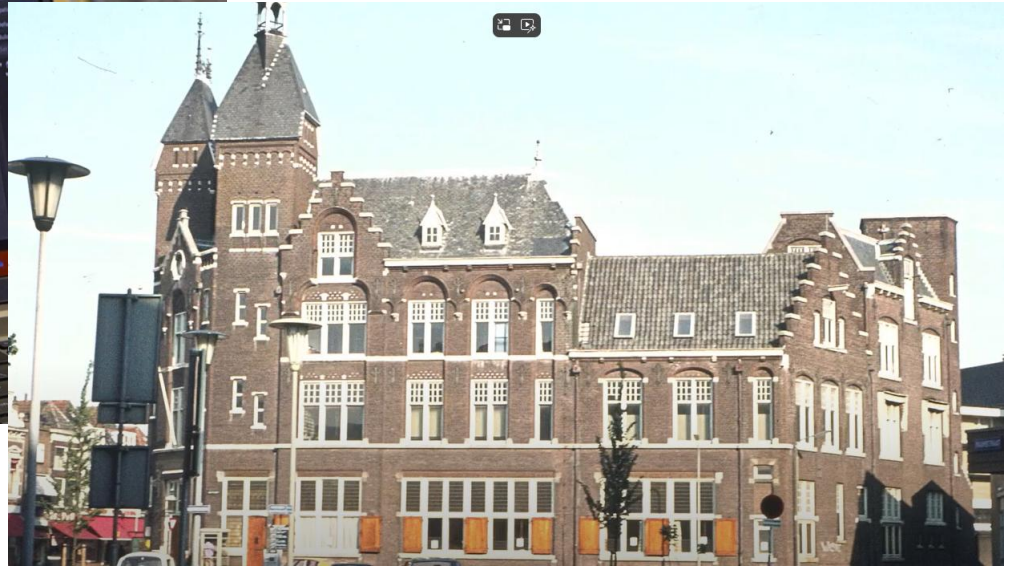
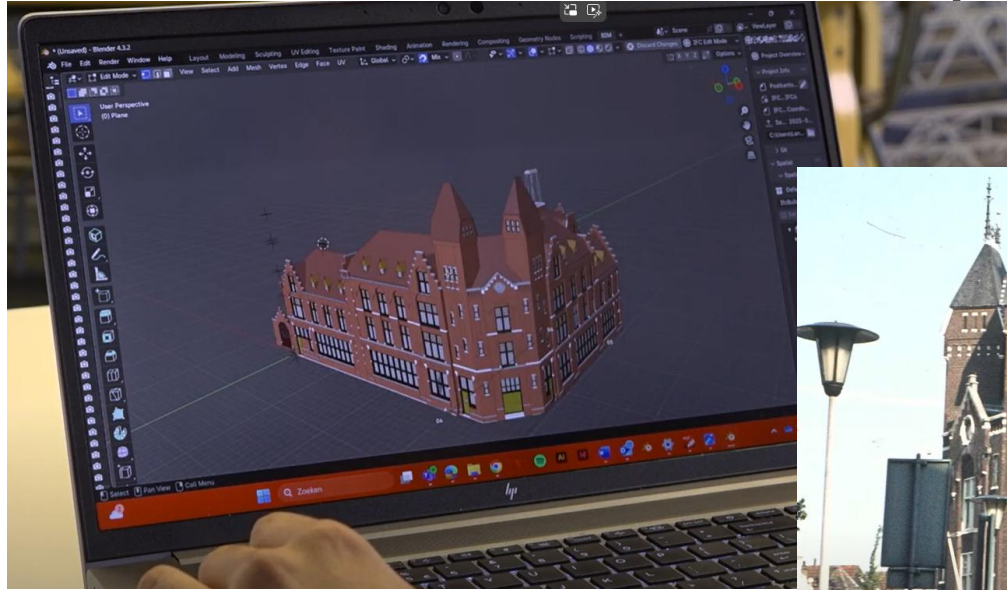
Previous projects



Previous projects



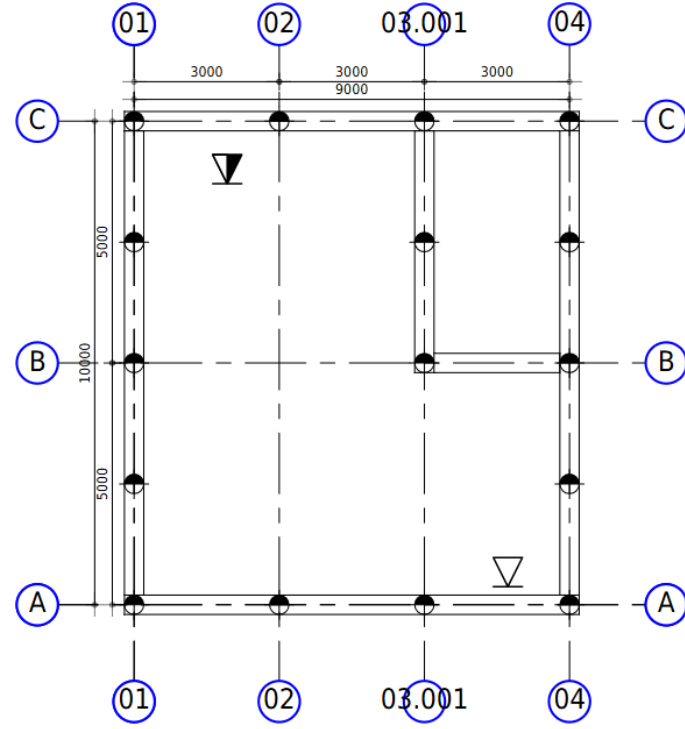
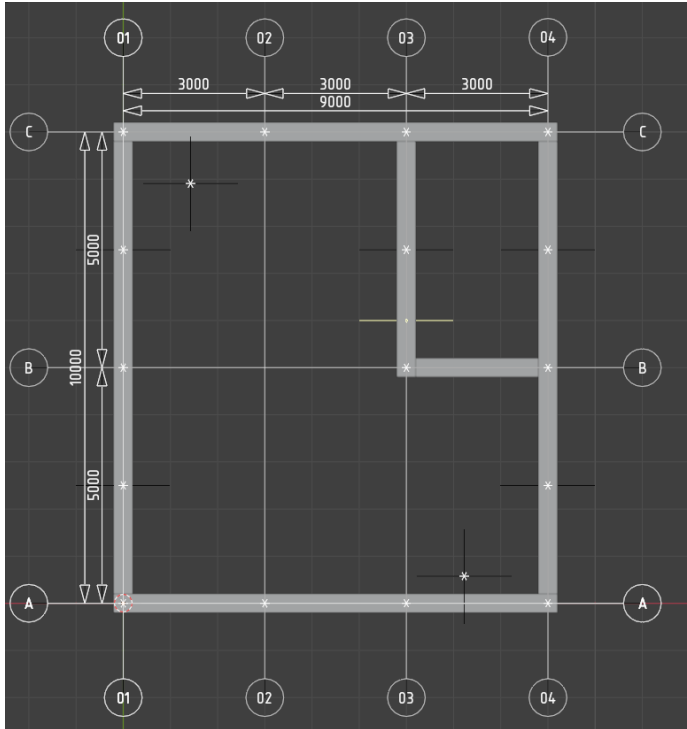
Previous projects



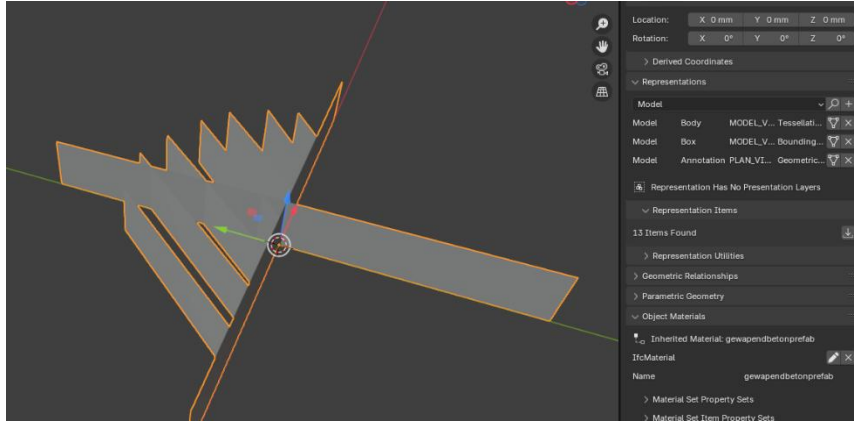
Previous projects



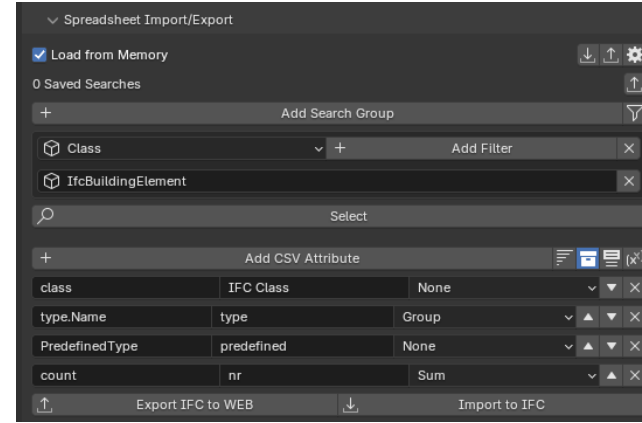
Pre-project testing



Pre-project testing



Symbols

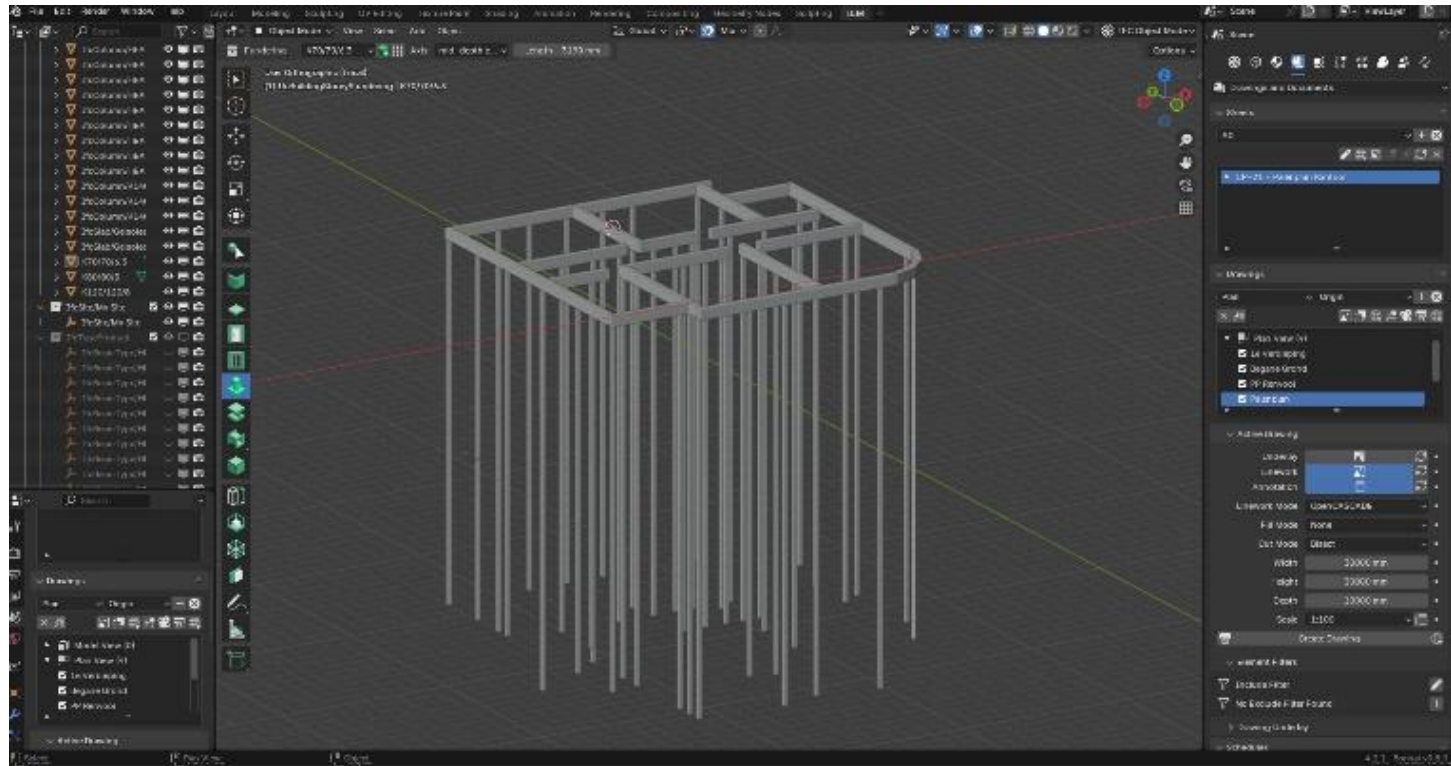


The screenshot shows a 'Schedules' tab with a table of data. The table has columns for 'IFC Class', 'type', 'predefined', and 'nr'. The data is organized into a summary row and a detailed row.

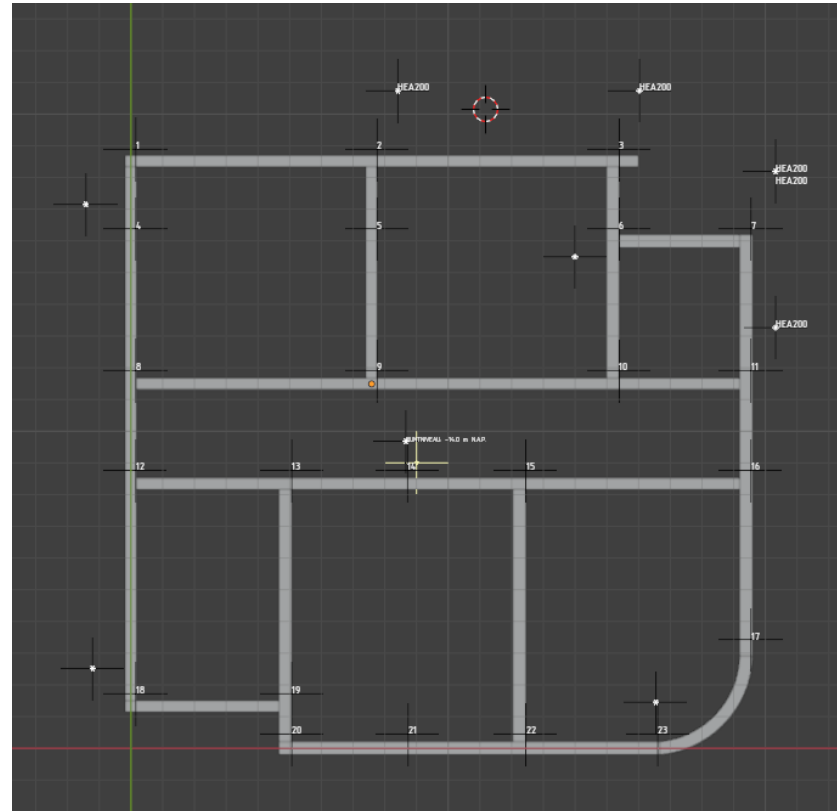
IFC Class	type	predefined	nr
sum: 0	sum: 0	sum: 0	sum: 8
IfcBeam	FB 400x500	N/A	6.0
IfcPile	N/A	BORED	2.0

Schedules

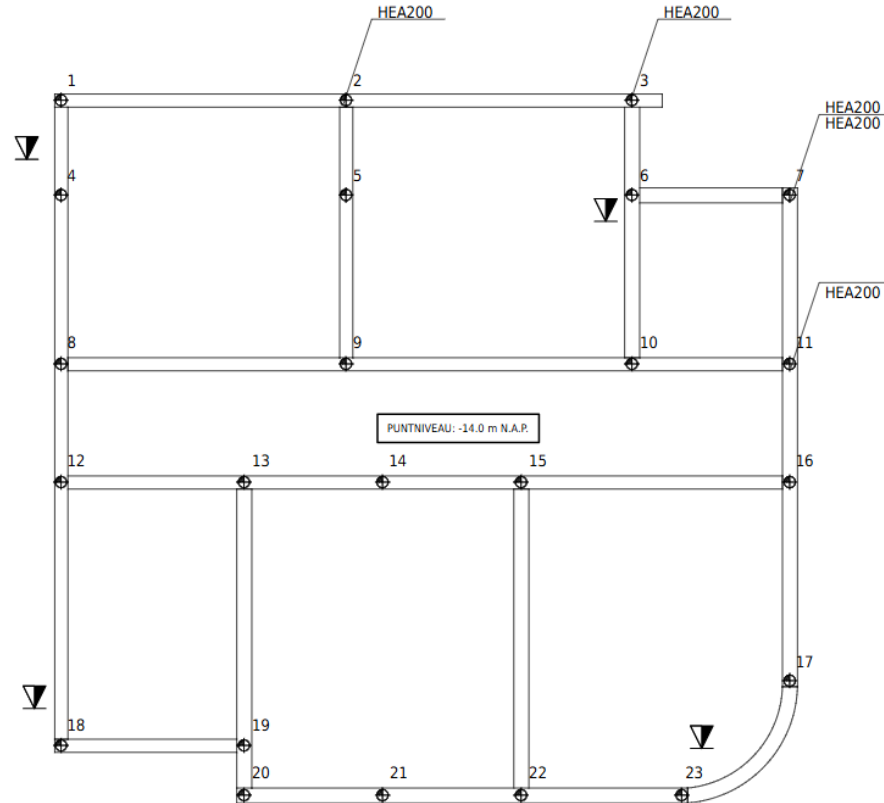
Pre-project testing



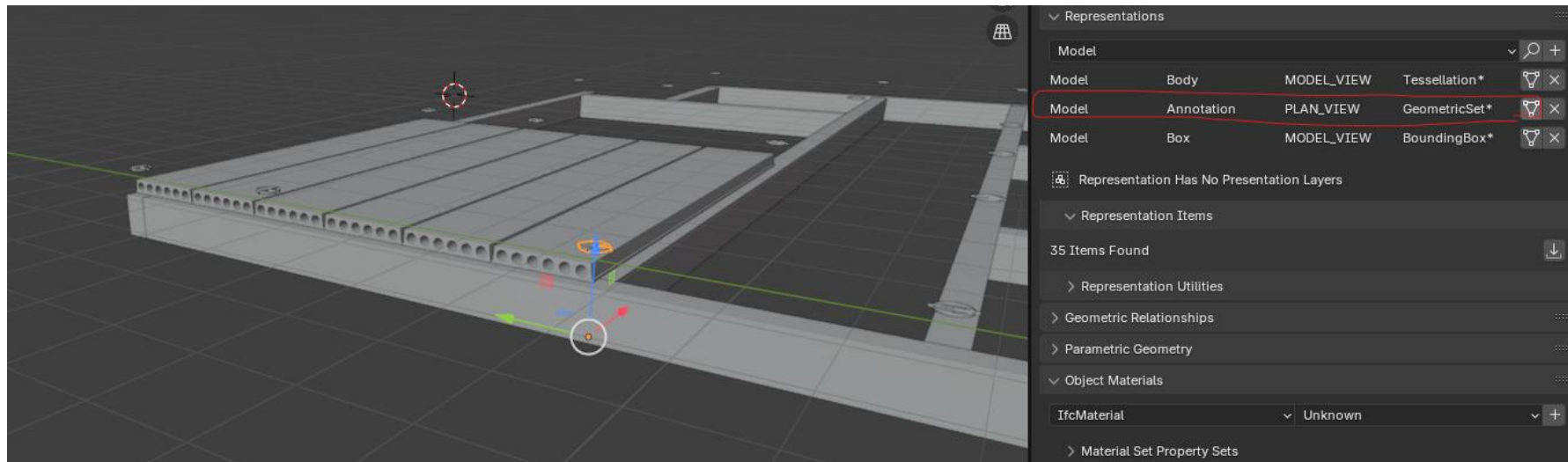
Pre-project testing



Pre-project testing



Pre-project testing



```
</g>  
<g id="kol_ond" transform="translate(0, -4) scale(1.5)">  
  <path  
    d="M12 0H-1L-6 15"  
    style="fill:none;stroke:#000000; stroke-width:0.15;"  
    id="path1" />  
</g>
```

Technical drawing of a building floor plan showing four rooms (01, 02, 03, 04) with dimensions and furniture placement.

Room 01: Located in the bottom right corner. It contains a bed (01) and a desk (02). The room has a curved wall on the right side.

Room 02: Located in the bottom left corner. It contains a desk (03) and a chair (04). The room has a curved wall on the left side.

Room 03: Located in the top left corner. It contains a desk (05) and a chair (06). The room has a curved wall on the left side.

Room 04: Located in the top right corner. It contains a desk (07) and a chair (08). The room has a curved wall on the right side.

Dimensions:

- Overall width: 7200 mm (divided into 3600 mm and 3600 mm).
- Overall height: 4000 mm (divided into 2000 mm and 2000 mm).
- Room 01 width: 3600 mm.
- Room 02 width: 3600 mm.
- Room 03 width: 3600 mm.
- Room 04 width: 3600 mm.
- Room 01 height: 2000 mm.
- Room 02 height: 2000 mm.
- Room 03 height: 2000 mm.
- Room 04 height: 2000 mm.

Furniture:

- 01: Bed
- 02: Desk
- 03: Desk
- 04: Chair
- 05: Desk
- 06: Chair
- 07: Desk
- 08: Chair

Legend:

- 01: Bed
- 02: Desk
- 03: Desk
- 04: Chair
- 05: Desk
- 06: Chair
- 07: Desk
- 08: Chair

REKENINGGEVENIS
 Houtwerkzaamheden uitvoeren volgens NEN-EN 1907 & NEN-EN 12699.
 Deskundig heibezocht is aangebevolen.
 Sandoerrapport: XXXXXX, DO, DOOR
 Funderingswaaier: XXXXXX, DO, DOOR

Na aanbrengen funderingspalen alle omliggende meten.

-Wapening volgens ophaal fabrikant

PAALSTAAT PREFAB BETONPAAL

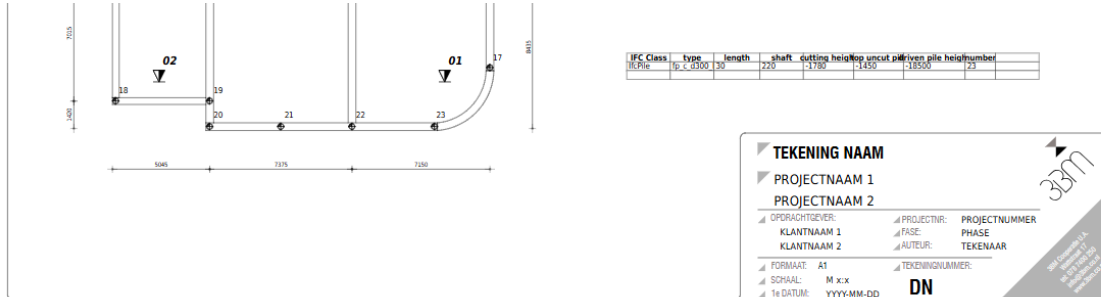
IFC Class	type	length	shaft	cutting height	top uncut pile	driven pile height	number
IfcPile	fp_c_d300	30	220	-1780	-1450	-18500	23



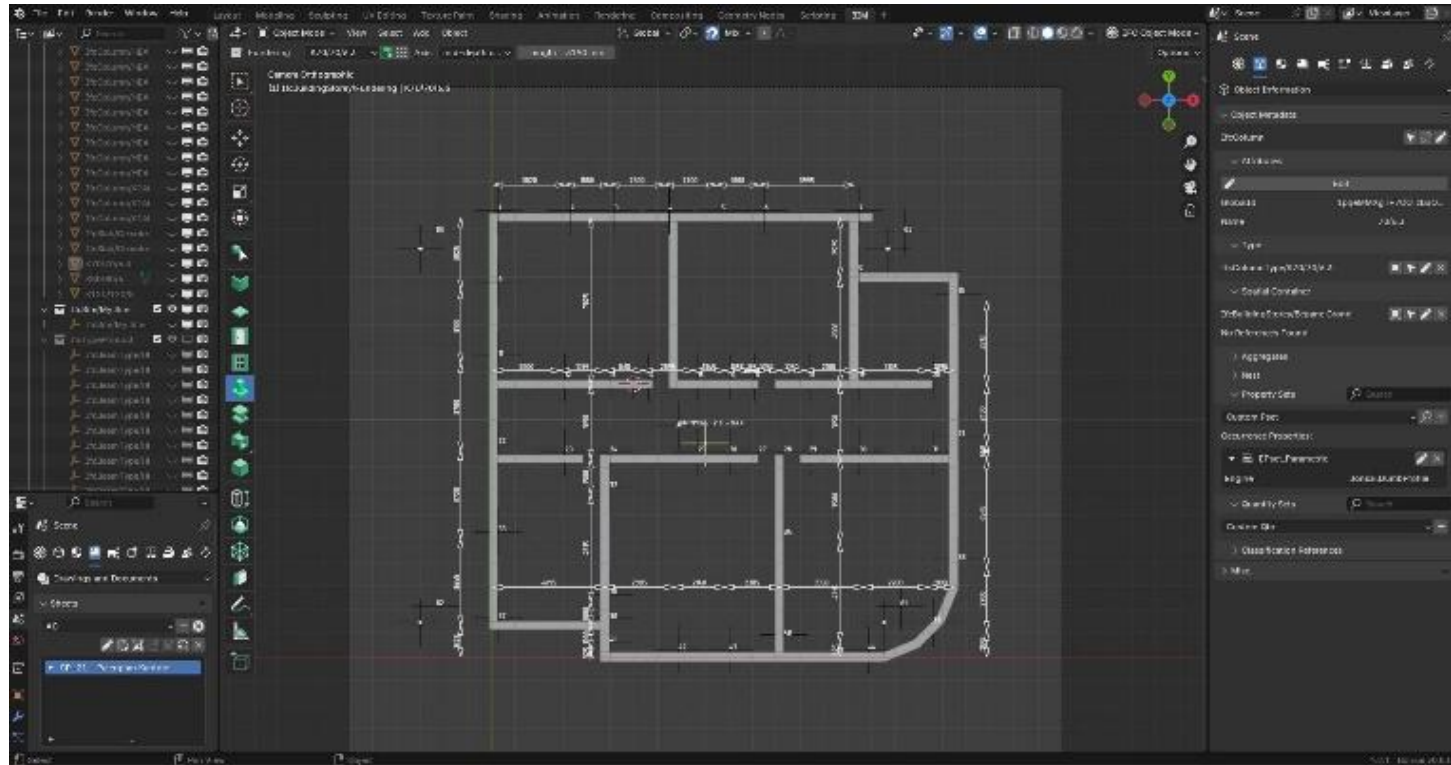
Pre-project testing



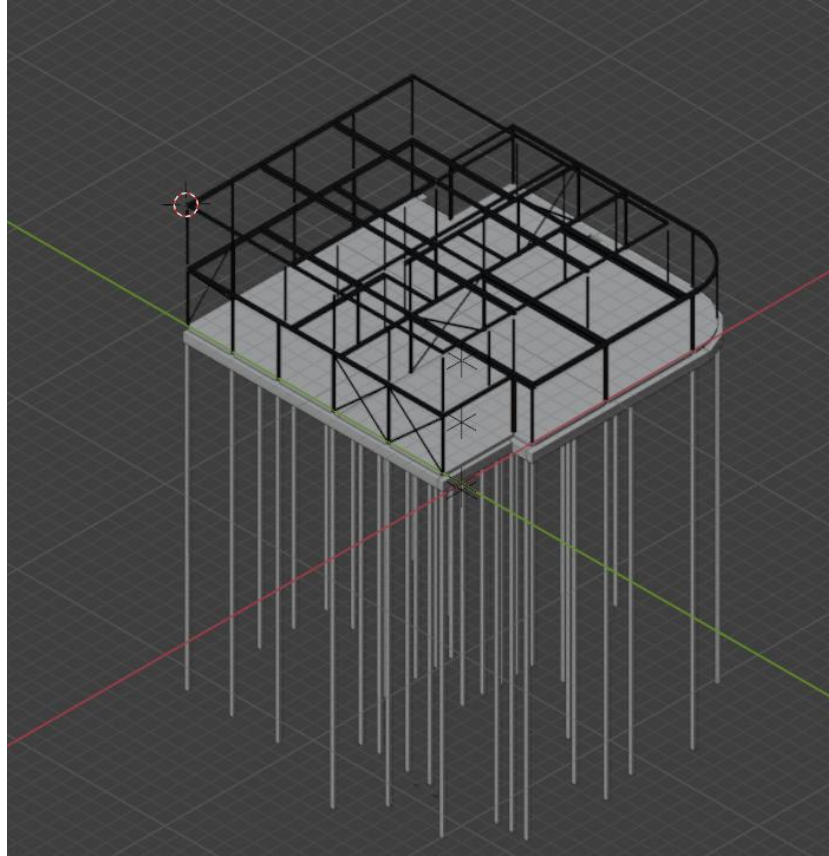
IFC Class	type	length	shaft	cutting height	top uncut pile	driven pile height	number	
lfcPile	fp_c_d300	30	220	-1780	-1450	-18500	23	



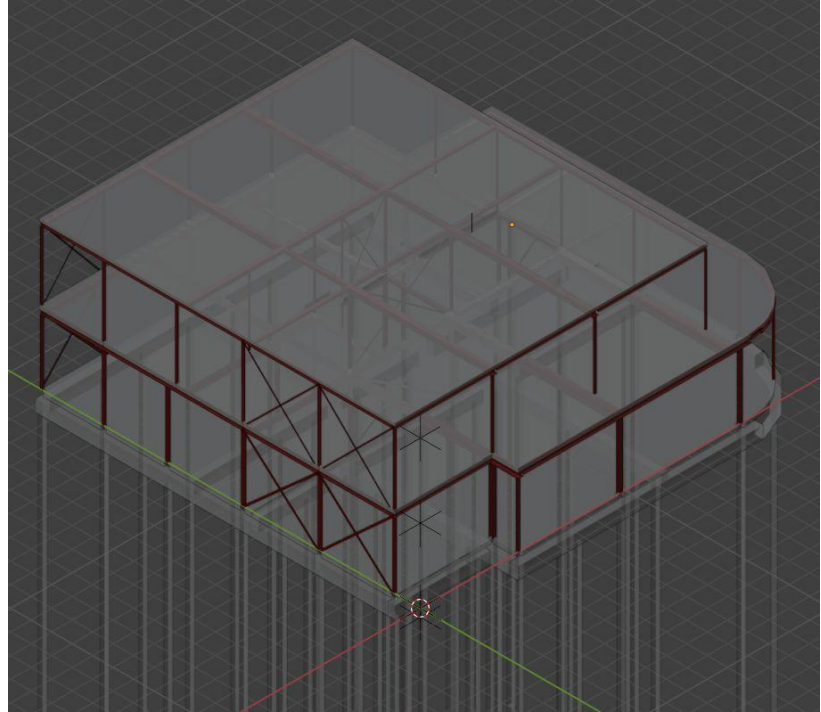
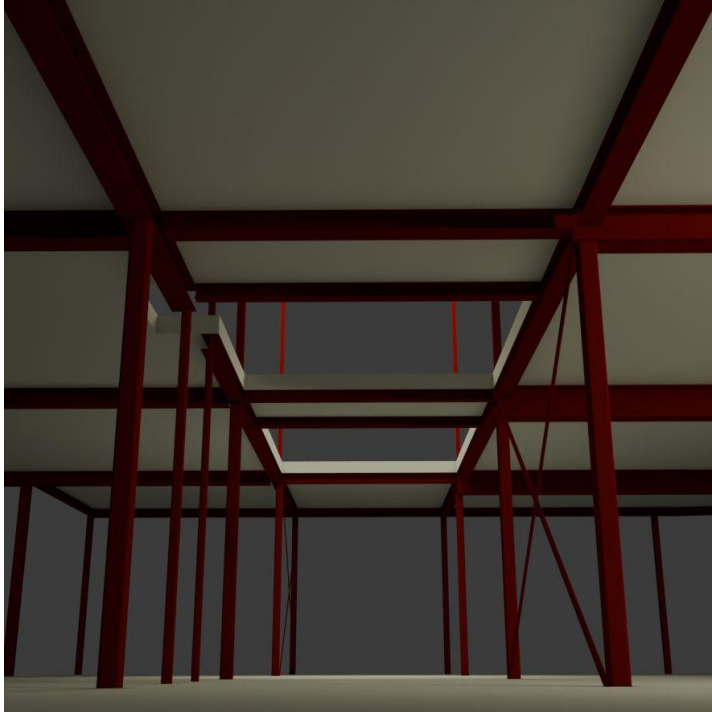
Modeling & Drawing Generation



Modeling & Drawing Generation



Modeling & Drawing Generation



Modeling & Drawing Generation

RENVOOI PREFAB BETONPALEN

REKENGEGEVENS

Reiwerkzaamheden uitvoeren volgens NEN-EN 1997 & NEN-EN 12699
Deskundig heitoezicht is aanbevelen.

Sondeerrapport: XXXXX, DD, DOOR
Funderingsadvies: XXXXX, DD, DOOR

Peil t.o.v. N.A.P. in het werk te controleren. Bij afwijkingen overleg met hoofdconstructeur.

Alle palen kalenderen.

Kalenderingen opsturen naar de hoofdconstructeur.

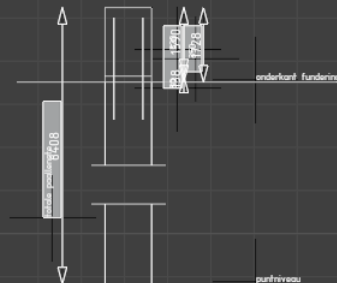
Statische berekening van de heipaal (op sterkte) aan te leveren door aannemer ter controle aan hoofdconstructeur.

Na aanbrengen funderingspalen alle paalposities inmeten.

PAALGEVENS

Prefab betonpaal

-Wapening volgens opgaaf fabrikant



PAALSTAAT PREFAB BETONPAAL

	[mm]		[m]	[m tov NAP]	[mm tov PEIL]	[mm tov PEIL]	[mm tov PEIL]
Symbol	SCHACHT	AANTAL	LENGTE	INHEINIVEAU	O.K. BALK	B.K. NIET GESNELDE PAAL	AFHAKHOOGTE
	220	16	22.33	-21.00	-1020	-670	-1000
	220	28	20.33	-19.00	-1020	-670	-1000
44							

RENVOOI PREFAB BETONPALEN

REKENGEGEVENS

Heiwerkzaamheden uitvoeren volgens NEN-EN 1997 & NEN-EN 12699
Deskundig heitoezicht is aanbevelen.

Sondeerrapport: XXXXX, DD, DOOR
Funderingsadvies: XXXXX, DD, DOOR

Peil t.o.v. N.A.P. in het werk te controleren. Bij afwijkingen overleg met hoofdconstructeur.

Alle palen kalenderen.

Kalenderingen opsturen naar de hoofdconstructeur.

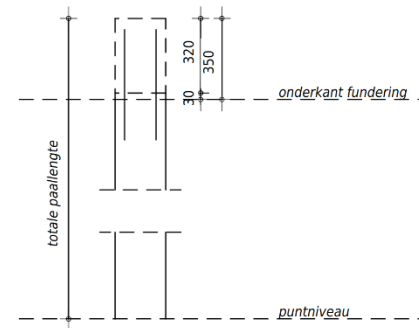
Statische berekening van de heipaal (op sterkte) aan te leveren door aannemer ter controle aan hoofdconstructeur.

Na aanbrengen funderingspalen alle paalposities inmeten.

PAALGEVENS

Prefab betonpaal

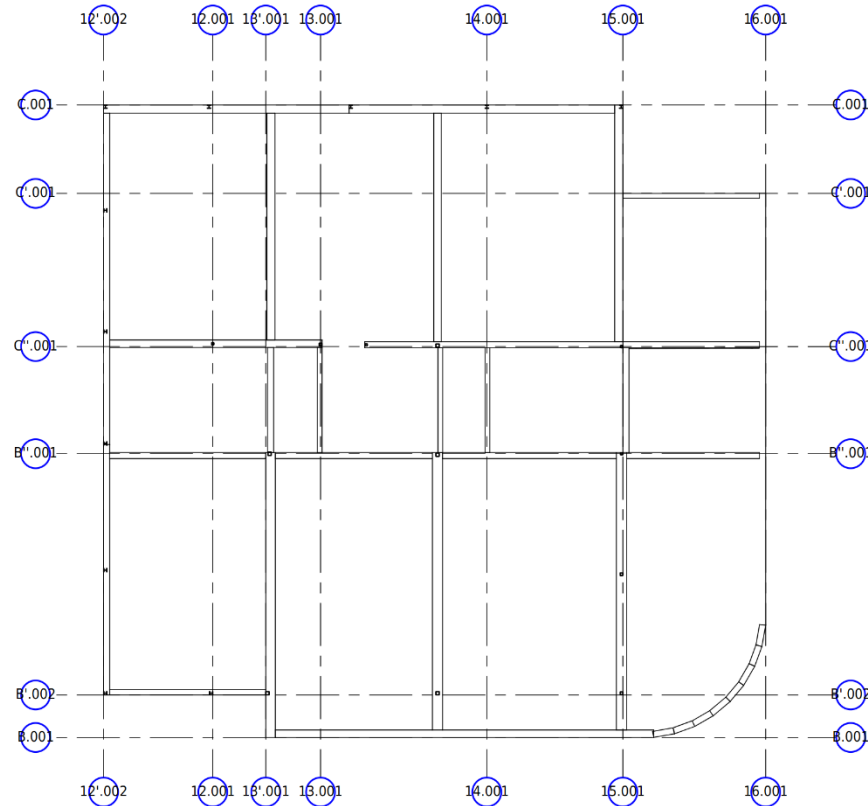
-Wapening volgens opgaaf fabrikant



PAALSTAAT PREFAB BETONPAAL

	[mm]		[m]	[m tov NAP]	[mm tov PEIL]	[mm tov PEIL]	[mm tov PEIL]
Symbol	SCHACHT	AANTAL	LENGTE	INHEINIVEAU	O.K. BALK	B.K. NIET GESNELDE PAAL	AFHAKHOOGTE
	220	16	22.33	-21.00	-1020	-670	-1000
	220	28	20.33	-19.00	-1020	-670	-1000
44							

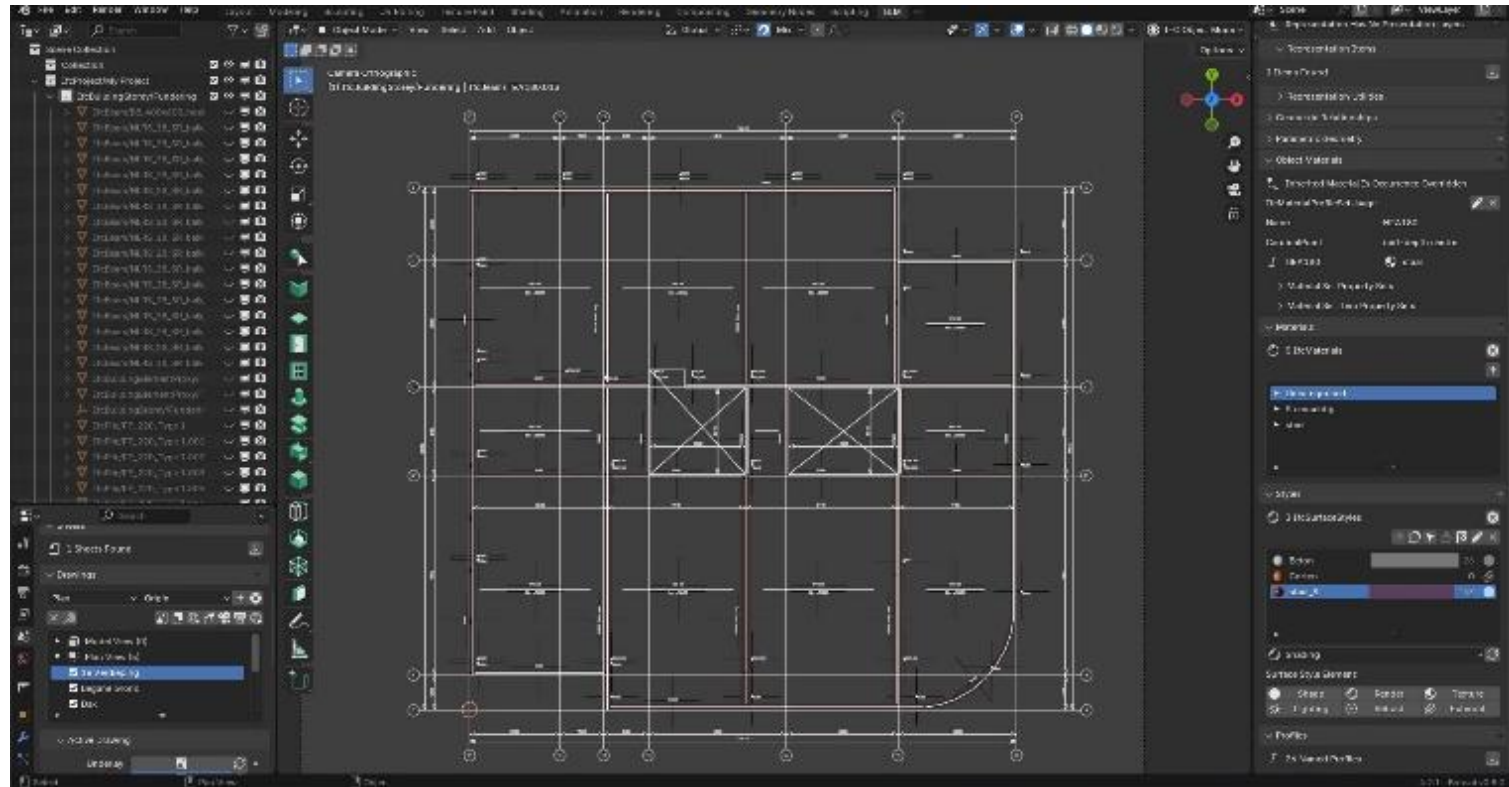
Modeling & Drawing Generation



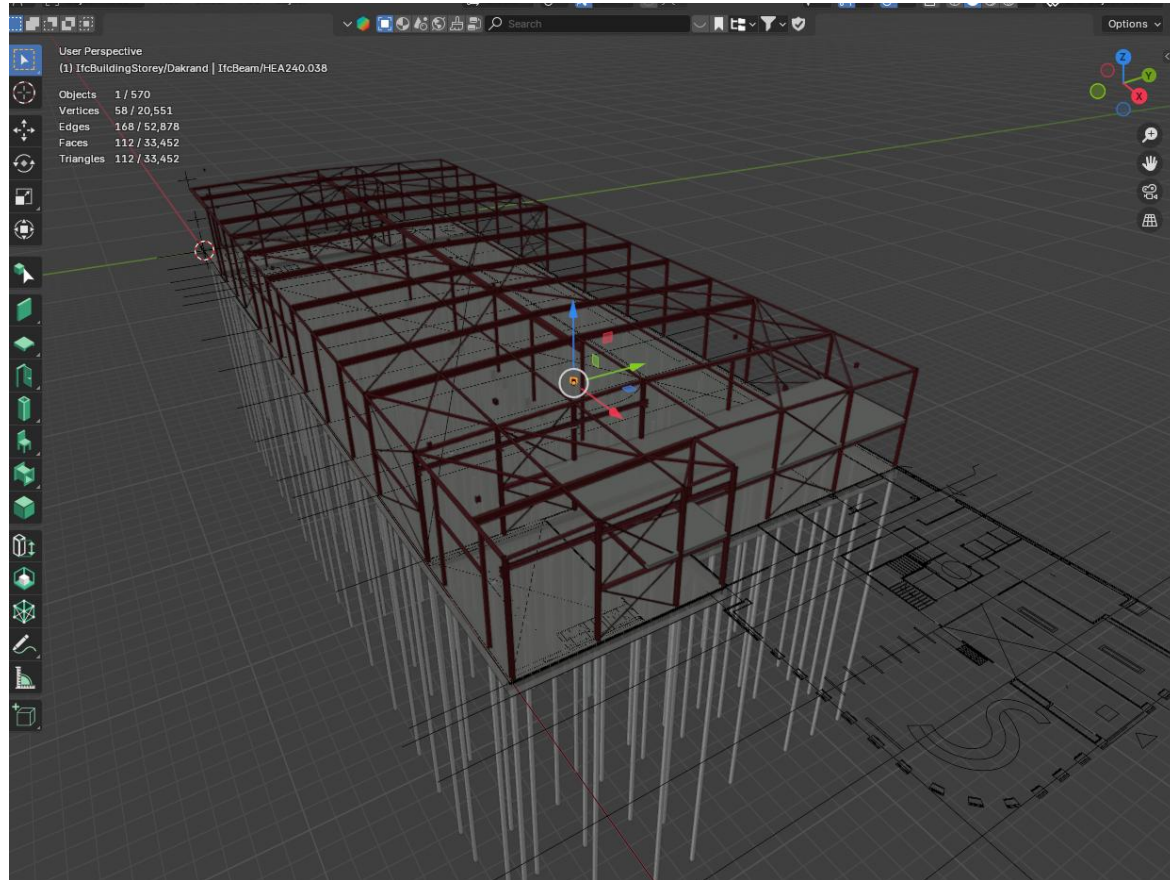
Second storey ->
problems!

(spatial container fixed
it)

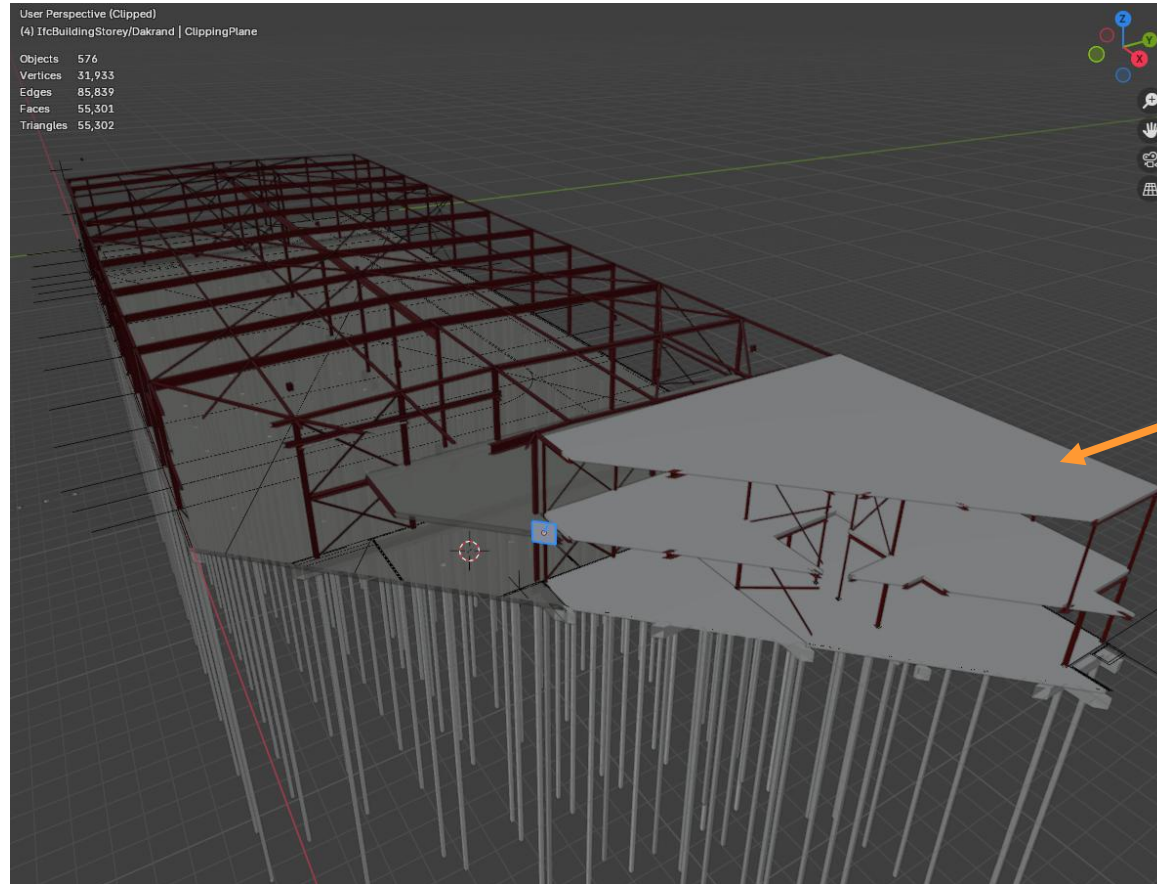
Modeling & Drawing Generation



Modeling & Drawing Generation

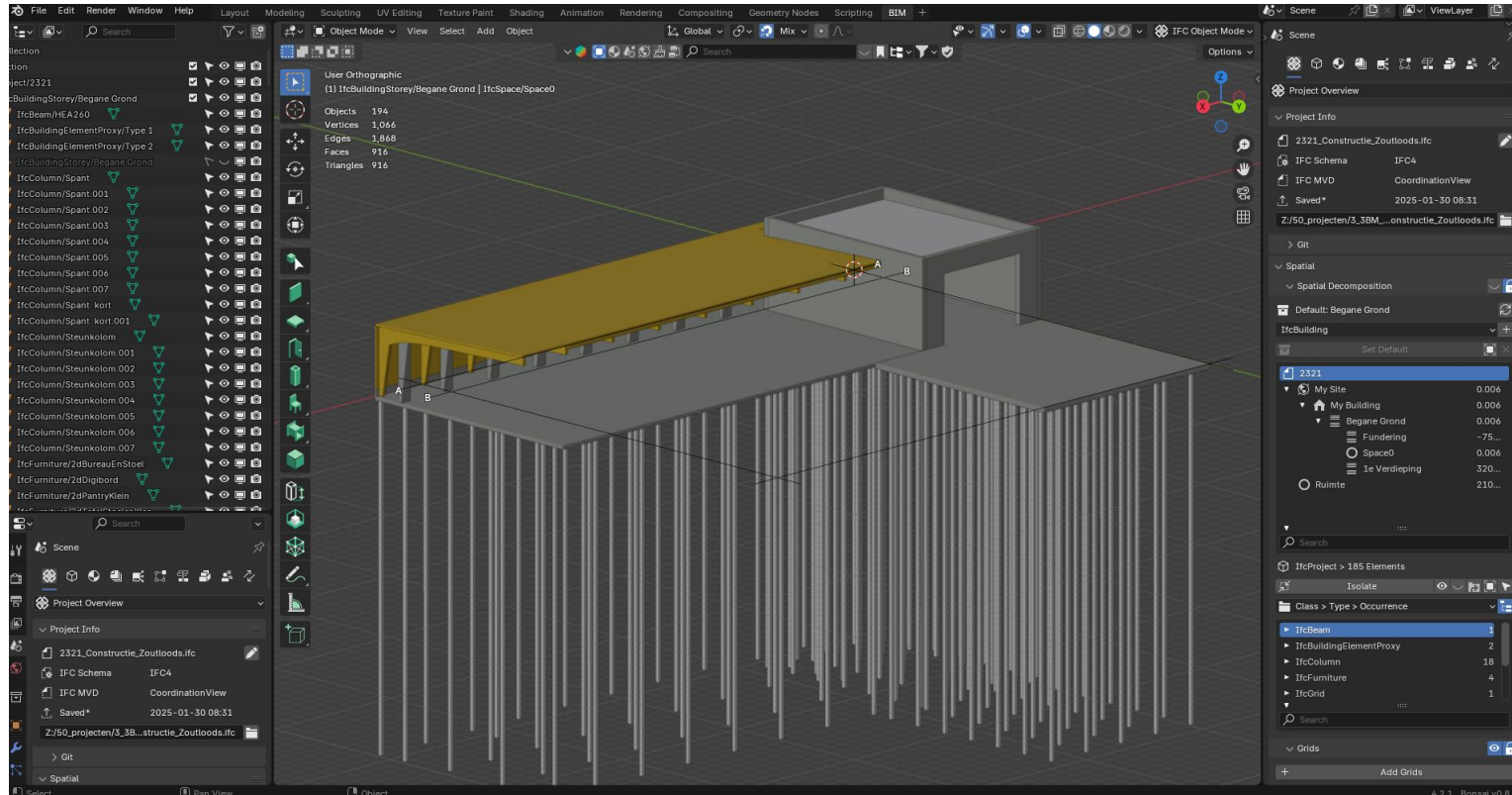


Modeling & Drawing Generation

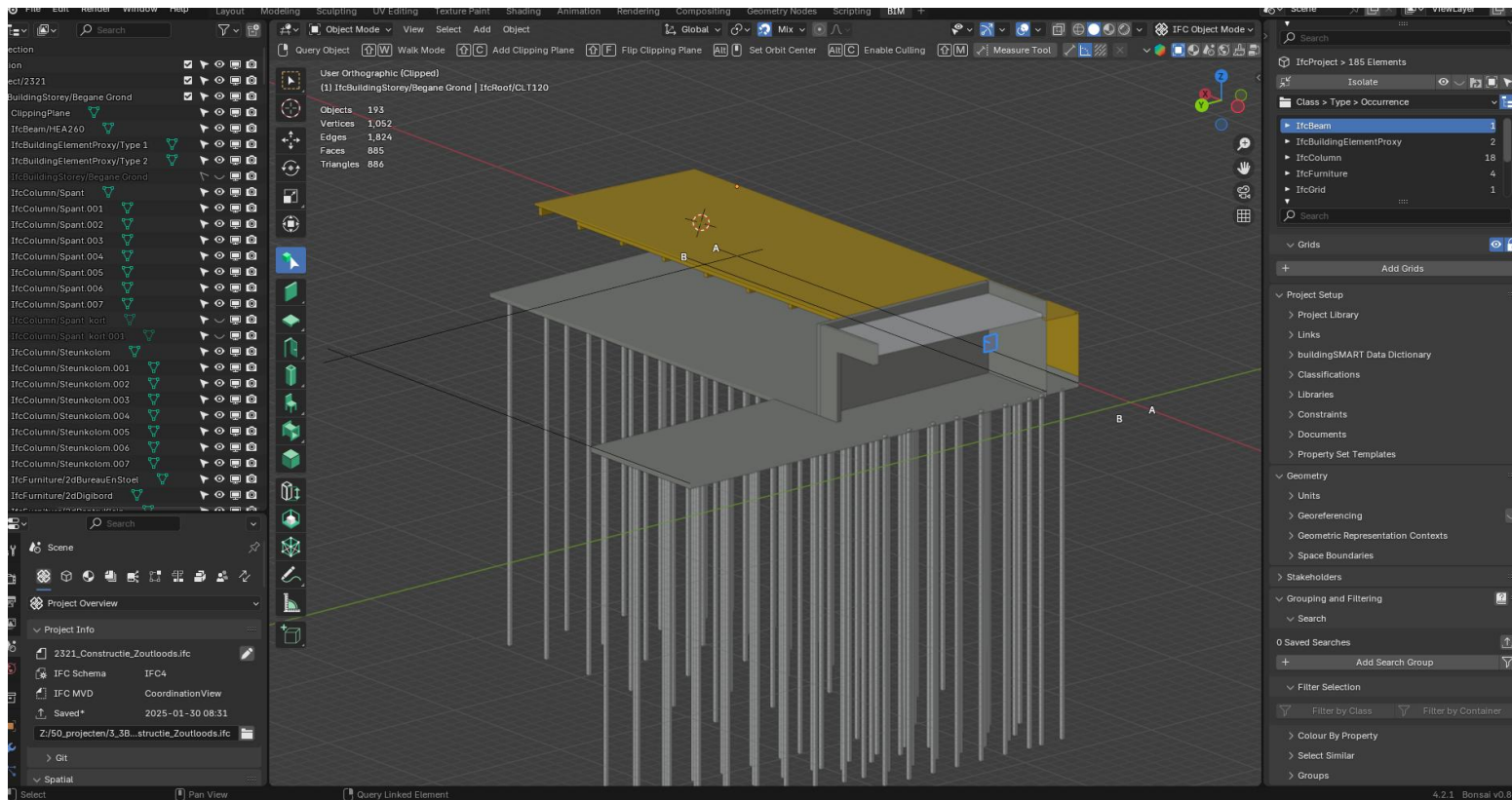


Linked IFC

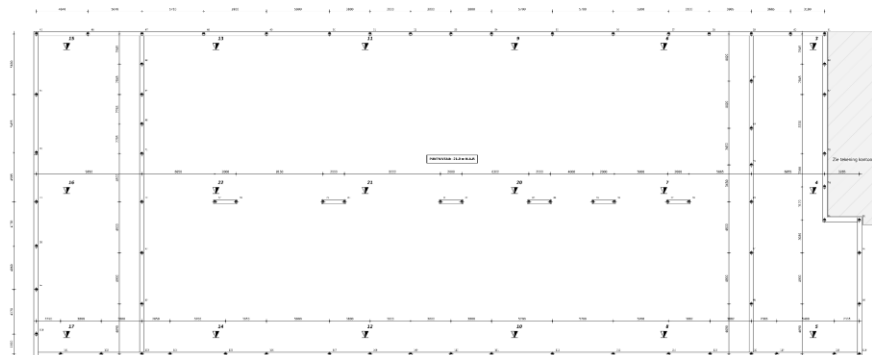
Modeling & Drawing Generation



Modeling & Drawing Generation



Palenque Kantoa



Palencia, H.

RENVOSI PREFAB BETONPALEN

FAHRGEBÜHRE
 Mehrere Ausfahrten in einem einzigen Jahr: 24,00 € & 48,00 € (24,00 €
 einmündige Fahrer, 48,00 € ab 18 Jahren).

Lehrerschein: 1000,- € (200,- €
 für die Theorie, 800,- € für die Praxis).

Wird von der 1. Fahrt an ein zweites, 40-facheingesparter, zweites
 Ausfahrtstermin
 480,- € (200,- € für die Theorie, 280,- € für die Praxis) angesetzt. Diese
 Gebühr wird demjenigen, der die 1. Fahrt antritt, bei der ersten
 Ausfahrt zur zweiten oder vierten Fahrt
 in vollqualifizierender Ausbildung (z.B. Berufsausbildung) halbiert.

FAHRGEBÜHRE



ANALYST PRIMA BETHUNGA

[illegible]

4. RENVOOI BELASTINGEN EN UITGANGSPUNTEN

1. <http://www.who.int>
2. <http://www.who.int>
3. <http://www.who.int>
4. <http://www.who.int>

[illegible][illegible]

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SIC CODE	VENDOR NUMBER	BILLING MONTH END DATE	CUSTOMER BILL TO VENDOR DUE DATE	PAY TO ORDER NO.

RENVOOI BOUWKUNDIGE BASIS

Ergebnis	Ja	Nein
1. Die Ergebnisse der Analyse sind für die Praxis relevant.		
2. Die Ergebnisse der Analyse sind für die Praxis relevant.		
3. Die Ergebnisse der Analyse sind für die Praxis relevant.		
4. Die Ergebnisse der Analyse sind für die Praxis relevant.		
5. Die Ergebnisse der Analyse sind für die Praxis relevant.		
6. Die Ergebnisse der Analyse sind für die Praxis relevant.		
7. Die Ergebnisse der Analyse sind für die Praxis relevant.		
8. Die Ergebnisse der Analyse sind für die Praxis relevant.		
9. Die Ergebnisse der Analyse sind für die Praxis relevant.		
10. Die Ergebnisse der Analyse sind für die Praxis relevant.		

RENVOOI VERANKERINGSLENGTEN

ONDERDEEL	Bedrijven	ONDERAPPLICATIES	
REFORMULATIE	C2025	ontwikkelen lokale ondernemers	
REFORMULAAT	8500	Realiseren van regionale ontwikkelings	
WELKEKLASSEN	KL4, KL3, KL2		
DEKINGEN ONDER	40		
DOVEN	35		
DOVEN	30		

BENYOOI STAAL

[illegible]

4. BENVOOL KANAALPLAATSDIENEN

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4 SONDERINGEN

[illegible]

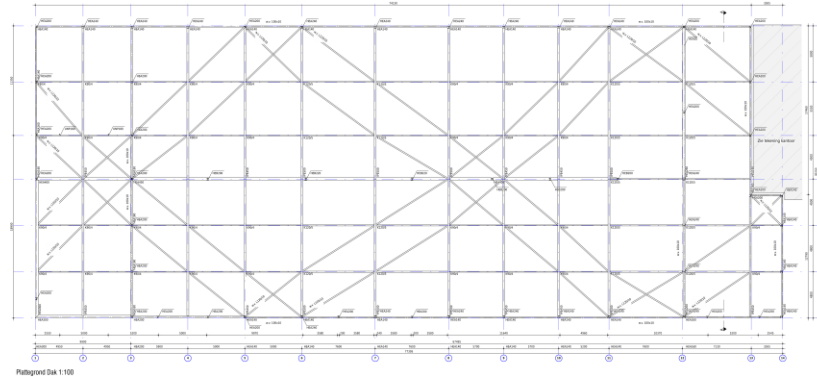
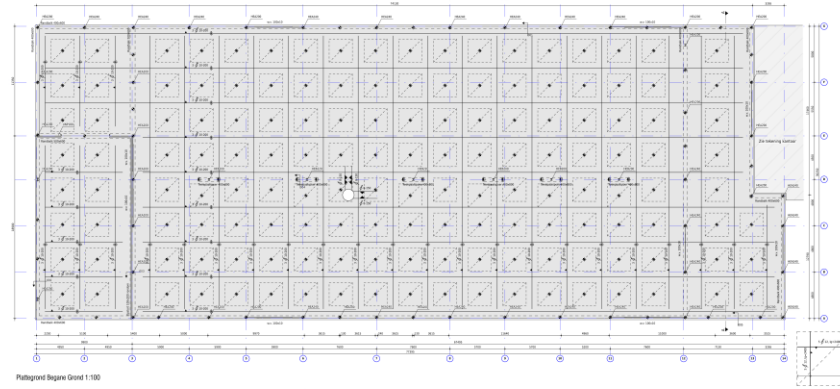
RENVOOI CHEMISCHE ANKERS

RENVOL LASSEN

Author Contributions: Conceptualization: J. C. S. and J. A. S.; Data curation: J. C. S.; Formal analysis: J. C. S.; Funding acquisition: J. C. S.; Investigation: J. C. S.; Methodology: J. C. S.; Project administration: J. C. S.; Resources: J. C. S.; Supervision: J. C. S.; Validation: J. C. S.; Visualization: J. C. S.; Writing—original draft: J. C. S.; Writing—review and editing: J. C. S. and J. A. S.

Example 1: `var myObj = {
 prop1: 1,
 prop2: 2,
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 prop306: 306,
 prop3`

Modeling & Drawing Generation



[illegible][illegible][illegible]

Detail C06 1:50

REVENUE BELASTINGEN EN UITGANGSPUNTEN	
Revenuen van de Staat	100,00
Revenuen van de Gemeenten	100,00
Revenuen van de Provincies	100,00
Revenuen van de Landelijke Overheid	100,00
Revenuen van de Gemeenschappelijke Regelingen	100,00
Revenuen van de Europese Unie	100,00
Revenuen van de Internationale Gemeenschap	100,00
Revenuen van de Private Sector	100,00
Revenuen van de Publieke Sector	100,00
Revenuen van de Sociale Sector	100,00
Revenuen van de Culturele Sector	100,00
Revenuen van de Sportsector	100,00
Revenuen van de Media Sector	100,00
Revenuen van de Kunstsector	100,00
Revenuen van de Wetenschap Sector	100,00
Revenuen van de Technologie Sector	100,00
Revenuen van de Milieu Sector	100,00
Revenuen van de Energie Sector	100,00
Revenuen van de Transport Sector	100,00
Revenuen van de Infrastructuur Sector	100,00
Revenuen van de Bouw Sector	100,00
Revenuen van de Industrie Sector	100,00
Revenuen van de Handel Sector	100,00
Revenuen van de Dienstverlening Sector	100,00
Revenuen van de Financiële Sector	100,00
Revenuen van de Media Sector	100,00
Revenuen van de Kunstsector	100,00
Revenuen van de Wetenschap Sector	100,00
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Revenuen van de Handel Sector	100,00
Revenuen van de Dienstverlening Sector	100,00
Revenuen van de Financiële Sector	100,00

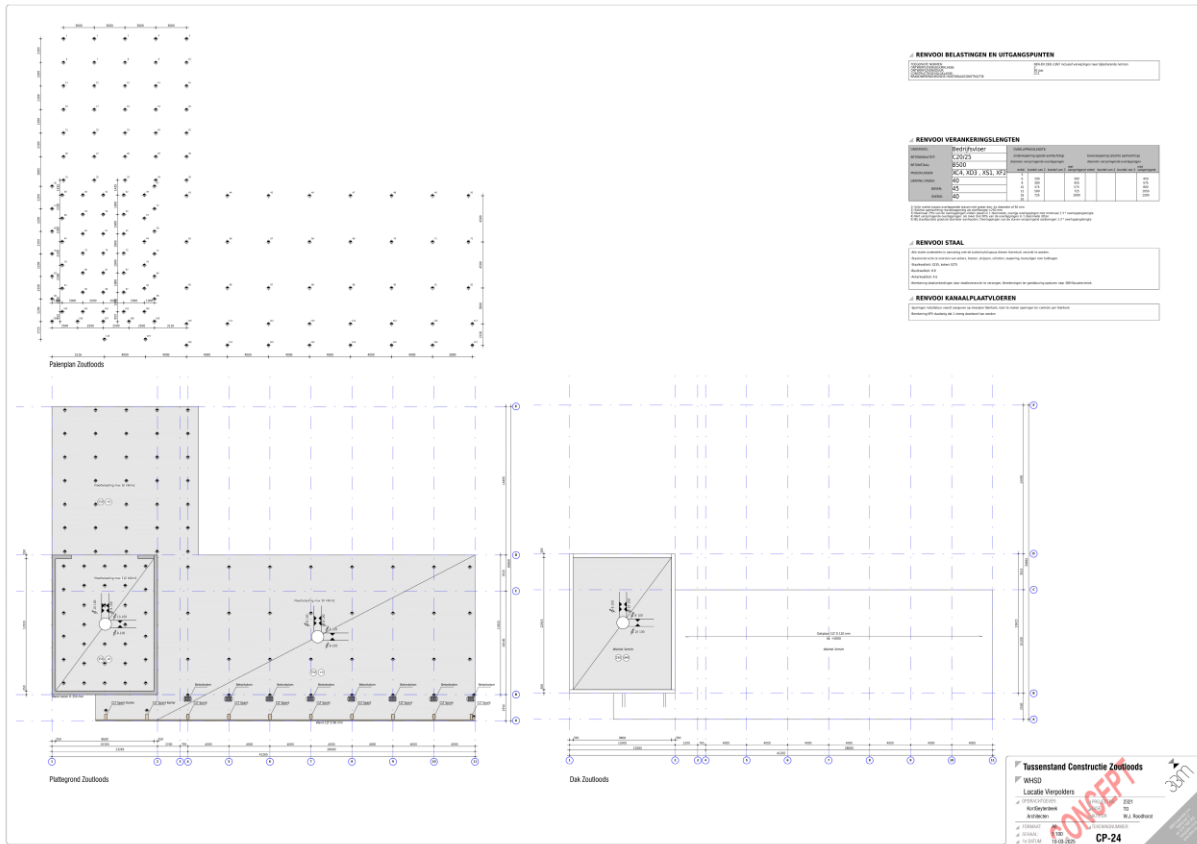
RENVOLP BOUWKUNDIGE BASIS

[illegible]

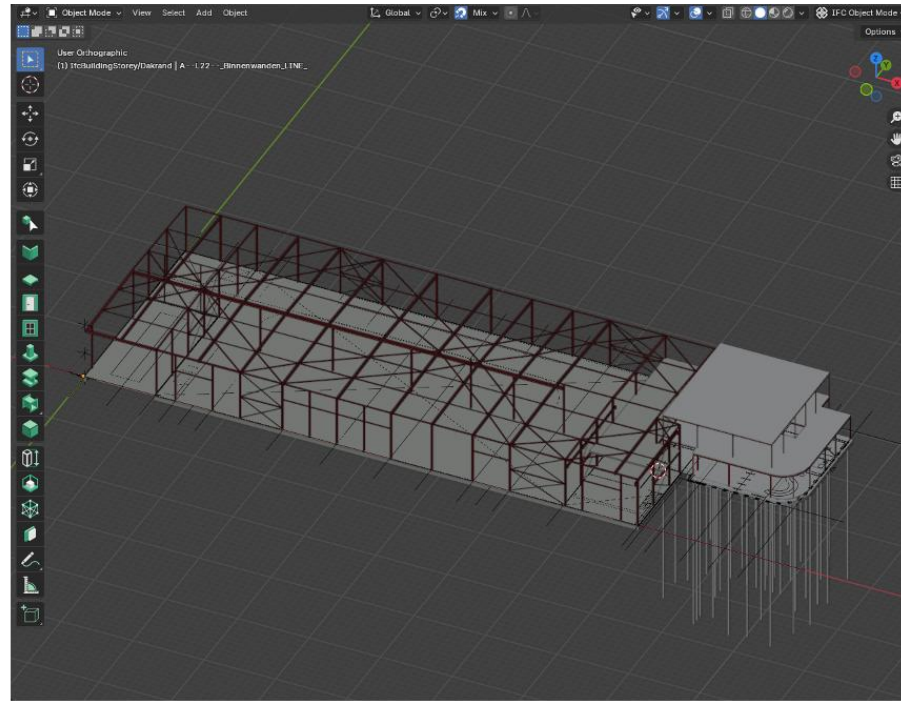
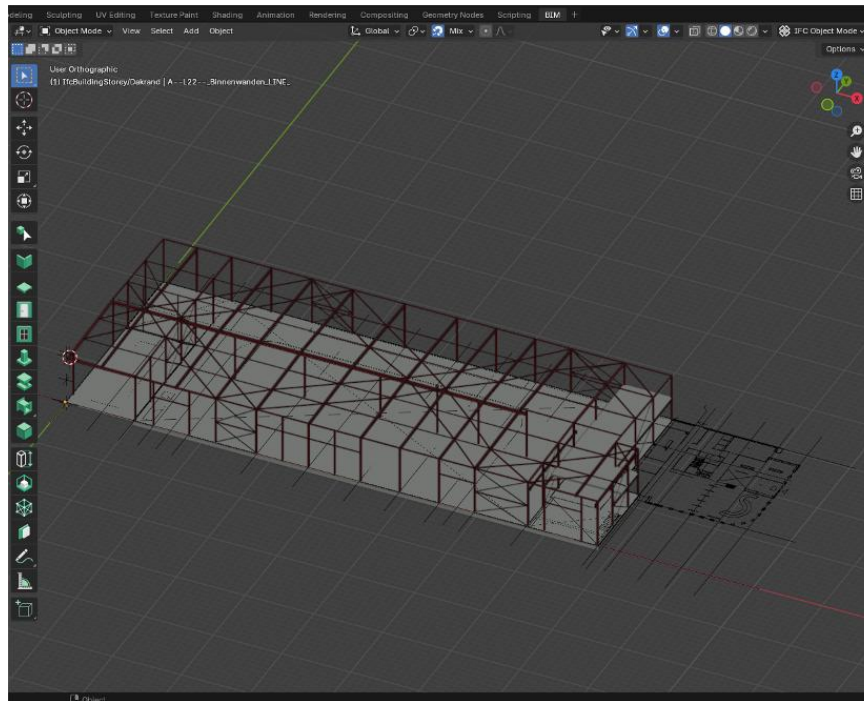
RENVOOI STAAL

RENVOOI KANAALPLAATVLOEREN

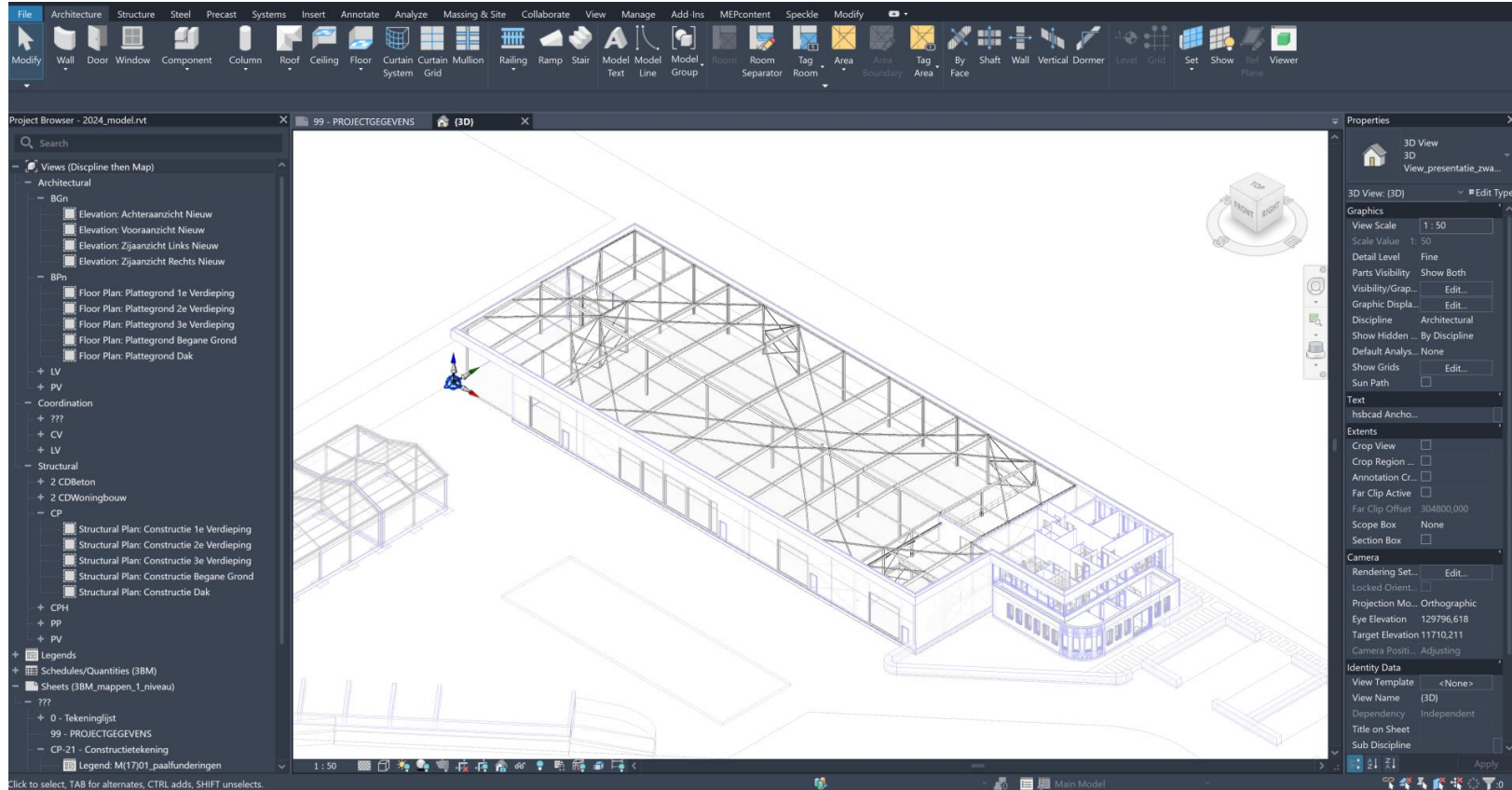
Modeling & Drawing Generation



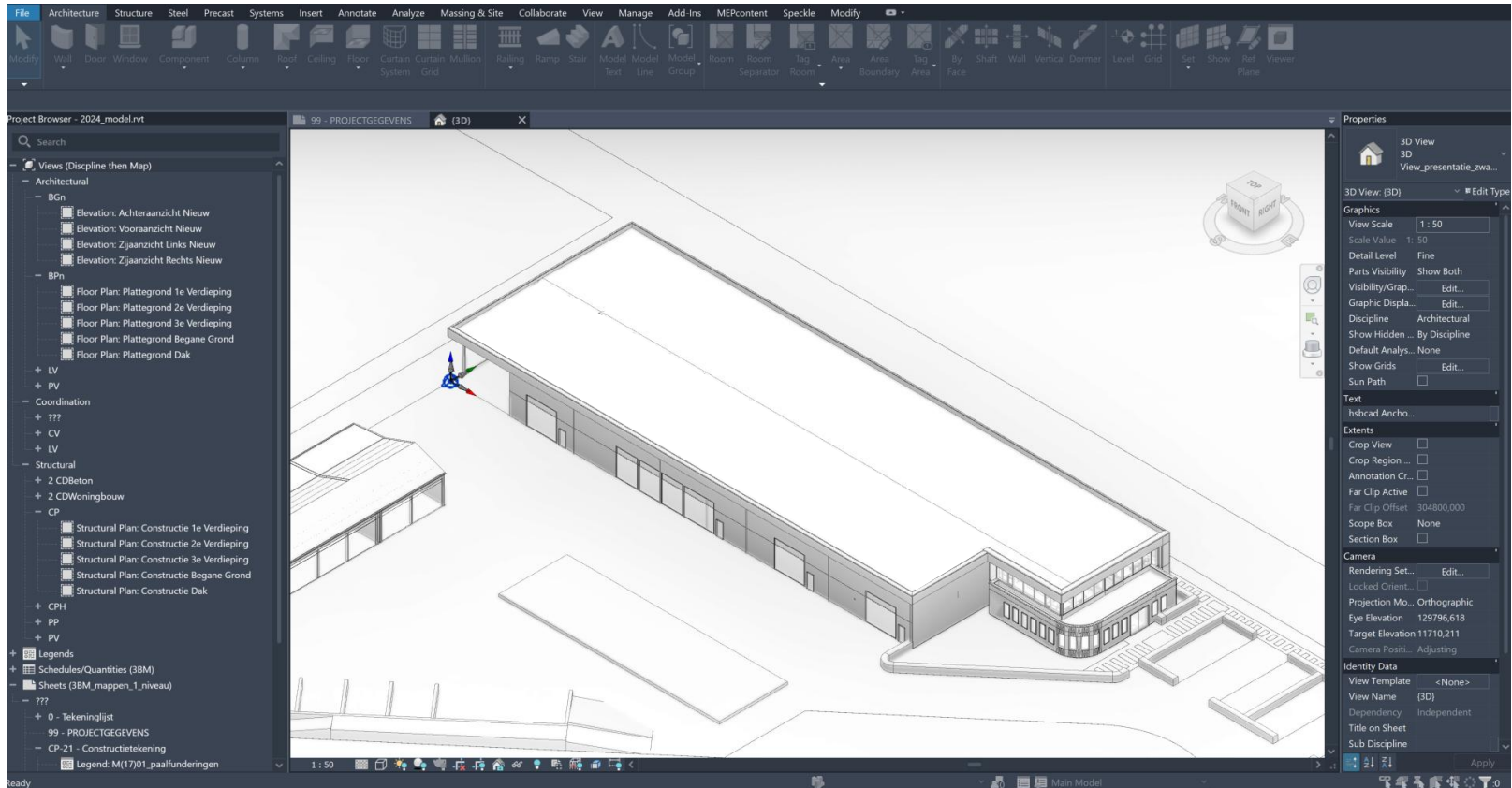
Collaboration



Collaboration



Collaboration



Problems & Solutions

“Problems”

A LOT of small problems

Annotations: a lot manual, time consuming trial-and-error, can't see what you do unless printing

Exporting to PDF: we couldn't export the normal way, time consuming, svg error

Elements behind other elements invisible on drawing

Missing functionality: Missing text sizes, linetypes etc.

“Solutions”

Get more familiar with software, ask on forum

write script for simple parts, started donation proposal for complex parts

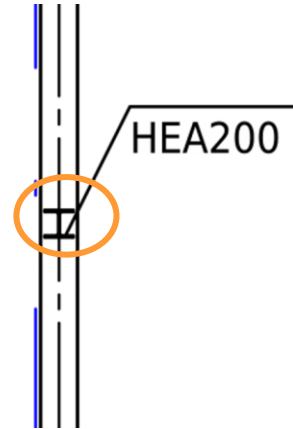
Import into Inkscape, export as PDF -> unfortunately large file sizes, but works!

Manually tracing outlines of upper element, hiding the element itself on drawing

Added them ourself to Bonsai / our template!

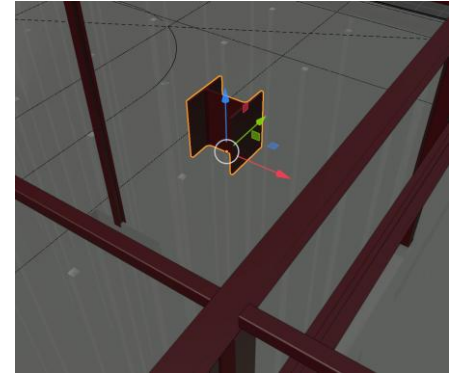
Problems & Solutions

“Problems”



Elements behind other elements
invisible on drawing

“Solutions”



Manually tracing outlines of
upper element, hiding the
element itself on drawing

OR

Problems & Solutions

See drawing as they print in the Bonsai viewport

PROJECT

Onderdeel van: [OSArch](#) - Open Source in AEC | Fiscal Host: Open Source Collective

See drawing as they print in the Bonsai viewport

BUDGET OVER ONS BIJDRAGEN GELD BIJDRAGEN ACTIES

Budget

Transparante en open financiën.

Alle Uitgaven Verlichtingen Bekijk alle verrichtingen →

FF	Financial contribution to See drawing as they print in th... Credit van FGSS-to-the-future naar See drawing as they print in the Bonsai viewport • 27 maart 2025	+ \$100,00 USD VOLTOOID
CA	Financial contribution to See drawing as they print in th... Credit van carlkip naar See drawing as they print in the Bonsai viewport • 25 maart 2025	+ \$50,00 USD VOLTOOID
MA	Financial contribution to See drawing as they print in th... Credit van Marcal naar See drawing as they print in the Bonsai viewport • 25 maart 2025	+ \$20,00 USD VOLTOOID

BIJDRAGE #842015 Details bekijken

BIJDRAGE #841079 Details bekijken

BIJDRAGE #841076 Details bekijken

§ VANDAAG'S SALDO	\$144,31USD
≈ TOTAAL OPGEHAALD	\$144,31USD
≈ TOTAAL UITBETAALD	≈ -USD
§ GESCHAT JAARLIJKS BUDGET	\$170,00USD

Issue #5130 fix + sheet all expand and contract, based on #4771 #6379

Merged Andrej730 merged 1 commit into IfcOpenShell:v0.8.0 from jes-r:v0.8.0 on Mar 17

Conversation 6 Commits 1 Checks 3 Files changed 1

jes-r commented on Mar 17 Contributor

No description provided.

Issue IfcOpenShell#5130 fix + sheet all expand and contract d3214ac

jes-r commented on Mar 17 Contributor Author

Dont know how to fix the mathutils error as i havent used it?

jes-r mentioned this pull request on Mar 17

expand/collapse all drawing contexts at one time. #5130

Closed

Successfully merged

Reviewers: sboddy

Assignees: No one assigned

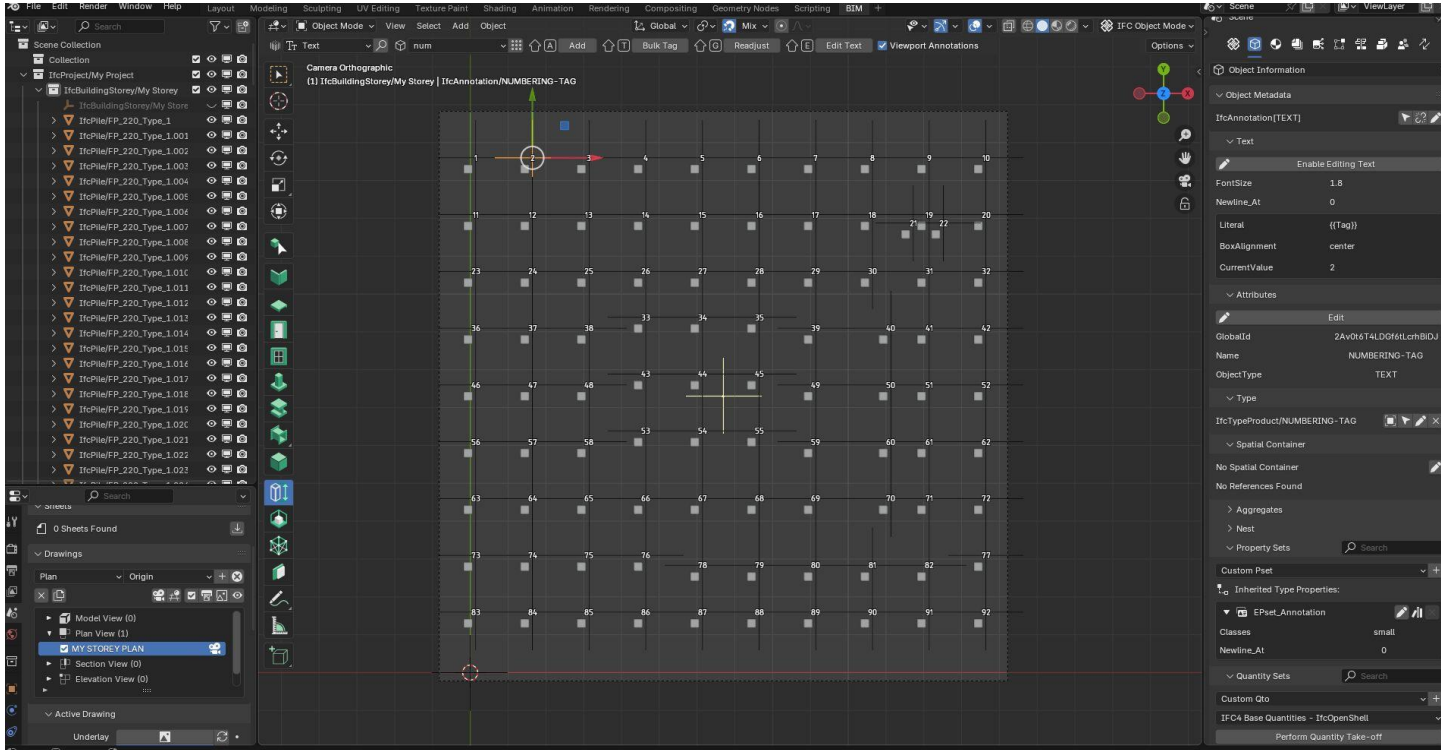
Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merged

Problems & Solutions



Pileplan
tedious work.
1 change -> again!

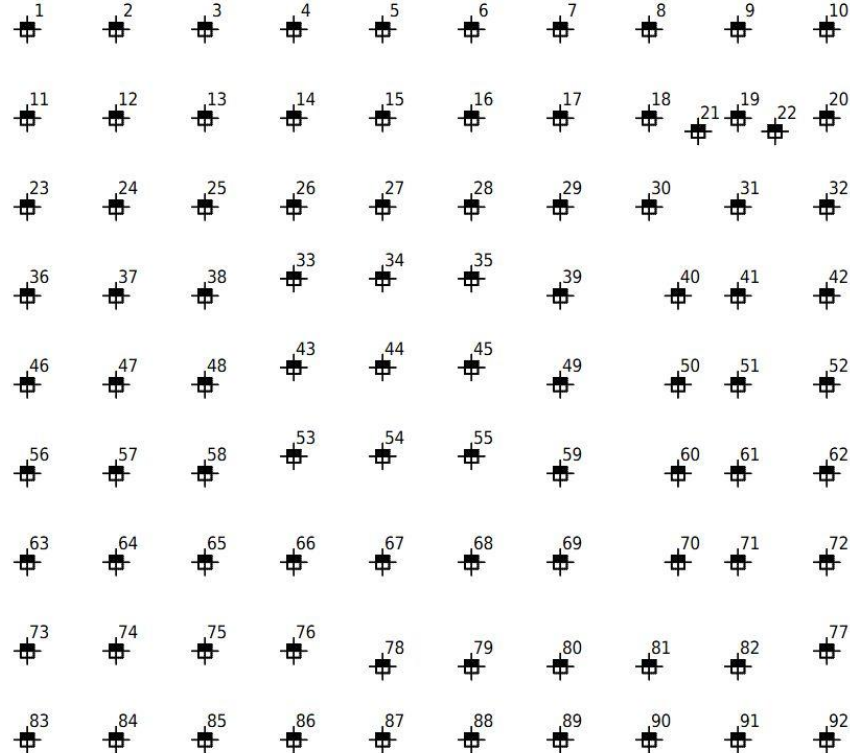
Problems & Solutions

```
26 # Sort the coordinates from top-left to bottom-right (highest y first, lowest x second)
27 sorted_coordinates = sorted(coordinates, key=lambda p: (-p[1], p[0]))
28
29 # Set the starting number for numbering
30 start_number = 1 # You can change this value to whatever starting number you want
31
32 # Number the sorted IfcFile elements
33 for i, (x, y, pile) in enumerate(sorted_coordinates):
34     # Create the tag (number) for the pile
35     tag_value = str(start_number + i)
36
37     # Assign the tag value to the pile's 'Tag' attribute (if it exists)
38     if hasattr(pile, 'Tag'):
39         pile.Tag = tag_value
40         print(f"Updated {pile.GlobalId} with Tag: {tag_value}")
41     else:
42         print(f"Error: 'Tag' attribute not found for {pile.GlobalId}")
43
44 # Optionally, save the updated IFC file under a new name
45 ifc_file.write(file_path + "_numbered.ifc")
46 print("IFC Saved to: ", file_path + "_numbered.ifc")
47 print(f"--- %s seconds ---" % (time.time() - start_time))
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Updated 3YH4Z4EbLBQx_lPgQd5xXa with Tag: 86
Updated 1zgrXxVt54dAKYzFkv3y7Q with Tag: 87
Updated 2kpK5jpc99tvoudbS98qeG with Tag: 88
Updated 3sumymj8rCdByZbtgcrKf with Tag: 89
Updated 0Bh4Gxwuv5xP1AeB$h_43S with Tag: 90
Updated 1GrVuBBGb4EvTii1502zGm with Tag: 91
Updated 0ndQ_Fz91340EhJzQC0yub with Tag: 92
IFC Saved to:          pilenumber_numbered.ifc
--- 0.18461227416992188 seconds ---
PS C:\Blender Foundation\Blender\4.2\extensions\local\lib\python3.11\site-packages\bonsai >
```

Problems & Solutions



Future of Bonsai



home

download

community

documentation

search ifc class

\$3118 / \$5,000 funded



Bonsai

An add-on for beautiful, detailed, and data-rich OpenBIM with Blender



Free, open source, native IFC authoring platform

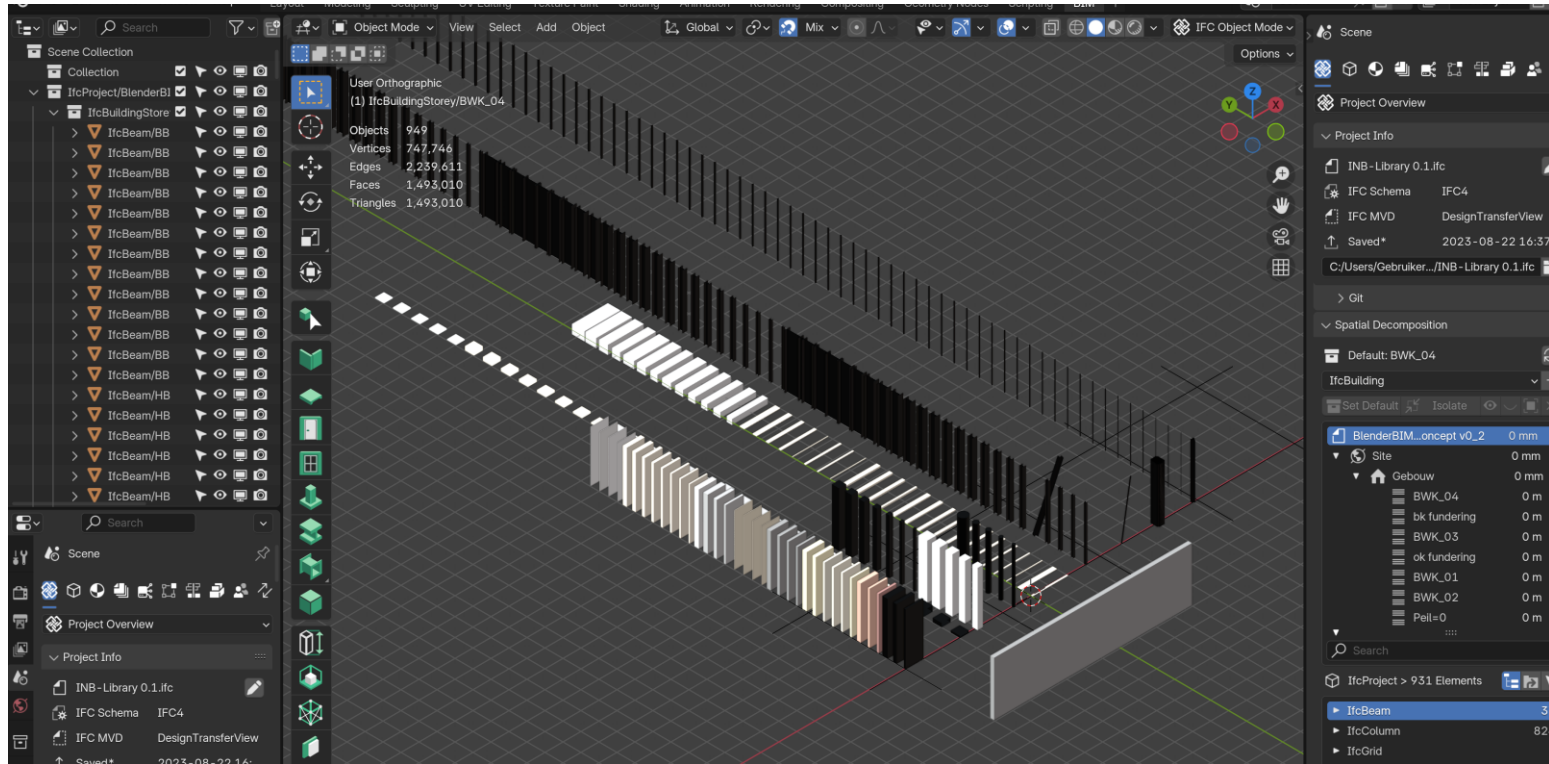
Future of Bonsai

The screenshot shows the GitHub interface for the IfcOpenShell repository. The top navigation bar includes links for Code, Issues (948), Pull requests (40), Discussions, Actions, Projects (2), Wiki, Security, and Insights. A search bar is located on the right. Below the navigation bar, the 'Issues' section is active, displaying a list of open issues. The 'Open' count is circled in orange. The issues listed include bugs and features related to coordinate errors, Blender integration, build errors, wall flipping, door instance deletion, resource editing, and annotation position.

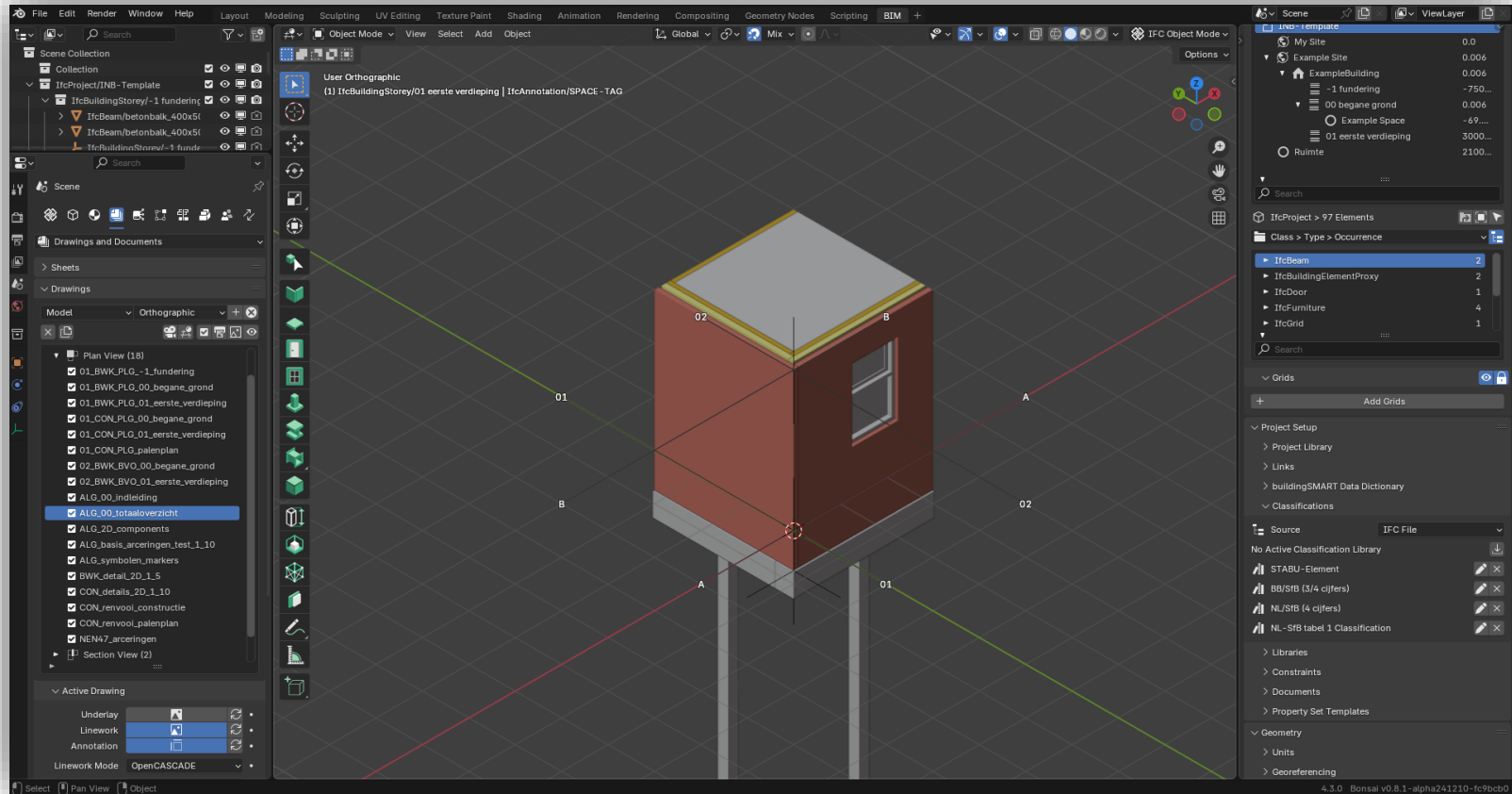
Issue list details:

- Unhelpful error message when creating coordinates** (Bug #6567 - zaesur opened 1 hour ago)
- Error getting Bonsai v08.2 in Blender 4.4.1. DLL load failed while importing bsdf** (Bug #6566 - Rosmai opened 2 hours ago)
- build error in XmlSerializer.cpp** (Bug #6565 - brunopostle opened 10 hours ago)
- Flip start/end of wall** (Feature #6561 - theoryshaw opened 12 hours ago)
- if you select the door type and delete via the outliner, it seems to delete the instance, which is fine, but when you open the file again, the door instance is there, but it's an empty.** (Bug #6560 - theoryshaw opened 13 hours ago)
- Can't edit "Resource"** (Bug #6559 - Ave-5D opened 13 hours ago)
- Take into account z position when adding Annotation PLAN_LEVEL** (Feature #6558 - SavuGust27 opened 17 hours ago)

Future of Bonsai | INB-Template

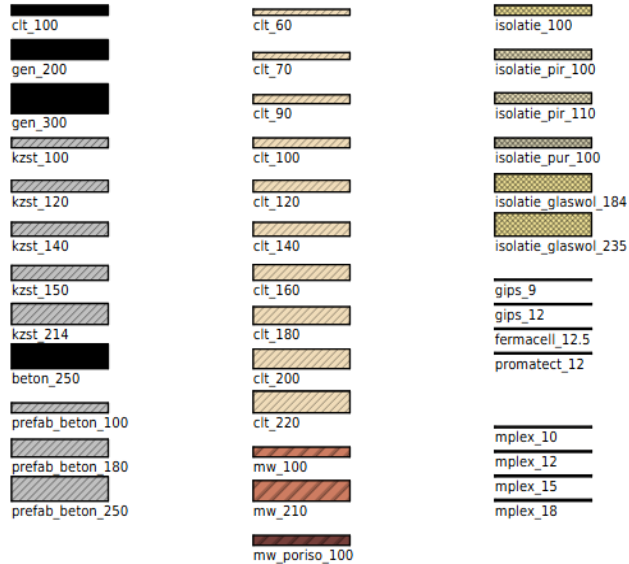


Future of Bonsai | INB-Template

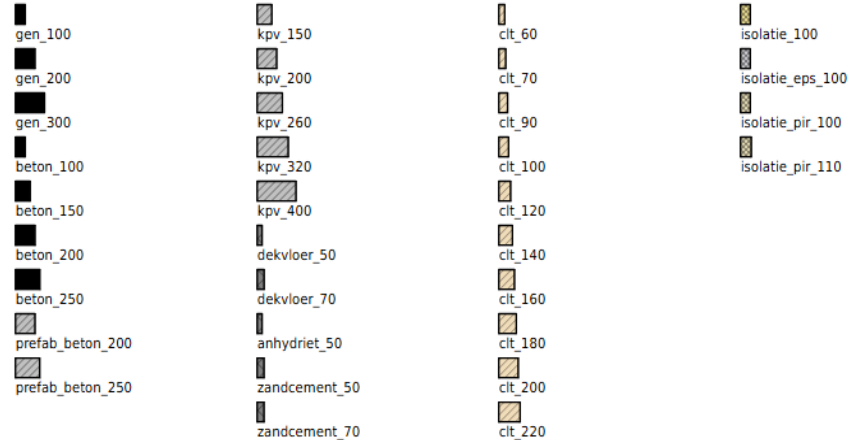


Future of Bonsai | INB-Template

lfcWallTypes



lfcSlabTypes



Future of Bonsai | INB-Template

Arceringen Test 1:10

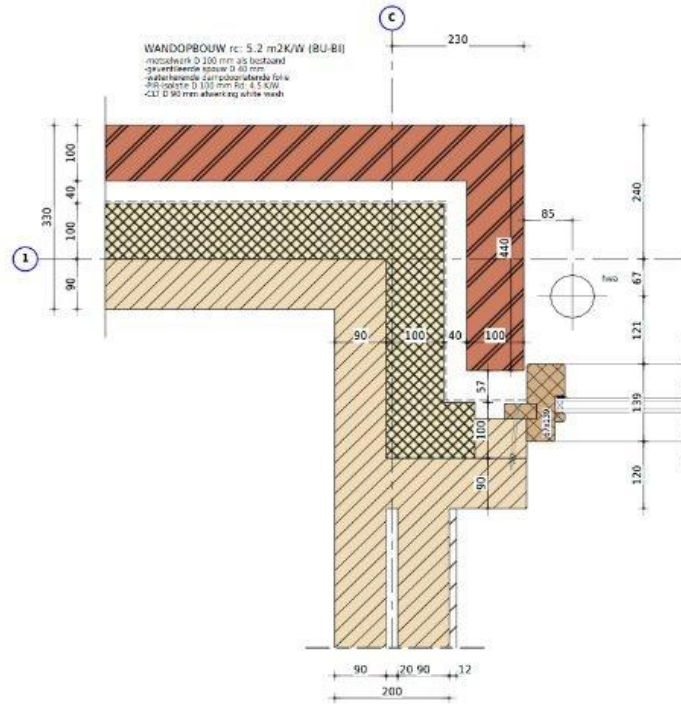
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isolatie	
sierbeton	
metselwerknietgebakkenkunststeen	
gewapendbetonprefab	
naaldhout	
loofhout	
naaldloofhoutlangsarcering	
bekledingsplaat	
nietdragendelichtescheidingswanden	
lood	
afdichtingsmiddel	
samengesteldewandvloerafwerking	
natuursteen	

```

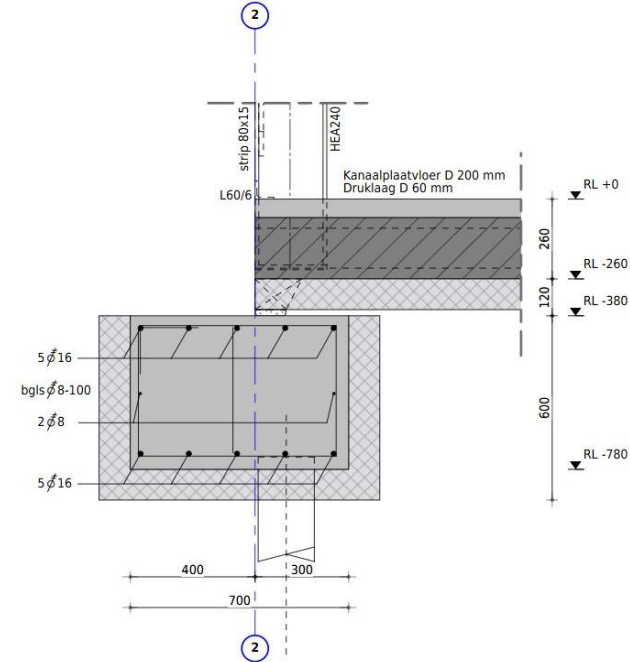
158 .PredefinedType-naaldhoutredcedar{ fill: url( #CUST_NEN47
159 .PredefinedType-naaldhoutiroko{ fill: url( #CUST_NEN47_12_
160 .PredefinedType-naaldhoutniove{ fill: url( #CUST_NEN47_12_
161 .PredefinedType-naaldhoutbangkirai{ fill: url( #CUST_NEN47
162 .PredefinedType-naaldhoutazobe{ fill: url( #CUST_NEN47_12_
163 .PredefinedType-naaldhoutbamboe{ fill: url( #CUST_NEN47_12_
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166 .PredefinedType-bekledingsplaatOSB{ fill: url( #CUST_NEN47
167 .PredefinedType-bekledingsplaatMDF{ fill: url( #CUST_NEN47
168 .PredefinedType-bekledingsplaatwatervast{ fill: url( #CUST
169 .PredefinedType-bekledingsplaatkoume{ fill: url( #CUST_NE
170 .PredefinedType-bekledingsplaatrodespaan{ fill: url( #CUST
171 .PredefinedType-bekledingsplaatgroenespaan{ fill: url( #CU
172 .PredefinedType-bekledingsplaatunderlayment{ fill: url( #C
173 .PredefinedType-isolatieglaswol{ fill: url( #CUST_NEN47_17
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175 .PredefinedType-isolatiePIR{ fill: url( #CUST_NEN47_17_iso
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177 .PredefinedType-isolatiePUR{ fill: url( #CUST_NEN47_17_iso
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187 .material-NEN471metselwerkbaksteen{ fill: url( #NEN47_1_me
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189 .material-NEN473metselwerknietgebakkenkunststeen{ fill: url
190 .material-NEN474nietdragendelichtescheidingswanden{ fill: u
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194 .material-NEN479natuursteen{ fill: url( #NEN47_9_natuurstee
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196 .material-NEN4711samengesteldewandvloerafwerking{ fill: url
197 .material-NEN4712naaldhout{ fill: url( #NEN47_12_naaldhout

```

Future of Bonsai | INB-Template



Horizontaal Detail , Schaal: 1:5



Future of Bonsai | INB-Template

The screenshot shows the GitHub repository page for 'INB-Template' by user '3BMLabs'. The repository is public and has 8 stars, 1 watch, and 0 forks. The main branch is 'main'. The repository description is 'IFC-template voor de bouw-sector in Nederland (experimentele fase)'. The file list includes folders like '.idea', '02_scripts', '03_user example projects', '04_development projects', '05_classifications', 'drawings', 'img', 'layouts', and 'sheets', as well as files like 'INB-Template 0.3.ifc', 'INB-Template 0.31.ifc', 'LICENSE', and 'README.md'. The right sidebar shows the 'About' section with a README, MIT license, and activity, and the 'Releases' section with no published releases.

INB-Template Public

Edit Pins Watch 1 Fork 0 Star 8

main 1 Branch 0 Tags

Go to file Add file Code

About

IFC-template voor de bouw-sector in Nederland (experimentele fase)

Readme MIT license Activity Custom properties 8 stars 1 watching 0 forks Report repository

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Contributors 2

jes-r Jesse Boonhorst

File/Folder	Description	Last Commit
.idea	tekeningen	7 months ago
02_scripts	materials verwijderd zonder corrupt	yesterday
03_user example projects	Handleiding, dubbele types verwijderd	3 days ago
04_development projects	materials verwijderd zonder corrupt	yesterday
05_classifications	Symbolen, palenplan, tijdelijke template erbij	4 months ago
drawings	v0.4	42 minutes ago
img	Img update	2 months ago
layouts	v0.4	42 minutes ago
sheets	v0.4	42 minutes ago
INB-Template 0.3.ifc		
INB-Template 0.31.ifc		
LICENSE	Initial commit	7 months ago
README.md	Img update	2 months ago

github.com/3BMLabs/INB-Template



Future of Bonsai



OSArch - Open Source in AEC

COLLECTIVE | community | aec | Open-source | + 8 meer

       Fiscal Host: Open Source Collective

OSArch helps support a built environment created with free software, increased transparency, and a more ethical approach.

 BIJDRAGEN  OVER ONS  BUDGET  VERBINDEN

 CONTACT

A solid foundation
\$ 10.000
per year

Estimated Annual Budget
\$ 5.066

Huidig saldo
\$ 6.986,09

Future of Bonsai

Projecten

Support the following initiatives from OSArch - Open Source in AEC.

PROJECT

Bonsai: Open multiple drawings in viewport

D&D Phase 1 Item 10 - Open multiple drawings in viewport

Meer zien

LAATSTE ACTIVITEIT DOOR

FF

PROJECT

Bonsai: Join projection planes that are coplanar

D&D Phase 1 Item 09 - Join projection planes that are coplanar

Meer zien

LAATSTE ACTIVITEIT DOOR

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PROJECT

Bonsai: Saveable view - clipping box equivalent

D&D Phase 1 Item 08 - Saveable view specific clipping box equivalent?

Meer zien

LAATSTE ACTIVITEIT DOOR

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PROJECT

Bonsai: editing sheets/layouts easier from bonsai

D&D Phase 1 Item 07 - placing and editing sheets / layouts easier from bonsai

Meer zien

LAATSTE ACTIVITEIT DOOR

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PROJECT

Bonsai: Edit text using tab, enter etc

D&D Phase 1 Item 06 - Edit text using tab, enter etc

Meer zien

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PROJECT

Bonsai: Element visibility options

D&D Phase 1 Item 04 - Element visibility options

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PROJECT

Bonsai: seeing elements behind other elements

D&D Phase 1 Item 03 - Element visibility options

Meer zien

LAATSTE ACTIVITEIT DOOR

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What to do?

- 01 Start **today** with Blender & Bonsai for IFC viewing and editing;
- 02 **Support** Blender and IfcOpenShell financially!;
- 03 Be ready to switch:
 - Start with 1 Architect/Engineer;
 - Start making the library;

Big things

1. Modelling like playing on the piano
2. Parametric Design in Blender and Bonsai
3. Improved drawing functionality
4. Improved IfcGit for collaboration